

Euclidean Three-Space over the Surreal Numbers

Coordinate geometry, spherical trigonometry,
and the rotation group $\mathrm{SO}(3, \mathbf{No})$

22 September 2026

Abstract

We develop Euclidean geometry in $\mathbf{No}^3$ in two parts. Every construction is carried out in a set-sized real closed subfield of $\mathbf{No}$, or in a set-sized real-closed Hahn field when analytic evaluation is needed. Distances, areas, volumes and sphere radii may be infinite or infinitesimal. Angles between directions always have canonical finite surreal representatives, so spatial trigonometry needs no oscillatory value at an infinite argument.

Part I is coordinate geometry. It covers affine incidence, projections, skew lines, sphere intersections, quadrics and rigid motions. On spheres it proves the sine and cosine laws, polar duality, and sharp existence and congruence criteria, including an exceptional continuous SSA family. It also gives spherical centers, Ceva, a branch-correct solid-angle formula, and a finitely additive area for geodesic polygons. Parallel transport has quaternionic holonomy equal to the spherical excess, and a relative planar-limit area estimate holds for arbitrarily thin infinitesimal triangles.

Part II studies the class group $G = \mathrm{SO}(3, \mathbf{No})$. Rotation-matrix entries are bounded, and standard part gives a split extension $G = K \rtimes \mathrm{SO}(3, \mathbb{R})$. The infinitesimal kernel K is uniquely divisible, and each of its elements is a single commutator of elements of K . Every finite subgroup is conjugate to its standard part by an infinitesimal rotation. We treat strong exponential and logarithm maps and the valuation-graded cross-product algebra. For the global rotation exponential we give its fibers and differential; its critical radii are $2\pi\mathbf{Oz}$. Finally, three topologies are separated.

The rotation algebra shared by the two parts is printed once in Part I: axes, Rodrigues, quaternions, and the Cayley chart. The quaternion double cover is cited from the collection's surquaternion report.

Status and provenance. An AI-assisted research report merged from two manuscripts of 22 September 2026 (sources 04 and 09 of the batch that placed them); see [section 1.1](#). It has not been refereed, and no statement in it has a Lean formalization. Classical identities, the Bays–Peterzil perfectness precedent and imported analytic foundations are credited, not claimed. No priority claim or resolution of a named open problem is made. The accompanying symbolic checks test selected finite identities only. Repository pin of this merge: 50cb709.

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1 Scope, sources, and conventions

An infinite coordinate is not a point at infinity, and an infinitesimal distance is not zero. Those two distinctions are enough to change many geometric interpretations, but they do not destroy Euclidean coordinate algebra. The appropriate starting point is an ordered field equipped with positive square roots, rather than the completeness of the real line.

The central principle of Part I is

Coordinates and lengths can have arbitrary scale; geometric angles can always be finite.

For example, the vectors $(\omega, 1, 0)$ and $(\omega, 0, 0)$ make the nonzero angle $\text{atan}(\omega^{-1})$. The large coordinate does not require evaluation of $\sin \omega$. Likewise, a great-circle arc on a sphere of radius $R = \omega$ can have an infinite length Ra while its central angle a lies in $[0, \pi]$. The dimensionless quantity entering trigonometric functions is a , not Ra .

Part II asks what the group of rotations of this space is. Let $G(F) = \text{SO}(3, F) = \{A \in M_3(F) : A^T A = I, \det A = 1\}$ for a real closed field F ; the group of interest is $G(\mathbf{No})$, read with the size conventions of [section 2](#). There are three complementary answers:

$$G(\mathbf{No}) \cong \text{Sp}(1, \mathbf{No})/\{1, -1\}, \quad (1.1)$$

$$G(\mathbf{No}) \cong K \rtimes \text{SO}(3, \mathbb{R}), \quad (1.2)$$

$$K \cong (\mathfrak{m}^3, \star), \quad t \star u = \frac{t + u + t \times u}{1 - t \cdot u}. \quad (1.3)$$

Here $\text{Sp}(1, \mathbf{No})$ is the unit group of the Hamilton quaternion algebra, $\mathfrak{m}$ is the class of infinitesimal surreals, and K consists of the rotation matrices infinitesimally close to the identity. The denominator in [equation \(1.3\)](#) is a unit infinitesimally close to one, so the formula is valid on all of $\mathfrak{m}^3$. It is a noncommutative group law, not vector addition. The first isomorphism is the algebraic double cover of the collection's surquaternion report; the second and third are [theorem 18.1](#) and [proposition 18.2](#).

The geometry of a rotation still has three scalar degrees of freedom. What changes is the range of those scalars and the meaning of analysis. Infinitesimal changes of axis and angle give genuinely different rotations even when their standard parts agree. A small rotation can move an infinitely distant point by a finite or infinite distance. Ordinary real paths, compactness arguments, and power-series convergence cannot simply be imported into the full fine topology.

1.1 Two manuscripts, one report

This report is merged from two manuscripts, both dated 22 September 2026 and written independently against different commits of the repository `VladimirReshetnikov/Surreal`. The source numbers are those of the batch in which they arrived.

Source	Manuscript	Pin	Contribution here
04 (base)	<i>Analytic Geometry and Trigonometry in Surreal Three-Space: finite angles, arbitrary scales, spherical duality, solid angles, rotations, and curvature</i> (34 pages)	d22a5b3	Sections 2 to 15 , that is, section 2 and all of Part I; sections 28.1 and 28.3 ; the first proof of theorem 2.2 . Files prefixed <code>04-three-space-</code> .

Source	Manuscript	Pin	Contribution here
09	<i>Rotations of Surreal Three-Space: Algebra, Spin, Infinitesimal Structure, and Topology</i> (31 pages, 19 sections, 26 numbered statements)	dcf8666	All of Part II (sections 16 to 27); in Part I the localization proposition and three-part decomposition of section 2, the hat identities of section 3, and the algebraic-angle, spin-lift, bounds and Cayley-chart statements of section 5; sections 28.2 and 28.4. Files prefixed 09-so3-.

Why one report. The two manuscripts overlap in the finite algebra of rotations and are otherwise complementary. Source 04 is a geometry of points, lines, spheres and spherical triangles, for which rotations are one tool; source 09 is a structure theory of the rotation group. Source 04 is the base: it is the wider development and fixes the scalar conventions. Source 09 forms Part II, after the geometry.

Printed once. The following results are proved in both sources and appear once here. We say which source's statement and proof are printed.

- (a) The finite phase theorem, [theorem 2.2](#): source 04's statement and proof, with source 09's logarithmic proof kept as a marked second route.
- (b) The classification of distance-preserving maps, [theorem 5.1](#). We print source 04's affine statement, add source 09's orientation clause and its remark on field automorphisms, and give one proof; the two proofs were the same polarization argument.
- (c) The axis theorem and Rodrigues formula, [theorem 5.3](#). The statement is source 09's algebraic-angle form, which contains source 04's finite-angle form [equation \(5.2\)](#). Both proofs were the same determinant argument.
- (d) Finite axis-angle representatives, [theorem 5.4](#). This combines a paragraph of source 04 with the second half of source 09's theorem on finite angles, whose proof is printed.
- (e) The Hamilton product and the quaternion rotation formula, [equations \(5.6\) and \(5.7\)](#). The two sources give these identically.
- (f) The two-to-one spin map $\mathrm{Sp}(1, F) \rightarrow \mathrm{SO}(3, F)$ with kernel $\{\pm 1\}$, [theorem 5.5](#). Both sources reprove it by the argument that the surquaternion report already gives as the first route of `squat:thm:rotations`. We cite that theorem and do not repeat the proof; only source 09's explicit lift formula is printed.
- (g) The rational Cayley chart, [theorem 5.6](#). Source 04's vector formula [equation \(5.15\)](#) and source 09's matrix formula are the same map, because $\mathbf{t} \times (\mathbf{t} \times \mathbf{x}) = \mathbf{t}(\mathbf{t} \cdot \mathbf{x}) - \|\mathbf{t}\|^2 \mathbf{x}$. The theorem is source 09's; source 04's remark on infinite chart coordinates is kept.
- (h) Scalar foundations. The finite and infinitesimal elements, the standard part, the Hahn valuation, strong summability and the commutative positive-support lemma appear once, in [section 2](#). The noncommutative all-lengths version needed for BCH is [lemma 22.1](#).
- (i) The symmetric spectral theorem over a real closed field. It is proved in [section 4.3](#) (source 04); source 09's proof of [lemma 20.1](#) used the same eigenvector argument and now refers to it.

Where the merge had to choose. Symbols were renamed only where the two sources, or a source and the rest of the collection, used one symbol for two things. [Section 1.4](#) lists each renaming.

The largest is that source 04’s scalar workspace K is written F here, because K is source 09’s infinitesimal kernel. The word *fine* also changes meaning: source 04 used it for the order topology of one set-sized workspace, source 09 for the full surreal topology. Here *fine* always has the second meaning, in agreement with the surquaternion and foundations reports. Source 04’s usage is written *intrinsic*. Labels from source 04 carry the prefix `e3:`, and labels from source 09 carry `e3:so:`. Text written for the merge is confined to this section, the framing of [section 5](#), [section 29](#), the ledgers, and short cross-reference remarks. It adds no mathematical claim ([section 29.3](#)).

1.2 Relation to the collection

Trigonometry. The trigonometry report [11], in `docs/surcomplex/trigonometry`, develops canonical finite phases, arbitrary-scale plane geometry and the spherical sine and cosine laws. [Theorem 2.2](#) is the Hahn-workspace form of its `trigonometry:thm:polar`. The choice of representatives is its `trigonometry:cor:representatives`, and the planar half-angle chart is its `trigonometry:thm:cayley`. The spherical laws of [theorem 8.2](#) rederive its `trigonometry:thm:spherical` from Gram data, adding polar duality, reconstruction and area. The Ehrlich–Kaplan phase of [section 24](#) is its `trigonometry:def:globaltrig` and `trigonometry:thm:globalexp`. A treatment of $\mathrm{SO}(2, \mathbf{No})$ is being added to that report at the same time as this merge. This report cites only labels present at its pin 50cb709.

Surquaternions. The report *Surquaternions* (`docs/surquaternions/surquaternions`) [13] already contains the following, cited by label with its theorem numbers at pin 50cb709:

- the rotation double cover, and the isomorphism of $\mathrm{SO}(3, F)$ with $\mathrm{Aut}_F(\mathbb{H}_F)$ (`squat:sub:spin`; `squat:thm:rotations`, Theorem 4.4);
- the quaternion Cayley and stereographic chart (`squat:prop:charts`, Proposition 4.5);
- the standard-part splitting of the unit group (`squat:sub:unitreduction`, §4.3);
- infinitesimal exponential and logarithm, and the filtered group of infinitesimal rotations (`squat:thm:principal-log`, `squat:thm:rotation-log`, `squat:thm:graded`; Theorems 8.1, 8.5, 8.6);
- the all-lengths positive-support lemma (`squat:lem:neumann`, Lemma 7.1) and set discreteness of the fine topology (`squat:thm:discrete`, Theorem 7.3);
- the differential of the four-dimensional radial exponential (`squat:thm:exp-derivative`, Theorem 11.2).

Part II works instead with 3×3 rotation matrices and matrix Cayley coordinates. The two descriptions differ by the factor two of the spin differential ([proposition 21.1](#)). In particular the rank-one critical spheres of the rotation exponential in [theorem 25.2](#) are not the rank-two critical spheres of `squat:thm:exp-derivative`. At its pin dcf86662b574, source 09 records these surquaternion results as starting points, not as its own discoveries. At its pin d22a5b35d5b3, source 04 inspected only the trigonometry report and one Lean file. It does not mention the surquaternion report, although that report already contained the double cover and the quaternion Cayley chart at that commit. The citations in [section 5](#) correct this omission.

Foundations. The foundations report (`docs/foundations-and-computation/foundations`) [14] proves set discreteness of the fine topology (`found:thm:discrete`) and separates intrinsic from full-class subspace topologies (`found:prop:twotopologies`). [Theorem 26.1](#) specializes the first to rotation coordinates.

Lean. `Surreal/Algebra/Geometry.lean` develops two-dimensional dot products, determinants, Cauchy–Schwarz, the Ptolemy inequality and the area form of Heron’s identity over ordered fields with nonnegative square roots, for labels of the trigonometry report [12]. It explicitly distinguishes these field-valued lengths from the real-valued distances used by conventional metric-space interfaces. The repository contains no three-dimensional Euclidean or rotation-group Lean development, and no label of this report is mapped to a Lean declaration.

Vector and tensor fields. The report `docs/surreal/vector-and-tensor-fields`, placed in the same commit as this one, develops finite tensor algebra, metrics and Levi-Civita connections over set-sized real closed surreal fields. The explicit sphere connection, curvature and transport of [section 12](#) are computed directly here and are not derived from that framework.

1.3 Three levels of assumptions

Level	Scalar assumptions	What the assumptions support
Algebraic	Ordered field with positive square roots; real closed when roots of general polynomials are used	Incidence, norms, cross products, rigid motions, Gram matrices, algebraic sine–cosine data, the finite algebra of $\mathrm{SO}(3, F)$
Standard part	A real closed field $F \supseteq \mathbb{R}$	The split extension, the infinitesimal kernel, finite subgroups
Trigonometric	A real-closed Hahn workspace with its canonical restricted analytic structure	Finite angle representatives, inverse trigonometric functions, triangle identities, bounded analytic formulas, strong exponential and logarithm
Differential	Explicit restricted-analytic charts and differentiation in an external parameter, or the fine Fréchet derivative	Metric tensors, sphere curvature, finite-angle geodesics, parallel transport, the differential of the rotation exponential

The trigonometric and differential levels do not imply an unrestricted integration theory. Our spherical polygon area is explicitly constructed and proved finitely additive. The distinction matters even for apparently harmless real arguments; [section 15](#) gives concrete counterexamples to naive transfers of calculus.

Source 09 states its hypotheses in the same spirit. The proofs in [sections 5, 16](#) and [17](#) are finite algebra over a real closed field. Standard-part results require $F \supseteq \mathbb{R}$. Strong sums are constructed either in the full surreal normal form or in a named divisible Hahn workspace. Global infinite angles use the specific Ehrlich–Kaplan normalization. Semialgebraic homotopy statements are made over set-sized real closed fields. These are different hypotheses; none is silently substituted for another.

1.4 Notation, and the renamings made by the merge

F denotes a real closed scalar field. In Part I it is a set-sized workspace inside **No**, chosen in [section 2](#); in Part II it is any real closed field unless $F \supseteq \mathbb{R}$ is stated. Bold letters denote vectors in F^3 in Part I; Part II, following source 09, writes vectors in plain type. Points can be subtracted to give vectors, and an origin is chosen whenever coordinates are used. The physical radius of a sphere is $R > 0$. For a spherical triangle with unit direction vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, the opposite *angular* side lengths are a, b, c , and the interior angles are α, β, γ . Physical side lengths are Ra, Rb, Rc . We

consistently use

$$x = \mathbf{v} \cdot \mathbf{w} = \cos a, \quad y = \mathbf{w} \cdot \mathbf{u} = \cos b, \quad z = \mathbf{u} \cdot \mathbf{v} = \cos c. \quad (1.4)$$

The determinant $\tau = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ is oriented; $D = \tau^2$ is the Gram determinant; $d = \sqrt{D} = |\tau|$ is nonnegative. These symbols will not also denote area or a side length. In [section 16](#) the letters a, b, c, d are instead the coordinates of a quaternion $a + bi + cj + dk$.

The merge fixed one meaning per symbol as follows.

Here	Source notation	Reason
$F, F^3, \text{SO}(3, F)$ $\mathbb{T}(F)$	source 04: $K, K^3, \text{SO}(3, K)$ source 04: $U(K)$, the unit circle	K is the infinitesimal kernel of Part II U denotes elements of K ; $\mathbb{T}$ as in the trigonometry report
$\text{Rot}(n; c, s)$ A for rotation matrices in section 5	source 09: $R(n; c, s)$; source 04: $Q_{n, \theta}$ source 04: Q	R is a sphere radius Q is also a point
Φ for an isometry $A_0 = \text{st}(A), B, B'$ real rotations	source 04: F source 09: R_0, R, S	F is the field R is a radius; S_u a reflection
$\mathcal{E}, \mathcal{J}(w)$	source 09: $E, J(w)$ for the global rotation exponential and its differential	E is the spherical excess, J a Jacobian
dual number κ , $\kappa^2 = 0$	source 09: τ	τ is an oriented determinant
Gn, St	source 04: G, H for the gnomonic and stereographic charts	$G(F) = \text{SO}(3, F)$; H_n is a half-turn
$\sin, \cos, \text{cis}$ on finite angles $\hat{x}$	source 09: $\sin_0, \cos_0, \text{cis}_0$ source 04: $[x]_\times$	one finite-angle convention; $\text{Sin}, \text{Cos}, \text{cis}_{\text{EK}}$ remain the global ones one name for the cross-product matrix
$f = \text{O}(g)$	source 04: $F = \text{O}(G)$	letters in use
<i>intrinsic</i> topology of a workspace	source 04: <i>fine</i>	<i>fine</i> means the full surreal topology

The local matrix $E = P - I$ in source 09's positive-root lemma is written Y . Two further symbols keep a double use from their sources. The letter t is the Hahn monomial variable in t^γ ; the vector $\mathbf{t}$, or $t \in F^3$ in Part II, is a Cayley coordinate. $\text{Exp}_P, \text{Log}_P$ with a point subscript are the exponential map of a sphere and its inverse, while $\text{Exp}_m, \text{Log}_m$ are the strong matrix exponential and logarithm.

2 Scalar foundations and canonical finite trigonometry

2.1 Set-sized workspaces inside the surreal numbers

The surreal numbers form a real-closed ordered *class* field, not a set. We use their Conway normal forms but perform each construction in a set-sized field. Let $\Gamma \subset \mathbf{No}$ be a set-sized divisible ordered additive subgroup and put

$$F = \mathbb{R}((t^\Gamma)), \quad \sum_{\gamma} a_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} a_{\gamma} \omega^{-\gamma} \in \mathbf{No}. \quad (2.1)$$

Supports in the left-hand expression are well ordered. This identifies F with a real-closed subfield of $\mathbf{No}$. Its canonical restricted-analytic structure is an elementary extension of $\mathbb{R}_{\text{an}}$; these structures are compatible under enlargement of Γ [[1](#), §2].

Every finite configuration of surreal coordinates belongs to such a workspace: collect all exponents occurring in its finitely many normal forms and take their rational span. More generally the same argument works for any set of inputs. Neither the support set nor this span is a proper class. All operations used below stay in a workspace or in a specified larger one. Consequently the theorems apply to arbitrary individual surreal coordinates without treating $\mathbf{No}$ as a set or forming a proper-class power set.

Source 09 makes the same point for the rotation group. There is no set containing all surreal numbers, so $\mathbf{No}^3$, $M_3(\mathbf{No})$ and $G(\mathbf{No})$ are proper classes. One may use a class theory such as NBG, or a universe-relative formulation in which the surreal carrier is large relative to the universe containing its individual sign sequences. A matrix still has only nine entries, and every formula for its multiplication, determinant, inverse, or action is finite. Statements about all elements of this class group are schemes of ordinary finite algebra. Statements about a finite subgroup, a sequence, or a set-indexed family retain their ordinary size restriction. We do not form an unrestricted set of all subsets of $\mathbf{No}$, or invoke ordinary compactness for such a nonexistent set-sized space. For finite algebra alone a smaller field suffices.

Proposition 2.1 (Localization of finite algebra). *Every set-sized family of surreal coordinates lies in a set-sized real closed subfield $F \subseteq \mathbf{No}$. One can require $\mathbb{R} \subseteq F$. Thus every set-sized configuration of vectors and matrices can be studied inside an honest real closed field without changing its finite algebraic identities.*

Proof. Adjoin the coordinates to $\mathbb{R}$, obtaining a set-sized ordered subfield. Take its real closure inside the real closed field $\mathbf{No}$: equivalently, take the elements of $\mathbf{No}$ algebraic over that subfield. They form a set and a real closed field. The inclusion preserves the arithmetic and order. It does not assert that the subfield is closed under arbitrary strong sums or inherits the same topology as its intrinsic order topology. $\square$

The Hahn workspace [equation \(2.1\)](#) is larger than this subfield; its purpose is to contain strong sums and analytic evaluations. An arbitrary birthday cutoff is not a substitute for either construction. For example, the finite-birthday surreals are dyadic rationals and are not closed under division. A workspace must have the closure properties actually needed.

We write

$$\mathcal{O} = \{a \in F : |a| < n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m} = \{a \in F : |a| < 1/n \text{ for every } n \geq 1\}.$$

These are respectively the finite elements and the infinitesimals. Every $a \in \mathcal{O}$ has a unique decomposition

$$a = \text{st}(a) + \varepsilon, \quad \text{st}(a) \in \mathbb{R}, \quad \varepsilon \in \mathfrak{m}.$$

The standard-part map is an ordered ring homomorphism on $\mathcal{O}$, with kernel $\mathfrak{m}$. In a Hahn workspace it is the coefficient of t^0 for finite series. The same definitions apply to any ordered field $F \supseteq \mathbb{R}$, with subscripts $\mathcal{O}_F, \mathfrak{m}_F$ where the field matters; $\mathcal{O}$ and $\mathfrak{m}$ without subscript refer to the workspace, or to all of $\mathbf{No}$. In that generality the real cut determined by a bounded a has a real supremum, whose difference from a is infinitesimal. Uniqueness follows because no nonzero real is infinitesimal (source 09).

A surreal has a set-supported normal form $x = \sum_{a \in S} r_a \omega^a$ with $r_a \in \mathbb{R} \setminus \{0\}$ and S reverse well ordered. Splitting the exponents into positive, zero, and negative parts gives

$$x = x^\uparrow + r + \varepsilon. \tag{2.2}$$

The first part is purely infinite, the second is real, and the third is infinitesimal. Purely infinite does not mean positive: a purely infinite number may have either sign. Normal-form and analytic foundations are treated in [1, 3]; only set-supported sums are used.

For $a \neq 0$, let $v(a)$ be the least exponent in its Hahn support, and set $v(0) = +\infty$. In normal-form terms, $v(x) = -\max \text{supp}_\omega(x)$, the negative of the leading exponent, so $v(\omega^{-\gamma}) = \gamma$; the two sources use this one convention, which is also the collection's. For a vector or matrix, v is the minimum of the valuations of its entries. Thus $v(t^\gamma) = \gamma$, positive valuation means infinitesimal magnitude, and $v(ab) = v(a) + v(b)$. If $a, b \neq 0$, the notation $a \sim b$ means $a/b = 1 + \eta$ with $\eta \in \mathfrak{m}$. Later, $f = O(g)$ means $|f| \leq C|g|$ for a positive *real* constant C in the stated range; this is not an assertion about convergence of a sequence of ordinary partial sums.

2.2 What an infinite Taylor sum means

A family of Hahn series is *strongly summable* when the union of its supports is well ordered and every exponent occurs in only finitely many members. The sum is then defined coefficientwise. The positive-support lemma of Hahn-series algebra says that if S is a well-ordered subset of $\Gamma_{>0}$, its additive monoid is well ordered and every exponent has only finitely many decompositions as an unordered finite sum from S . This is the summability mechanism used below; one must not replace it by a claim that $n\delta$ is cofinal in Γ for every $\delta > 0$. The latter claim is false in higher-rank ordered groups. See [4, 2] for the relevant support theory. The all-lengths version for ordered finite words, needed for noncommutative matrix series, is [lemma 22.1](#), which is `squat:lem:neumann` of the surquaternion report.

If f is real analytic near a real number r and $h \in \mathfrak{m}$, define

$$f^F(r + h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(r)}{n!} h^n. \quad (2.3)$$

The positive-support lemma proves strong summability. It also justifies multiplication, substitution, and differentiation of these expressions when used in the corresponding analytic germs. This is consistent with the restricted-analytic structure in [equation \(2.1\)](#).

For a nonzero ordinary real argument, the usual Taylor series about zero need not be a strongly summable Hahn family: infinitely many nonzero real terms have exponent zero. One first evaluates the real analytic function at its real center, and only then evaluates its infinitesimal Taylor expansion. This convention is essential, not a stylistic choice.

2.3 Sine, cosine, and inverse tangent

For $a = r + h \in \mathcal{O}$, define $\sin a$ and $\cos a$ by [equation \(2.3\)](#). In particular,

$$\sin h = h - \frac{h^3}{3!} + \frac{h^5}{5!} - \cdots, \quad (2.4)$$

$$\cos h = 1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \cdots. \quad (2.5)$$

The formal addition identities, together with the ordinary real identities, give

$$\sin(a + b) = \sin a \cos b + \cos a \sin b, \quad \cos(a + b) = \cos a \cos b - \sin a \sin b, \quad (2.6)$$

and $\cos^2 a + \sin^2 a = 1$. In particular $\sin h = h(1 + O(h^2))$. The real sign and monotonicity properties on any fixed real bounded interval hold with surreal inputs as well. They follow either from the local expansions and endpoint checks, or from restricted-analytic elementarity.

Define $\text{atan } a$ by real-centered Taylor evaluation for finite a , and for infinite a by

$$\text{atan } a = \text{sgn}(a) \frac{\pi}{2} - \text{atan}(1/a). \quad (2.7)$$

It is the unique angle in $(-\pi/2, \pi/2)$ whose tangent is a . Thus inverse tangent accepts arbitrary surreal inputs and always returns a finite surreal angle. Tangent itself is used only where its cosine denominator is nonzero.

Theorem 2.2 (Every direction has a finite phase). *Let $\mathbb{T}(F) = \{(c, s) \in F^2 : c^2 + s^2 = 1\}$, with multiplication $(c, s)(c', s') = (cc' - ss', sc' + cs')$. The map*

$$\theta \longmapsto (\cos \theta, \sin \theta)$$

induces an isomorphism $\mathcal{O}/(2\pi\mathbb{Z}) \simeq \mathbb{T}(F)$. Consequently every unit-circle direction has a unique representative in $[0, 2\pi)$, and also a unique representative in $(-\pi, \pi]$.

Proof. Both coordinates of a unit-circle point are finite. Its standard part is an ordinary real unit-circle point. Rotate this standard part to $(1, 0)$ using an ordinary real phase. The resulting point (C, S) has $C = 1 + \mathfrak{m}$, $S \in \mathfrak{m}$, and $C > 0$. Put $h = \text{atan}(S/C) \in \mathfrak{m}$. Since the unit point with positive first coordinate and slope S/C is unique, $(C, S) = (\cos h, \sin h)$. This proves surjectivity. Addition follows from [equation \(2.6\)](#).

For the kernel, the real standard part of a phase must be $2\pi k$ for an ordinary integer k . Subtract it. An infinitesimal h with $\sin h = 0$ must vanish because $\sin h = h(1 + O(h^2))$ and the second factor is a unit. Thus the kernel is exactly $2\pi\mathbb{Z}$, not an unspecified group of infinite periods. Finally a finite element can be reduced into either stated interval by an ordinary integer multiple of 2π . $\square$

Second proof (source 09). Write $\text{cis } \theta = \cos \theta + i \sin \theta$ in $F(i)$. For $\theta = r + \varepsilon$ with $r \in \mathbb{R}$ and $\varepsilon \in \mathfrak{m}$, real-centered evaluation gives $\text{cis } \theta = e^{ir} \sum_{k \geq 0} (i\varepsilon)^k / k!$; this is [equation \(24.1\)](#). For a unit surcomplex $z = c + is$, both coordinates are finite. Its standard part z_0 is an ordinary unit complex number. Choose a real angle r with $e^{ir} = z_0$. Then $z/z_0 = 1 + \delta$ for infinitesimal δ , so its strong logarithm is defined. Conjugation and inversion give

$$\overline{\log(z/z_0)} = \log(\overline{z/z_0}) = \log((z/z_0)^{-1}) = -\log(z/z_0).$$

It is therefore $i\varepsilon$ with real infinitesimal ε , and $z = \text{cis}(r + \varepsilon)$. This proves surjectivity. If $\text{cis}(r + \varepsilon) = 1$, standard part gives $r \in 2\pi\mathbb{Z}$; then logarithmic injectivity gives $\varepsilon = 0$. $\square$

The second proof is the argument of `trigonometry:thm:polar`, which proves the same isomorphism over all of **No** [11]. The choice of representatives, including the endpoint correction for infinitesimal excursions, is `trigonometry:cor:representatives`.

For $(X, Y) \neq (0, 0)$, define $\text{atan2}(Y, X) \in (-\pi, \pi]$ as the phase of $(X, Y)/\sqrt{X^2 + Y^2}$. With $r = \sqrt{X^2 + Y^2}$, an explicit formula is

$$\text{atan2}(Y, X) = 2 \text{atan} \frac{Y}{r + X} \quad \text{when } r + X > 0, \quad (2.8)$$

with value π on the negative X -axis. This definition works when either coordinate is infinite or infinitesimal.

For $-1 \leq x \leq 1$, let

$$\operatorname{acos} x = \operatorname{atan2}(\sqrt{1-x^2}, x) \in [0, \pi], \quad \operatorname{asin} x = \frac{\pi}{2} - \operatorname{acos} x. \quad (2.9)$$

Here the phase at $(1, 0)$ is zero. These inverse functions include their endpoints. For $-1 < x \leq 1$, another exact formula is $\operatorname{acos} x = 2 \operatorname{atan} \sqrt{(1-x)/(1+x)}$.

Remark 2.3 (Infinite input is a separate question). The present construction does not assert that no distinguished global sine or cosine exists. Ehrlich and Kaplan construct such functions using the initial integer part of **No** [3, §11]; their construction supplies additional structure and agrees with the finite theory used here. What is unnecessary for geometry is choosing a phase for every infinite input. The naive zero-centered sine series at ω is not a strongly summable Hahn series, regardless of whether a separate global definition is chosen. Part II uses that global phase explicitly, with its period class, in [section 24](#).

Part I

Coordinate geometry of **No**³

Part I is source 04. Rotations enter as one tool among several, and the shared rotation algebra of both sources is in [section 5](#).

3 Vectors, distances, orientation, and volume

For $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ set

$$\mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^3 x_j y_j, \quad \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}, \quad d(P, Q) = \|Q - P\|.$$

The cross product is given by the usual determinant formula. We use the orientation for which $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$.

Theorem 3.1 (Euclidean vector identities over the scalar field). *For all vectors in F^3 ,*

$$\|\mathbf{x} \times \mathbf{y}\|^2 = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2 - (\mathbf{x} \cdot \mathbf{y})^2, \quad (3.1)$$

$$\mathbf{x} \times (\mathbf{y} \times \mathbf{z}) = \mathbf{y}(\mathbf{x} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{x} \cdot \mathbf{y}), \quad (3.2)$$

$$(\mathbf{x} \times \mathbf{y}) \cdot (\mathbf{z} \times \mathbf{w}) = (\mathbf{x} \cdot \mathbf{z})(\mathbf{y} \cdot \mathbf{w}) - (\mathbf{x} \cdot \mathbf{w})(\mathbf{y} \cdot \mathbf{z}). \quad (3.3)$$

Moreover $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ and $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. For nonzero $\mathbf{x}, \mathbf{y}$, equality in the latter holds exactly when $\mathbf{y} = \lambda \mathbf{x}$ for some $\lambda > 0$.

Proof. The three displayed identities follow by expansion in coordinates over any commutative ring. Positivity of the left side of [equation \(3.1\)](#) and nonnegative square roots give Cauchy–Schwarz. Squaring the proposed triangle inequality reduces it to Cauchy–Schwarz. Equality requires $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\|$, hence $\mathbf{x} \times \mathbf{y} = 0$. The two vectors are then proportional, and the positive dot product makes the proportionality factor positive. The converse is direct. Source 09 proves Cauchy–Schwarz by a second finite route: for $\mathbf{y} \neq 0$, minimizing the quadratic $\|\mathbf{x} - t\mathbf{y}\|^2$ algebraically at $t = (\mathbf{x} \cdot \mathbf{y})/(\mathbf{y} \cdot \mathbf{y})$ gives the inequality. $\square$

Thus d is a positive-definite, symmetric, triangle-inequality distance with values in $F_{\geq 0}$. It is a *field-valued* distance, not an ordinary real-valued metric. The coordinate formulas do not implicitly make F^3 a Banach space over $\mathbb{R}$.

The cross product is represented by the matrix

$$\hat{x} = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix}, \quad \hat{x}y = x \times y. \quad (3.4)$$

Source 09 records the following polynomial identities, valid over every field of characteristic different from two:

$$\hat{x}^2 = xx^T - \|x\|^2 I, \quad \hat{x}^3 = -\|x\|^2 \hat{x}, \quad (3.5)$$

$$[\hat{x}, \hat{y}] = \widehat{x \times y}. \quad (3.6)$$

Applied to a vector, the first identity is the case of equal first two arguments of the vector triple product [equation \(3.2\)](#). No limiting process is involved.

For nonzero vectors define their unoriented angle by

$$\text{ang}(\mathbf{x}, \mathbf{y}) = \text{atan2}(\|\mathbf{x} \times \mathbf{y}\|, \mathbf{x} \cdot \mathbf{y}) \in [0, \pi]. \quad (3.7)$$

Lagrange's identity shows that its cosine is the normalized dot product and its sine is the normalized cross-product norm. If the vectors lie in a plane with specified unit normal $\mathbf{n}$, their oriented angle is instead $\text{atan2}(\mathbf{n} \cdot (\mathbf{x} \times \mathbf{y}), \mathbf{x} \cdot \mathbf{y})$.

3.1 Areas and tetrahedra

The area of a flat triangle PQR and the oriented volume of a tetrahedron $PQRS$ are

$$\text{Area}(PQR) = \frac{1}{2} \|(Q - P) \times (R - P)\|, \quad V_{\text{or}}(PQRS) = \frac{1}{6} \det[Q - P \ R - P \ S - P]. \quad (3.8)$$

The unoriented volume is $|V_{\text{or}}|$. These are algebraic definitions: finite dissection and linear change-of-variables identities follow from multilinearity. They preserve orientation and multiplicity information at infinitesimal scales.

If M is the matrix of the three edge vectors from P , then

$$\det(M^T M) = \det(M)^2 = 36 \text{Vol}(PQRS)^2.$$

In terms of the six pairwise distances d_{ij} , this becomes

$$288 \text{Vol}(PQRS)^2 = \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 \\ 1 & d_{01}^2 & 0 & d_{12}^2 & d_{13}^2 \\ 1 & d_{02}^2 & d_{12}^2 & 0 & d_{23}^2 \\ 1 & d_{03}^2 & d_{13}^2 & d_{23}^2 & 0 \end{pmatrix}. \quad (3.9)$$

Indeed $2M_i \cdot M_j = d_{0i}^2 + d_{0j}^2 - d_{ij}^2$; subtracting the corresponding rows and columns in the displayed determinant reduces it to eight times $\det(M^T M)$.

3.2 Finite-dimensional minimization does not need completeness

Let $V \subset F^3$ have an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_k$. Gram–Schmidt constructs one whenever a finite independent spanning family is given; it uses only finitely many nonzero divisions and positive square roots, so it constructs orthonormal frames in F^3 exactly as over $\mathbb{R}$, with no completeness assumption. The orthogonal projection

$$\text{proj}_V(\mathbf{x}) = \sum_{j=1}^k (\mathbf{x} \cdot \mathbf{e}_j) \mathbf{e}_j$$

satisfies, for $\mathbf{y} \in V$,

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x} - \text{proj}_V \mathbf{x}\|^2 + \|\text{proj}_V \mathbf{x} - \mathbf{y}\|^2.$$

It is therefore the unique nearest point. This proves a useful minimization theorem by a finite identity, without a compactness argument or an infimum construction. The same method gives least squares for full-column-rank matrices: the minimizer of $\|A\mathbf{x} - \mathbf{b}\|^2$ is $(A^\top A)^{-1} A^\top \mathbf{b}$.

4 Affine analytic geometry

4.1 Lines, planes, and incidence

A line is $L = P + F\mathbf{u}$ with $\mathbf{u} \neq 0$. A plane is $\Pi = \{X : \mathbf{n} \cdot X = h\}$ with $\mathbf{n} \neq 0$. A line meets this plane uniquely when $\mathbf{n} \cdot \mathbf{u} \neq 0$, at parameter

$$t = \frac{h - \mathbf{n} \cdot P}{\mathbf{n} \cdot \mathbf{u}}. \quad (4.1)$$

A zero denominator means that the line lies in the plane if the numerator also vanishes, and is disjoint from it otherwise. An infinitesimal nonzero denominator is not a reason to declare parallelism: it can produce an infinitely distant intersection.

Two nonparallel planes have line direction $\mathbf{n}_1 \times \mathbf{n}_2$. Three planes meet uniquely exactly when their normal determinant is nonzero. All remaining cases are ordinary finite-rank alternatives. For two lines $P + F\mathbf{u}$ and $Q + F\mathbf{v}$ with $\mathbf{u} \times \mathbf{v} \neq 0$, they intersect if and only if

$$(Q - P) \cdot (\mathbf{u} \times \mathbf{v}) = 0.$$

Otherwise they are skew.

The perpendicular foot and distance from P to Π are

$$P_\Pi = P - \frac{\mathbf{n} \cdot P - h}{\|\mathbf{n}\|^2} \mathbf{n}, \quad d(P, \Pi) = \frac{|\mathbf{n} \cdot P - h|}{\|\mathbf{n}\|}. \quad (4.2)$$

For a line $Q + F\mathbf{u}$, they are

$$P_L = Q + \frac{(P - Q) \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u}, \quad d(P, L) = \frac{\|(P - Q) \times \mathbf{u}\|}{\|\mathbf{u}\|}.$$

Reflection in Π sends P to $2P_\Pi - P$. These formulas are valid at every surreal scale.

Theorem 4.1 (Distance and closest points of nonparallel lines). *Let $\mathbf{r} = Q - P$ and put*

$$A = \mathbf{u}^2, \quad B = \mathbf{u} \cdot \mathbf{v}, \quad C = \mathbf{v}^2, \quad D_L = AC - B^2 > 0,$$

where $\mathbf{u}^2$ means $\mathbf{u} \cdot \mathbf{u}$. The unique closest parameters on $P + t\mathbf{u}$ and $Q + s\mathbf{v}$ are

$$t = \frac{C(\mathbf{r} \cdot \mathbf{u}) - B(\mathbf{r} \cdot \mathbf{v})}{D_L}, \quad s = \frac{B(\mathbf{r} \cdot \mathbf{u}) - A(\mathbf{r} \cdot \mathbf{v})}{D_L}. \quad (4.3)$$

Their distance is

$$d(L_1, L_2) = \frac{|\mathbf{r} \cdot (\mathbf{u} \times \mathbf{v})|}{\|\mathbf{u} \times \mathbf{v}\|}. \quad (4.4)$$

Proof. The difference $\mathbf{r} + s\mathbf{v} - t\mathbf{u}$ must be perpendicular to both direction vectors. Its two dot products give the linear system $At - Bs = \mathbf{r} \cdot \mathbf{u}$, $Bt - Cs = \mathbf{r} \cdot \mathbf{v}$, whose solution is [equation \(4.3\)](#). The remaining difference is parallel to $\mathbf{u} \times \mathbf{v}$, giving [equation \(4.4\)](#). Every other pair of points adds a vector in $\text{span}(\mathbf{u}, \mathbf{v})$ to this perpendicular difference. Pythagoras proves minimality and uniqueness of the two parameters. $\square$

For parallel lines use the point-to-line distance; the closest pair is then not unique. Angles between unoriented lines or planes are conventionally chosen acute, by taking an absolute value in the normalized dot product. Oriented normal vectors instead retain the supplementary-angle distinction. The acute line–plane angle θ for line direction $\mathbf{u}$ and plane normal $\mathbf{n}$ satisfies $\sin \theta = |\mathbf{u} \cdot \mathbf{n}| / (\|\mathbf{u}\| \|\mathbf{n}\|)$. The line–normal angle is its complement.

4.2 Spheres, circles, and discriminants

The sphere with center C and radius $R > 0$ is $\{X : \|X - C\|^2 = R^2\}$. For the line $P + t\mathbf{u}$, let t_0 be the parameter of its perpendicular foot from C , and let $h = d(C, L)$. Completing the square gives

$$\|\mathbf{u}\|^2 (t - t_0)^2 = R^2 - h^2. \quad (4.5)$$

There are two, one, or no intersections according as $R^2 - h^2$ is positive, zero, or negative. The two parameters, when distinct, are $t_0 \pm \sqrt{R^2 - h^2} / \|\mathbf{u}\|$. A plane at distance h from the center cuts a circle of radius $\sqrt{R^2 - h^2}$ when $h < R$, a point when $h = R$, and nothing when $h > R$.

For two spheres with distinct centers C_1, C_2 , let $d = \|C_2 - C_1\| > 0$ and $\mathbf{e} = (C_2 - C_1)/d$. Their intersection lies in the radical plane perpendicular to $\mathbf{e}$, at signed distance

$$q = \frac{R_1^2 - R_2^2 + d^2}{2d}$$

from C_1 . Its candidate circle has center $C_1 + q\mathbf{e}$ and squared radius $R_1^2 - q^2$. The sign of this expression gives the complete classification. Coincident centers must be handled separately. These formulas never compare an infinitesimal with zero by rounding it away.

Four affinely independent points determine a unique sphere. Translate the first point to zero and write the other three as columns of M . Its center $\mathbf{c}$ solves

$$M^T \mathbf{c} = \frac{1}{2} (\|M_1\|^2, \|M_2\|^2, \|M_3\|^2)^T. \quad (4.6)$$

The radius is $\|\mathbf{c}\|$. Small nonzero $\det M$ can make this center infinite even when all four input points are finite.

4.3 Quadrics and principal axes

A quadric has an equation

$$\mathbf{x}^\top A \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c = 0, \quad A = A^\top.$$

Over a real-closed field, a symmetric matrix has an orthonormal eigenbasis. Here is a proof that does not use compactness of a real unit sphere. A root λ of its characteristic polynomial lies in $F[i]$, and there is a nonzero vector z with $Az = \lambda z$. Since $\bar{z}^\top Az = \lambda \bar{z}^\top z$ is real and $\bar{z}^\top z = \sum |z_j|^2 > 0$, $\lambda \in F$. Its eigenspace has a nonzero real vector. Normalize it, restrict to its invariant perpendicular complement, and continue by induction. The argument works verbatim for symmetric $d \times d$ matrices over any real closed field; Part II uses it in [lemma 20.1](#).

After an orthogonal change of coordinates the quadric becomes $\sum \lambda_j y_j^2 + 2 \sum b'_j y_j + c = 0$. For $\lambda_j \neq 0$, completing the square removes the corresponding linear term. If all λ_j are nonzero this gives the familiar central forms, classified by coefficient signs and the remaining constant. Ellipsoids, hyperboloids, cones, empty loci, and point loci are distinguished exactly as over $\mathbb{R}$, with arbitrary surreal semiaxes. If a null direction retains a linear term, solving for that coordinate gives a paraboloid-type form. If all linear terms in the nullspace vanish, those directions are cylindrical. Thus the classification is algebraic, while the scales of the axes carry additional information.

4.4 Affine points are not projective ideal points

Homogeneous coordinates embed F^3 into $\mathbb{P}^3(F)$ by $\mathbf{x} \mapsto [x_1 : x_2 : x_3 : 1]$. An affine point such as $(\omega, 0, 0)$ is not $[1 : 0 : 0 : 0]$: the last homogeneous coordinate distinguishes them exactly. Projective completion is a separate incidence construction. It does not identify a very large finite-coordinate point with an ideal direction.

5 Rigid motions, rotations, quaternions, and screws

This section prints once the rotation algebra that both sources develop ([section 1.1](#)). Except for [theorem 5.4](#), everything here is finite algebra over a real closed field F . Source 04 stated it over its Hahn workspace, which is such a field; source 09 stated it over an arbitrary real closed field. We write $G(F) = \text{SO}(3, F)$ and use the cross-product matrix $\hat{x}$ of [equation \(3.4\)](#).

Theorem 5.1 (Classification of field-distance isometries). *Every map $\Phi : F^3 \rightarrow F^3$ preserving $d(P, Q) = \|P - Q\|$ has the form*

$$\Phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}, \quad A^\top A = I.$$

In particular, it is automatically onto. No continuity or affine assumption is needed. If Φ fixes the origin, it is F -linear with orthogonal matrix A , and it preserves the orientation of the standard basis exactly when $\det A = 1$. Conversely, every such matrix preserves the norm and fixes the origin.

Proof. Subtract $\Phi(0)$ and put $\mathbf{q}_j = \Phi(\mathbf{e}_j) - \Phi(0)$. Distances between $0, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and polarization,

$$(\Phi(\mathbf{x}) - \Phi(0)) \cdot (\Phi(\mathbf{y}) - \Phi(0)) = \frac{1}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2) = \mathbf{x} \cdot \mathbf{y},$$

show $\mathbf{q}_i \cdot \mathbf{q}_j = \delta_{ij}$. For arbitrary $\mathbf{x}$, the same identity with $\mathbf{y} = \mathbf{e}_j$ gives $(\Phi(\mathbf{x}) - \Phi(0)) \cdot \mathbf{q}_j = x_j$. The $\mathbf{q}_j$ form an orthonormal basis, so $\Phi(\mathbf{x}) - \Phi(0) = \sum x_j \mathbf{q}_j$. Its matrix is orthogonal and therefore invertible. Its determinant is 1 or -1 , and preserving the orientation of the given basis is precisely the first alternative. $\square$

The theorem excludes a potential ambiguity. A field automorphism applied to coordinates may carry squared distances to their images under that automorphism. That is a semilinear transformation of a different geometry, not an isometry preserving every surreal distance as an actual value. The rotations of F^3 are therefore exactly the elements of $G(F)$.

Proposition 5.2 (Bounds and equivariance). *For $A \in G(F)$, every entry satisfies $|A_{ij}| \leq 1$, and*

$$(Ax) \times (Ay) = A(x \times y), \quad A\hat{x}A^{-1} = \widehat{Ax}.$$

Thus no rotation matrix has an infinite entry, even when its vectors have infinite coordinates.

Proof. Each column has sum of squares one. The cross-product identity follows by pairing both sides with Az : both scalar triple products are multiplied by $\det A = 1$. The hat identity follows immediately. $\square$

5.1 Axis-angle representation

Theorem 5.3 (Axis theorem and Rodrigues formula). *Every $A \in G(F)$ fixes a unit vector n . There exist $c, s \in F$ with $c^2 + s^2 = 1$ such that*

$$A = \text{Rot}(n; c, s) = cI + (1 - c)nn^T + s\hat{n}. \quad (5.1)$$

If $A \neq I$, its fixed space is exactly Fn . Replacing (n, s) by $(-n, -s)$ leaves the rotation unchanged. Conversely $\text{Rot}(n; c, s) \in G(F)$ for every unit n and every (c, s) on the unit circle. When $(c, s) = (\cos \theta, \sin \theta)$ for a finite angle θ , which is always possible over a Hahn workspace by [theorem 5.4](#), the formula reads

$$\text{Rot}(n; \cos \theta, \sin \theta) x = (\cos \theta)x + (\sin \theta)(n \times x) + (1 - \cos \theta)(n \cdot x)n. \quad (5.2)$$

Proof. Since $A^T = A^{-1}$ and $\det A = 1$,

$$\det(A - I) = \det(A^T - I) = \det(A^{-1}(I - A)) = \det(I - A) = -\det(A - I).$$

Characteristic zero implies $\det(A - I) = 0$. Normalize a nonzero fixed vector to obtain n . Its perpendicular plane is invariant under A . In an oriented orthonormal basis $(u, n \times u, n)$, the restriction to that plane is an orthogonal 2×2 matrix of determinant one, necessarily $\begin{pmatrix} c & -s \\ s & c \end{pmatrix}$. This gives [equation \(5.1\)](#). Conversely, on $n^\perp$ the operator $Jx = n \times x$ satisfies $J^2 = -I$, so $cI + sJ$ is an orientation-preserving orthogonal map of that plane; on the axis the formula is the identity. Hence $\text{Rot}(n; c, s) \in G(F)$.

The determinant of the planar matrix minus the identity is $(c - 1)^2 + s^2 = 2(1 - c)$. It vanishes only when $c = 1, s = 0$, which gives the identity rotation. Otherwise the perpendicular plane contains no fixed vector. (Source 04's variant: if two independent vectors were fixed, their cross product would also be fixed, forcing $A = I$.) Expanding nn^T and $\hat{n}$ on a vector gives [equation \(5.2\)](#). $\square$

The pair (c, s) is an *algebraic angle*. It needs neither a sine function on all of F nor a real number measuring arc length. It is often the safest primary representation of a rotation over an arbitrary real closed field. In the basis used in the proof,

$$\text{tr } A = 1 + 2c, \quad -1 \leq \text{tr } A \leq 3, \quad \chi_A(X) = (X - 1)(X^2 - 2cX + 1). \quad (5.3)$$

Over $F(i)$, the eigenvalues are $1, c + is, c - is$. A half-turn is exactly a rotation with $c = -1, s = 0$, so

$$H_n = 2nn^\top - I, \quad H_n^2 = I, \quad \text{tr } H_n = -1. \quad (5.4)$$

Its axis is an unoriented line. These are all nonidentity involutions.

For a non-half-turn, the skew part is $A - A^\top = 2s\hat{n}$. At a half-turn it vanishes, although the rotation is not the identity. Algorithms that infer the axis only from the skew part must handle this case separately.

Theorem 5.4 (Every rotation has a finite axis–angle description). *Let F be a real-closed Hahn workspace as in section 2; by localization the statement applies to $G(\mathbf{No})$. Every $A \in G(F)$ has a representation*

$$A = \text{Rot}(n; \cos \theta, \sin \theta), \quad 0 \leq \theta \leq \pi.$$

For $A \neq I$, the number θ is unique. The axis is oriented uniquely when $0 < \theta < \pi$, by the requirement $\sin \theta > 0$, and is unoriented at $\theta = \pi$. Thus every rotation, even one involving infinitely separated spatial points, has a finite axis–angle description.

Proof. By theorem 5.3 and the finite phase theorem theorem 2.2, $A = \text{Rot}(n; \cos \theta, \sin \theta)$ for some finite θ . Reduce θ by an ordinary integral multiple of 2π to $(-\pi, \pi]$. To handle an infinitesimal excursion past an endpoint, shift once more by 2π ; the representative must lie in the actual surreal interval, not merely have standard part there. Reverse the axis if the angle is negative, obtaining $[0, \pi]$. On $(0, \pi)$ the finite sine is positive: away from the endpoints its standard part is positive, and at an infinitesimal endpoint displacement the leading Taylor term decides the sign. Equality of two circle values is exactly equality modulo $2\pi\mathbb{Z}$. With the axis-reversal convention, this gives uniqueness of θ and the stated axis ambiguity. $\square$

For unit vectors $\mathbf{u} \neq -\mathbf{v}$, put $c = \mathbf{u} \cdot \mathbf{v}$ and $\mathbf{k} = \mathbf{u} \times \mathbf{v}$. The shortest-angle rotation taking $\mathbf{u}$ to $\mathbf{v}$ is

$$A(\mathbf{u}, \mathbf{v}) = I + \hat{\mathbf{k}} + \frac{\hat{\mathbf{k}}^2}{1 + c}. \quad (5.5)$$

The formula also gives the identity when $\mathbf{u} = \mathbf{v}$. Using $\mathbf{k} \times \mathbf{u} = \mathbf{v} - c\mathbf{u}$ and $\mathbf{k} \times (\mathbf{k} \times \mathbf{u}) = -(1 - c^2)\mathbf{u}$ verifies the mapping. At an antipodal pair the formula is undefined for a genuine reason: there are many half-turns taking one vector to the other.

5.2 Quaternion representation

Write a quaternion as $(a, \mathbf{b}) \in \mathbb{H}_F = F \oplus F^3$ with multiplication

$$(a, \mathbf{b})(c, \mathbf{d}) = (ac - \mathbf{b} \cdot \mathbf{d}, a\mathbf{d} + c\mathbf{b} + \mathbf{b} \times \mathbf{d}). \quad (5.6)$$

The basis $(1, i, j, k)$ satisfies $i^2 = j^2 = k^2 = ijk = -1$. This is an associative algebra; associativity follows either from Hamilton's relations and the faithful matrix representation equation (16.4), or from the triple-product identity. Quaternion conjugation and norm square are

$$\overline{(a, \mathbf{b})} = (a, -\mathbf{b}), \quad N(a, \mathbf{b}) = a^2 + \|\mathbf{b}\|^2.$$

Conjugation reverses products, $q\bar{q} = N(q)$, and $N(pq) = N(p)N(q)$. Since a nonzero sum of squares is positive, $q^{-1} = \bar{q}/N(q)$ for every nonzero q . The center is exactly F . These are ordered-field facts, not real-analytic facts [8, 13].

Put $\mathrm{Sp}(1, F) = \{q \in \mathbb{H}_F : N(q) = 1\}$. Unit quaternions act on pure vectors by conjugation, $p(q)(\mathbf{x}) = q(0, \mathbf{x})q^{-1}$. Direct expansion gives

$$p(a, \mathbf{b}) \mathbf{x} = (a^2 - \|\mathbf{b}\|^2) \mathbf{x} + 2(\mathbf{b} \cdot \mathbf{x}) \mathbf{b} + 2a(\mathbf{b} \times \mathbf{x}), \quad (5.7)$$

which is `squat:eq:rodrigues` of the surquaternion report. It preserves the imaginary subspace and the norm. Its determinant is +1 without an appeal to connectedness. For $\mathbf{b} \neq 0$, put $n = \mathbf{b} / \|\mathbf{b}\|$ and compare with [equation \(5.1\)](#) using $c = a^2 - \|\mathbf{b}\|^2$ and $s = 2a \|\mathbf{b}\|$. If $\mathbf{b} = 0$, the map is the identity.

Theorem 5.5 (The spin exact sequence; `squat:thm:rotations`). *Conjugation gives a surjective group homomorphism with precisely two elements in every fiber:*

$$1 \longrightarrow \{1, -1\} \longrightarrow \mathrm{Sp}(1, F) \xrightarrow{p} G(F) \longrightarrow 1. \quad (5.8)$$

In particular, $\mathrm{Spin}(3, F) \cong \mathrm{Sp}(1, F)$.

This is Theorem 4.4 (`squat:thm:rotations`) of the surquaternion report [13], which also identifies $G(F)$ with the automorphism group of $\mathbb{H}_F$. Both sources reprove it by the argument of that theorem's first surjectivity route: a quaternion commuting with every pure vector is central, and a fixed axis gives a lift. The proof is not repeated here. Source 09's explicit lift of $\mathrm{Rot}(n; c, s)$ is used below. If $c \neq -1$,

$$a = \sqrt{(1+c)/2} > 0, \quad u = \frac{s}{2a} n \quad (5.9)$$

give $a^2 + \|u\|^2 = 1$, $a^2 - \|u\|^2 = c$ and $2au = sn$, so (a, u) is a lift. If $c = -1$, the unit imaginary quaternion $(0, n)$ lifts H_n . For a finite angle, source 04's lift is $(\cos(\theta/2), \sin(\theta/2)\mathbf{n})$, which gives [equation \(5.2\)](#). The standard Clifford-algebra spin group of the positive definite ternary form has this Hamilton realization; [equation \(5.8\)](#) supplies an entirely explicit model without Clifford terminology.

The phrase “double cover” here denotes this two-to-one algebraic group homomorphism, not an unproved topological covering assertion in the fine topology. The category in which it is a two-sheeted covering with the familiar fundamental groups is identified in [theorem 26.2](#).

5.3 The rational Cayley chart

Theorem 5.6 (The matrix Cayley chart). *For $t \in F^3$, define*

$$\mathrm{Cay}(t) = (I + \hat{t})(I - \hat{t})^{-1} = \frac{(1 - \|t\|^2)I + 2t\mathbf{t}^\top + 2\hat{t}}{1 + \|t\|^2}. \quad (5.10)$$

This is a bijection from F^3 onto the rotations which are not half-turns. Its inverse is

$$\hat{t} = (A - I)(A + I)^{-1}, \quad (5.11)$$

$$t = \frac{(A_{32} - A_{23}, A_{13} - A_{31}, A_{21} - A_{12})^\top}{1 + \mathrm{tr} A}. \quad (5.12)$$

Moreover, whenever $1 - t \cdot u \neq 0$,

$$\mathrm{Cay}(t) \mathrm{Cay}(u) = \mathrm{Cay}\left(\frac{t + u + t \times u}{1 - t \cdot u}\right). \quad (5.13)$$

If $1 - t \cdot u = 0$, the product is a half-turn.

Proof. The determinant of $I - \hat{t}$ is $1 + \|t\|^2 > 0$. Using [equation \(3.5\)](#) gives the displayed rational expression. Equivalently it is the rotation of the unit quaternion

$$q(t) = \frac{(1, t)}{\sqrt{1 + \|t\|^2}}.$$

Its scalar coordinate is nonzero, so the rotation is not a half-turn. Conversely, the lift [equation \(5.9\)](#) of a non-half-turn has nonzero scalar part, and dividing its imaginary part by that scalar gives t . Solving the matrix equation gives [equation \(5.11\)](#); taking skew coordinates gives [equation \(5.12\)](#). In particular,

$$\text{tr Cay}(t) = \frac{3 - \|t\|^2}{1 + \|t\|^2}, \quad 1 + \text{tr Cay}(t) = \frac{4}{1 + \|t\|^2}, \quad \det(\text{Cay}(t) + I) = \frac{8}{1 + \|t\|^2}. \quad (5.14)$$

Finally, before normalization, $(1, t)(1, u) = (1 - t \cdot u, t + u + t \times u)$. A nonzero scalar part gives the quotient in [equation \(5.13\)](#). If that scalar vanishes, the product is a nonzero imaginary quaternion and its rotation is a half-turn. $\square$

In vector form, and with $\mathbf{t} = \tan(\theta/2)\mathbf{n}$ for a rotation of finite angle θ away from half-turns, [source 04](#) writes the same chart as

$$\text{Cay}(\mathbf{t}) \mathbf{x} = \mathbf{x} + \frac{2}{1 + \|\mathbf{t}\|^2} (\mathbf{t} \times \mathbf{x} + \mathbf{t} \times (\mathbf{t} \times \mathbf{x})). \quad (5.15)$$

This rational chart can have infinite coordinates near a half-turn, although every entry of the rotation matrix remains between -1 and 1 ([proposition 5.2](#)). It is the three-dimensional analogue of the planar half-angle chart `trigonometry:thm:cayley`.

This is a chart on $G(F)$, not the quaternion stereographic chart $(1 - x)^{-1}(1 + x)$ on $\text{Sp}(1, F)$ of `squat:prop:charts`. The two are related, but their parameters have different half-angle factors. Confusing them changes the composition law and the leading infinitesimal rotation by a factor of two.

The half-turn locus is naturally $\mathbb{P}^2(F)$, by $[n] \mapsto 2nn^T - I$ after normalization. The chart and its translates show that the group has algebraic dimension three. After extending to $F(i)$, its standard algebraic form is PGL_2 ; thus the algebraic group is connected and absolutely simple of adjoint type. These are algebraic assertions. They do not imply that its F -point group is abstractly simple or that its fine topology is connected. For $F \supsetneq \mathbb{R}$, and for **No**, [sections 18](#) and [26](#) show that neither holds.

5.4 Screw motions

Let $\Phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with $A \in \text{SO}(3, F)$ nonidentity. Take its unit axis $\mathbf{n}$, put $h = \mathbf{b} \cdot \mathbf{n}$, and $\mathbf{b}_\perp = \mathbf{b} - h\mathbf{n}$. On $\mathbf{n}^\perp$, $I - A$ is invertible, so there is a unique $\mathbf{a} \perp \mathbf{n}$ satisfying $(I - A)\mathbf{a} = \mathbf{b}_\perp$. Therefore

$$\Phi(\mathbf{a} + t\mathbf{n}) = \mathbf{a} + (t + h)\mathbf{n}.$$

The motion is rotation about the line $\mathbf{a} + F\mathbf{n}$, followed by translation h along it. With $0 < \theta \leq \pi$,

$$\mathbf{a} = \frac{1}{2} (\mathbf{b}_\perp + \cot(\theta/2) \mathbf{n} \times \mathbf{b}_\perp). \quad (5.16)$$

This follows by inverting $(1 - \cos \theta)I - \sin \theta J$ on the perpendicular plane, with $J\mathbf{x} = \mathbf{n} \times \mathbf{x}$. An infinitesimal rotation with appreciable transverse translation can have an infinitely distant screw axis. The identity rotation is treated separately as a pure translation.

6 Coordinates and ordinary triangle trigonometry in space

6.1 Cylindrical and spherical coordinates

For $\mathbf{x} = (x, y, z)$ put $r = \sqrt{x^2 + y^2}$. When $r > 0$, cylindrical coordinates are

$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z, \quad \phi = \text{atan2}(y, x) \pmod{2\pi}.$$

At $r = 0$, azimuth is undetermined. There is no restriction on the scale of r or z .

For $\rho = \|\mathbf{x}\| > 0$, define colatitude $\theta = \arccos(z/\rho) \in [0, \pi]$. Away from the polar axis choose the same azimuth ϕ . Then

$$(x, y, z) = (\rho \sin \theta \cos \phi, \rho \sin \theta \sin \phi, \rho \cos \theta). \quad (6.1)$$

Every nonzero vector has this representation. At the poles the azimuth is free; at the origin both angles are free. The radial coordinate may be infinite, but both angular coordinates are finite. Choosing an azimuth branch introduces a coordinate seam, not a singularity of the sphere.

Differentiate the coordinate map with respect to its external variables. Its three coordinate derivatives are mutually perpendicular and have squared norms $1, \rho^2, \rho^2 \sin^2 \theta$. Therefore

$$ds^2 = d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2 \theta d\phi^2, \quad J = \rho^2 \sin \theta. \quad (6.2)$$

These are finite identities in restricted-analytic coordinate derivatives. They do not assert an integral transformation theorem for every function on F^3 . On a sphere of radius R , the local area density is $R^2 \sin \theta$ and the tangent metric is obtained by setting $d\rho = 0$.

6.2 Flat triangles retain their usual formulas

Every noncollinear triple of points lies in a unique affine plane. An orthonormal basis of that plane reduces its geometry to F^2 without changing lengths. Let its opposite side lengths be $A, B, C > 0$, its angles $\alpha, \beta, \gamma \in (0, \pi)$, and its area $\Delta > 0$. Then

$$\alpha + \beta + \gamma = \pi, \quad (6.3)$$

$$A^2 = B^2 + C^2 - 2BC \cos \alpha, \quad (6.4)$$

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma} = \frac{ABC}{2\Delta} = 2R_c, \quad (6.5)$$

$$\Delta^2 = s(s-A)(s-B)(s-C), \quad s = (A+B+C)/2. \quad (6.6)$$

These are the usual classical identities [6, §4.42(ii)], now with field-valued lengths.

For completeness, put the first two vertices at $(0, 0)$ and $(C, 0)$, and the third in the upper half-plane. If the ray from the first vertex to the third has phase α and the ray from the second to the third has phase δ , orientation gives $0 < \alpha < \delta < \pi$. The remaining angles are $\pi - \delta$ and $\delta - \alpha$, proving [equation \(6.3\)](#). Expanding the squared difference proves [equation \(6.4\)](#). Computing the same area from each pair of sides proves the first equalities in [equation \(6.5\)](#). The circumcenter is the solution of two perpendicular bisector equations, and the chord relation gives $A = 2R_c \sin \alpha$. Finally squaring $2\Delta = BC \sin \alpha$ and substituting the cosine law gives

$$16\Delta^2 = (A+B+C)(-A+B+C)(A-B+C)(A+B-C),$$

which proves Heron's formula.

The triangle inequalities are strict exactly in the noncollinear case. None of these statements requires the side lengths to be finite. For example, a triangle can have sides of scale ω and an altitude of scale $\omega^{-\omega}$; its nonzero area is the product scale, not automatically zero or infinite.

7 The surreal sphere and its intrinsic angular distance

Let

$$S_R^2(C) = \{C + Ru : \mathbf{u} \cdot \mathbf{u} = 1\}.$$

A great circle is its intersection with a plane through C . Translating and rescaling reduces all angular calculations to the unit sphere $S^2(F)$.

7.1 Minor arcs without a global sine function

For unit vectors $\mathbf{u} \neq -\mathbf{v}$, define the minor arc by the algebraic set

$$\left\{ \frac{(1-t)\mathbf{u} + t\mathbf{v}}{\|(1-t)\mathbf{u} + t\mathbf{v}\|} : 0 \leq t \leq 1 \right\}. \quad (7.1)$$

The denominator is nonzero: vanishing would make the endpoints antipodal unless they coincide. If $a = \text{ang}(\mathbf{u}, \mathbf{v}) \in (0, \pi)$ and

$$\mathbf{e} = \frac{\mathbf{v} - \cos a \mathbf{u}}{\sin a},$$

the same arc has the trigonometric parametrization

$$\mathbf{g}(s) = \cos s \mathbf{u} + \sin s \mathbf{e}, \quad 0 \leq s \leq a. \quad (7.2)$$

To see equality, work in the oriented plane basis $(\mathbf{u}, \mathbf{e})$. Normalized positive combinations of $(1, 0)$ and $(\cos a, \sin a)$ have exactly the phases between 0 and a , as is checked by the signs of their two oriented determinants with the endpoints. This also proves uniqueness of the minor arc. Antipodal points instead have a family of semicircles, indexed by unit directions perpendicular to either endpoint.

Define the angular distance on the unit sphere and the physical distance on $S_R^2(C)$ by

$$d_S(\mathbf{u}, \mathbf{v}) = \text{acos}(\mathbf{u} \cdot \mathbf{v}), \quad d_R(C + R\mathbf{u}, C + R\mathbf{v}) = R d_S(\mathbf{u}, \mathbf{v}). \quad (7.3)$$

The length assigned to a minor arc is Ra . Subarcs add because their finite phase differences add. One complete great circle has length $2\pi R$.

Theorem 7.1 (Angular triangle inequality). *d_S is a positive-definite F -valued distance on $S^2(F)$. For $a = d_S(\mathbf{u}, \mathbf{v})$,*

$$\|\mathbf{u} - \mathbf{v}\| = 2 \sin(a/2), \quad \frac{2}{\pi}a \leq \|\mathbf{u} - \mathbf{v}\| \leq a. \quad (7.4)$$

A minor great-circle arc minimizes the sum of arc lengths among all finite chains of minor great-circle arcs with the same endpoints.

Proof. Only the triangle inequality needs explanation. Set $a = d_S(\mathbf{u}, \mathbf{v})$ and $b = d_S(\mathbf{v}, \mathbf{w})$. If $a + b \geq \pi$ the claim follows from $d_S \leq \pi$. Otherwise write

$$\mathbf{u} = \cos a \mathbf{v} + \sin a \mathbf{e}, \quad \mathbf{w} = \cos b \mathbf{v} + \sin b \mathbf{f}$$

with unit perpendicular vectors, treating zero angles separately. Cauchy–Schwarz gives $\mathbf{e} \cdot \mathbf{f} \geq -1$, hence $\mathbf{u} \cdot \mathbf{w} \geq \cos(a + b)$. Cosine is decreasing on $[0, \pi]$, so $d_S(\mathbf{u}, \mathbf{w}) \leq a + b$. The chord identity follows by squaring. The inequalities follow from $2t/\pi \leq \sin t \leq t$ for $0 \leq t \leq \pi/2$, valid in the restricted-analytic field. Iterating the triangle inequality proves the finite-chain minimizing assertion. $\square$

This is an intrinsic distance construction with an explicit minimizing class. No supremum over arbitrary partitions, no real-valued length metric, and no unrestricted class of nondefinable curves has been silently introduced.

7.2 Gnomonic and stereographic charts

The gnomonic chart on the northern open hemisphere is

$$\text{Gn}(p, q) = \frac{(p, q, 1)}{\sqrt{1 + p^2 + q^2}}, \quad (p, q) = (u_1/u_3, u_2/u_3). \quad (7.5)$$

Great circles become straight lines because $\mathbf{n} \cdot \mathbf{u} = 0$ becomes $n_1p + n_2q + n_3 = 0$. Positive-cone spherical triangles become ordinary affine convex triangles. The chart is valid for arbitrary $p, q \in F$, including infinite values near the horizon. With $\mathbf{p} = (p, q)$ its metric on a sphere of radius R is

$$ds^2 = R^2 \frac{(1 + \|\mathbf{p}\|^2) \|d\mathbf{p}\|^2 - (\mathbf{p} \cdot d\mathbf{p})^2}{(1 + \|\mathbf{p}\|^2)^2}, \quad J_{\text{Gn}} = \frac{R^2}{(1 + \|\mathbf{p}\|^2)^{3/2}}. \quad (7.6)$$

Both follow by direct differentiation of [equation \(7.5\)](#).

Stereographic projection from the north pole has inverse

$$\text{St}(p, q) = \frac{(2p, 2q, p^2 + q^2 - 1)}{1 + p^2 + q^2}. \quad (7.7)$$

It covers the sphere minus that pole, with metric

$$ds^2 = \frac{4R^2(dp^2 + dq^2)}{(1 + p^2 + q^2)^2}.$$

The scalar multiple of $dp^2 + dq^2$ proves conformality, including at infinitesimal and infinite chart coordinates. A spherical circle cut out by $\mathbf{n} \cdot \mathbf{u} = h$ becomes

$$(n_3 - h)(p^2 + q^2) + 2n_1p + 2n_2q - (n_3 + h) = 0,$$

a circle or line. Infinite chart coordinates are not the omitted pole; they describe genuine points infinitesimally near it.

8 Spherical triangles and the Gram-matrix calculus

Definition 8.1 (Proper spherical triangle). Three linearly independent unit vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ determine the closed spherical triangle consisting of the normalized nonzero vectors in

$$\{\lambda\mathbf{u} + \mu\mathbf{v} + \nu\mathbf{w} : \lambda, \mu, \nu \geq 0\}.$$

Its edges are the minor great-circle arcs. Its interior angles are the angles between the inward-pointing edge tangents at each vertex.

Such a triangle lies in an open hemisphere. Indeed, if $P = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$, then $\mathbf{h} = P^{-\top}(1, 1, 1)^\top$ has positive dot product with every nonzero vector in the positive cone. This algebraic observation fixes which region and which angles are intended, even when its real standard part is degenerate. All sides and angles of a proper triangle lie strictly between 0 and π .

Use [equation \(1.4\)](#) and put

$$\mathcal{G} = P^\top P = \begin{pmatrix} 1 & z & y \\ z & 1 & x \\ y & x & 1 \end{pmatrix}, \quad D = \det \mathcal{G} = 1 + 2xyz - x^2 - y^2 - z^2 = \tau^2 > 0. \quad (8.1)$$

Write $s_a = \sqrt{1 - x^2} = \sin a$ and cyclically s_b, s_c . The unit tangents at $\mathbf{u}$ pointing toward $\mathbf{v}$ and $\mathbf{w}$ are

$$T_{uv} = \frac{\mathbf{v} - z\mathbf{u}}{s_c}, \quad T_{uw} = \frac{\mathbf{w} - y\mathbf{u}}{s_b}. \quad (8.2)$$

Consequently

$$\cos \alpha = \frac{x - yz}{s_b s_c}, \quad \sin \alpha = \frac{d}{s_b s_c}, \quad \alpha = \text{atan2}(d, x - yz), \quad (8.3)$$

with cyclic formulas for β, γ . The common positive denominator in the first two formulas is irrelevant to `atan2`.

Theorem 8.2 (Spherical sine and cosine laws). *Every proper spherical triangle satisfies*

$$\cos a = \cos b \cos c + \sin b \sin c \cos \alpha, \quad (8.4)$$

$$\frac{\sin \alpha}{\sin a} = \frac{\sin \beta}{\sin b} = \frac{\sin \gamma}{\sin c} = \frac{d}{\sin a \sin b \sin c}. \quad (8.5)$$

Cyclic permutations give the other cosine laws. A useful additional identity is

$$\sin a \cos \beta = \cos b \sin c - \sin b \cos c \cos \alpha. \quad (8.6)$$

Proof. The dot product of the two tangents in [equation \(8.2\)](#) is $(x - yz)/(s_b s_c)$. Their cross product is parallel to $\mathbf{u}$ and has norm $|\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|/(s_b s_c)$. This proves [equation \(8.3\)](#), and hence both main laws. For [equation \(8.6\)](#), substitute the cosine law into the right-hand side and reduce it to $(y - xz)/s_c$, which is the left-hand side by its cyclic tangent formula. $\square$

The sine and cosine laws are classical [6, §4.42(iii)]. Their proof here uses only finite vector identities plus the already constructed finite angle representatives. The same calculation over a field with square roots still gives the algebraic sine–cosine data even without naming angles.

8.1 Polar triangles and the cosine law for angles

Relabel the vertices so that $\tau > 0$ and define inward unit side normals

$$\mathbf{n}_a = \frac{\mathbf{v} \times \mathbf{w}}{s_a}, \quad \mathbf{n}_b = \frac{\mathbf{w} \times \mathbf{u}}{s_b}, \quad \mathbf{n}_c = \frac{\mathbf{u} \times \mathbf{v}}{s_c}. \quad (8.7)$$

For example $\mathbf{n}_a \cdot \mathbf{u} = d/s_a > 0$, and its dot products with the other two vertices vanish. These normals are independent and positively oriented: their determinant is $d^2/(s_a s_b s_c) > 0$.

Theorem 8.3 (Polar duality). *The triangle with vertices $\mathbf{n}_a, \mathbf{n}_b, \mathbf{n}_c$ has sides and angles*

$$a^* = \pi - \alpha, \quad b^* = \pi - \beta, \quad c^* = \pi - \gamma, \quad \alpha^* = \pi - a, \quad \beta^* = \pi - b, \quad \gamma^* = \pi - c. \quad (8.8)$$

Its polar is the original triangle. In particular,

$$\cos \alpha = -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cos a, \quad (8.9)$$

and cyclically.

Proof. By equation (3.3), $\mathbf{n}_b \cdot \mathbf{n}_c = (yz - x)/(s_b s_c) = -\cos \alpha$. The side is in $(0, \pi)$, so it equals $\pi - \alpha$. Also $(\mathbf{w} \times \mathbf{u}) \times (\mathbf{u} \times \mathbf{v}) = \tau \mathbf{u}$; after normalization the corresponding inward polar normal is $\mathbf{u}$. This proves involutivity and the angle relations. Apply the side cosine law to the polar triangle and substitute equation (8.8). $\square$

8.2 Half-angle, right-triangle, and haversine formulas

Let $s = (a + b + c)/2$. The identity

$$D = 4 \sin s \sin(s - a) \sin(s - b) \sin(s - c) \quad (8.10)$$

follows by factoring

$$D = (\cos(b - c) - \cos a)(\cos a - \cos(b + c)).$$

For a proper triangle $s < \pi$ and $s - a, s - b, s - c > 0$, as proved in the next section. The half-angle formulas therefore have unambiguous positive roots:

$$\sin(\alpha/2) = \sqrt{\frac{\sin(s - b) \sin(s - c)}{\sin b \sin c}}, \quad (8.11)$$

$$\cos(\alpha/2) = \sqrt{\frac{\sin s \sin(s - a)}{\sin b \sin c}}. \quad (8.12)$$

They follow by substituting the cosine law into $\sin^2(\alpha/2) = (1 - \cos \alpha)/2$ and $\cos^2(\alpha/2) = (1 + \cos \alpha)/2$. Their ratio gives the half-angle tangent formula. Applying the same formulas to the polar triangle gives half-side formulas in terms of the angles.

For a right triangle with $\gamma = \pi/2$, the Napier relations can be written in the particularly safe form

$$\cos c = \cos a \cos b, \quad \sin a = \sin c \sin \alpha, \quad \sin b = \sin c \sin \beta, \quad (8.13)$$

$$\cos \alpha = \cos a \sin \beta, \quad \cos \beta = \cos b \sin \alpha, \quad \cos c = \cot \alpha \cot \beta. \quad (8.14)$$

They follow from the side and angle cosine laws and the sine law. The last expression is defined since $\sin \alpha, \sin \beta > 0$. Formulas involving $\tan a$, $\tan b$, or $\tan c$ require additional nonzero-cosine checks. For instance a proper right spherical triangle can itself have a side of length $\pi/2$.

Define $\text{hav } t = \sin^2(t/2)$. Rearranging the cosine law gives

$$\text{hav } a = \text{hav}(b - c) + \sin b \sin c \text{ hav } \alpha. \quad (8.15)$$

This formulation separates small positive quantities rather than obtaining them by subtracting two almost equal cosines.

9 Existence, reconstruction, and ambiguous data

Theorem 9.1 (Complete side-data criterion and reconstruction). *For $a, b, c \in (0, \pi)$ the following are equivalent:*

1. *They are the angular sides of a proper spherical triangle.*
2. *The matrix $\mathcal{G}$ in equation (8.1) is positive definite.*

3. $D > 0$.

4. $a < b + c$, $b < c + a$, $c < a + b$, and $a + b + c < 2\pi$.

When these conditions hold, a positively oriented realization is

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (z, s_c, 0), \quad \mathbf{w} = \left(y, \frac{x - yz}{s_c}, \frac{\sqrt{D}}{s_c} \right). \quad (9.1)$$

It is unique up to an orthogonal transformation; the positively oriented realizations form one $\text{SO}(3, F)$ orbit.

Proof. A genuine triple has $\mathcal{G} = P^\top P$ with P invertible, hence positive definiteness. Conversely $1 > 0$ and $1 - z^2 > 0$ are its first two leading principal minors, so $D > 0$ is exactly the remaining condition in Sylvester's criterion. Alternatively [equation \(9.1\)](#) directly constructs a realization: the identity $D = (1 - y^2)(1 - z^2) - (x - yz)^2$ makes the last vector unit and its dot products are y, x . Its determinant is $\sqrt{D} > 0$.

To prove the inequality criterion, use [equation \(8.10\)](#). Here $0 < s < 3\pi/2$ and each $s - a, s - b, s - c$ belongs to $(-\pi/2, \pi)$. At most one of these three numbers is nonpositive. If $s \geq \pi$, all three are positive because $a, b, c < \pi$, whereas $\sin s \leq 0$, giving $D \leq 0$. If $s < \pi$, positivity of D is equivalent to all three differences being positive. This is exactly the stated set of inequalities.

Finally, if P and P' are two realizations, then $Q = P'P^{-1}$ satisfies $Q^\top Q = I$ because their Gram matrices agree. The sign of its determinant is the ratio of their orientation signs. $\square$

The perimeter restriction is essential. The three inequalities alone are not sufficient: sides all equal to $3\pi/4$ satisfy the ordinary triangle inequalities but have perimeter exceeding 2π .

9.1 SAS, AAA, and ASA

Any two sides $b, c \in (0, \pi)$ and included angle $\alpha \in (0, \pi)$ determine exactly one proper triangle up to isometry. An explicit realization is

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (\cos c, \sin c, 0), \quad \mathbf{w} = (\cos b, \sin b \cos \alpha, \sin b \sin \alpha).$$

Its determinant is $\sin b \sin c \sin \alpha > 0$. Compute a from the dot product and the remaining angles from [equation \(8.3\)](#). This avoids any ambiguous inverse sine.

By polar duality, angles $\alpha, \beta, \gamma \in (0, \pi)$ occur exactly when

$$\alpha + \beta + \gamma > \pi, \quad \beta + \gamma - \alpha < \pi, \quad (9.2)$$

with the last inequality also imposed cyclically. They then determine the triangle up to isometry. In particular, *three angles fix the shape and angular size* on a fixed sphere; there is no nontrivial similarity scaling of angular sides. The condition that the angle sum exceed π alone is not sufficient. Two angles and their included side (ASA) reduce to SAS for the polar triangle and likewise determine a unique proper triangle for inputs in $(0, \pi)$.

9.2 The spherical SSA ambiguity has an exceptional family

Suppose that $a, b, \alpha \in (0, \pi)$ are given, with a opposite α . Set

$$P = \cos b, \quad Q = \sin b \cos \alpha, \quad X = \cos a.$$

The unknown side c must solve

$$PC + QS = X, \quad C^2 + S^2 = 1, \quad S > 0, \quad (C, S) = (\cos c, \sin c). \quad (9.3)$$

Every solution gives a proper triangle via the SAS construction, whose opposite side is the prescribed a .

Proposition 9.2 (Exact SSA alternatives). *If $L = P^2 + Q^2 > 0$, SSA data have zero, one, or two solutions up to isometry. The candidate unit-circle pairs are*

$$(C, S) = \frac{X}{L}(P, Q) \pm \frac{\sqrt{L - X^2}}{L}(-Q, P), \quad (9.4)$$

provided $L - X^2 \geq 0$; only candidates with $S > 0$ are retained, and duplicate candidates are counted once.

If $P = Q = 0$, there are no solutions unless $a = \pi/2$. If $a = \pi/2$, then $b = \alpha = \pi/2$ as well, and every $c \in (0, \pi)$ gives a solution, with $\beta = \pi/2$ and $\gamma = c$.

Proof. For $L > 0$, intersect the line with the unit circle by decomposing a point into components parallel and perpendicular to (P, Q) . This gives equation (9.4); a line has at most two intersections. The upper-half condition is exactly $c \in (0, \pi)$. If $P = Q = 0$, the original equation is $0 = X$. Since $b \in (0, \pi)$ and $\sin b > 0$, the vanishing of P, Q means $b = \alpha = \pi/2$. When also $a = \pi/2$, take $\mathbf{u} = (1, 0, 0)$, $\mathbf{v} = (\cos c, \sin c, 0)$, $\mathbf{w} = (0, 0, 1)$. Their determinant is $\sin c > 0$. The angle formulas give the claimed remaining angles. $\square$

Thus a blanket assertion that spherical SSA always has at most two solutions misses a genuine continuous exceptional family. The polar construction gives the corresponding AAS ambiguity. The effect is already present over $\mathbb{R}$; surreal coordinates add arbitrarily small perturbations of the exceptional data, for which branch-sensitive computation is especially important.

9.3 Latitude, longitude, and bearings

For latitude $\varphi \in [-\pi/2, \pi/2]$ and longitude λ modulo 2π , the unit direction is

$$(\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi).$$

Two such points have separation a given by

$$\cos a = \sin \varphi_1 \sin \varphi_2 + \cos \varphi_1 \cos \varphi_2 \cos(\lambda_2 - \lambda_1), \quad (9.5)$$

$$\text{hav } a = \text{hav}(\varphi_2 - \varphi_1) + \cos \varphi_1 \cos \varphi_2 \text{hav}(\lambda_2 - \lambda_1). \quad (9.6)$$

At a nonpolar starting point and for a distinct nonantipodal endpoint, the initial bearing measured from north toward east is

$$\text{atan2}(\cos \varphi_2 \sin \Delta\lambda, \cos \varphi_1 \sin \varphi_2 - \sin \varphi_1 \cos \varphi_2 \cos \Delta\lambda). \quad (9.7)$$

Indeed these are the east and north components of the projected tangent to the target. Undefined bearings at poles, coincident points, or antipodes are geometric ambiguities, not numerical failures. The physical great-circle distance remains Ra for any positive surreal radius.

10 Spherical centers, medians, and Ceva

Assume $\tau = d > 0$ and retain the matrix P and inward normals of [equation \(8.7\)](#).

10.1 Circumcenter and incenter

Put $\mathbf{h} = P^{-\top}(1, 1, 1)^\top$. Then

$$\mathbf{o} = \frac{\mathbf{h}}{\|\mathbf{h}\|}, \quad \rho_c = \arccos \frac{1}{\|\mathbf{h}\|} \quad (10.1)$$

is the center and angular radius of the circumcircle with radius less than $\pi/2$. All three dot products with $\mathbf{o}$ equal $1/\|\mathbf{h}\| > 0$. Cauchy–Schwarz makes this number at most one, and distinct vertices make it strictly less than one. Every equidistant unit center solves $P^\top \mathbf{o} = k(1, 1, 1)^\top$, so the two possibilities are exactly $\mathbf{o}$ and $-\mathbf{o}$, with radii ρ_c and $\pi - \rho_c$. The chosen circumcenter need not lie inside the triangle.

The incenter has the particularly simple weighted-vector expression

$$\mathbf{j} = \frac{s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}}{\|s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}\|}, \quad r_i = \arcsin \frac{d}{\|s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}\|}. \quad (10.2)$$

To prove this, take dot products of its numerator with the three inward normals. Each equals d . Hence $\mathbf{j}$ has equal positive signed distances from the three side great circles, and it lies inside the positive cone. Those signed angular distances are the inverse sines in [equation \(10.2\)](#). Uniqueness follows by solving the nonsingular linear system $N^\top \mathbf{j} = k(1, 1, 1)^\top$ for the inward normal matrix N . Equivalently, the inside angle bisectors below concur at this point. The physical circumradius and inradius along the sphere are $R\rho_c$ and Rr_i , not ρ_c and r_i themselves.

10.2 Medians and angle bisectors

The midpoint of a minor edge from $\mathbf{v}$ to $\mathbf{w}$ is $(\mathbf{v} + \mathbf{w})/\|\mathbf{v} + \mathbf{w}\|$. The three vertex-to-opposite-midpoint great circles concur at

$$\frac{\mathbf{u} + \mathbf{v} + \mathbf{w}}{\|\mathbf{u} + \mathbf{v} + \mathbf{w}\|}.$$

This follows by writing the common numerator as a vertex plus the sum of the other two. It is a median center, not an assertion that it is the surface-area centroid.

At $\mathbf{u}$, an internal angle bisector has tangent direction $T_{uv} + T_{uw}$. Its opposite-side intersection is proportional to $s_b \mathbf{v} + s_c \mathbf{w}$. If that point is D_a on the edge vw , then

$$\frac{\sin d_S(\mathbf{v}, D_a)}{\sin d_S(D_a, \mathbf{w})} = \frac{\sin c}{\sin b}. \quad (10.3)$$

This is a ratio of sines of angular subarcs, not in general a ratio of their angular lengths.

Theorem 10.1 (Spherical Ceva). *For interior points D_a, D_b, D_c on the minor edges vw, wu, uv , the three cevians concur inside the triangle if and only if*

$$\frac{\sin d_S(\mathbf{v}, D_a)}{\sin d_S(D_a, \mathbf{w})} \frac{\sin d_S(\mathbf{w}, D_b)}{\sin d_S(D_b, \mathbf{u})} \frac{\sin d_S(\mathbf{u}, D_c)}{\sin d_S(D_c, \mathbf{v})} = 1. \quad (10.4)$$

Proof. Write D_a as the normalization of $\lambda \mathbf{v} + \mu \mathbf{w}$ with $\lambda, \mu > 0$. Cross products with the endpoints show that its first sine ratio is μ/λ . Do the same on the other two edges. A positive linear combination $A\mathbf{u} + B\mathbf{v} + C\mathbf{w}$ belongs to the first cevian exactly when $C/B = \mu/\lambda$. The cyclic conditions have a positive solution precisely when their product is one. Normalizing that solution gives the interior concurrence point. This is ordinary barycentric Ceva expressed in spherical sine ratios. $\square$

11 Spherical excess, solid angles, and polygon area

Define

$$M = 1 + x + y + z, \quad B_0 = \sqrt{M^2 + D}. \quad (11.1)$$

An important positive identity is

$$M^2 + D = 2(1+x)(1+y)(1+z) > 0. \quad (11.2)$$

The sign of M must not be discarded. It determines whether the triangle's area is smaller or larger than a quarter of the full sphere.

Theorem 11.1 (Branch-correct spherical excess). *For a proper spherical triangle,*

$$E := \alpha + \beta + \gamma - \pi = 2 \operatorname{atan2}(d, M), \quad 0 < E < 2\pi. \quad (11.3)$$

In particular, $E < \pi$, $E = \pi$, or $E > \pi$ according as $M > 0$, $M = 0$, or $M < 0$. The associated half-angle data are

$$\cos(E/2) = \frac{M}{B_0}, \quad \sin(E/2) = \frac{d}{B_0}. \quad (11.4)$$

Proof. The polar triangle has perimeter less than 2π by [theorem 9.1](#), so $3\pi - (\alpha + \beta + \gamma) < 2\pi$. Each interior angle is less than π . Hence $0 < \alpha + \beta + \gamma - \pi < 2\pi$.

For an algebraic evaluation, work in $F[i]$, and put $A_1 = x - yz$, $A_2 = y - zx$, $A_3 = z - xy$ and $T = (1-x)(1-y)(1-z)$. Direct multiplication, using $d^2 = 1 + 2xyz - x^2 - y^2 - z^2$, gives

$$2(A_1 + id)(A_2 + id)(A_3 + id) + T(M + id)^2 = 0. \quad (11.5)$$

This can be checked by separately comparing real and imaginary parts; they reduce respectively to

$$2(A_1 A_2 A_3 - D(A_1 + A_2 + A_3)) + T(M^2 - D) = 0$$

and $A_1 A_2 + A_2 A_3 + A_3 A_1 - D + TM = 0$. By [equation \(8.3\)](#), the product of the three angle phases is the product of the first three factors in [equation \(11.5\)](#), divided by $(1-x^2)(1-y^2)(1-z^2)$. Since that denominator equals $T(M^2 + D)/2$, we obtain

$$\operatorname{cis}(\alpha + \beta + \gamma) = -\frac{(M + id)^2}{M^2 + D}.$$

It follows that E and $2 \operatorname{atan2}(d, M)$ have the same phase. Both lie in $(0, 2\pi)$, where phase is injective, so they are equal. The sign alternatives and [equation \(11.4\)](#) follow immediately. $\square$

When $M > 0$, [equation \(11.3\)](#) can be written $2 \operatorname{atan}(d/M)$. When $M < 0$, the same uncorrected one-argument arctangent would give a negative number instead of the correct value in $(\pi, 2\pi)$. This is a geometric branch error, not merely a floating-point issue.

11.1 L'Huilier's formula

For $s = (a + b + c)/2$, all four half-arguments below lie in $(0, \pi/2)$. We have the exact formula

$$\tan(E/4) = \sqrt{\tan(s/2) \tan((s-a)/2) \tan((s-b)/2) \tan((s-c)/2)}. \quad (11.6)$$

Here the root is positive, even for infinitesimal excess or for excess infinitesimally close to 2π .

To prove it, use $\tan(E/4) = d/(B_0 + M)$ from [equation \(11.4\)](#). The product-to-sum identity

$$B_0 + M = 8 \cos(s/2) \cos((s-a)/2) \cos((s-b)/2) \cos((s-c)/2)$$

follows by expansion, using $B_0 = 4 \cos(a/2) \cos(b/2) \cos(c/2) > 0$ from [equation \(11.2\)](#). Replace the four full-angle sines in [equation \(8.10\)](#) by twice the corresponding sine–cosine products. After division by $(B_0 + M)^2$, one obtains the square of [equation \(11.6\)](#). Positivity supplies the indicated root.

11.2 Area as a precise finite-polygon functional

Definition 11.2 (Spherical polygon area). For a proper triangle on $S_R^2(C)$ define its area to be $R^2 E$. For a convex geodesic polygon contained in an open hemisphere, triangulate it by minor great-circle diagonals and add these triangle areas.

Theorem 11.3 (Consistency, positivity, and finite additivity). *The definition is independent of triangulation. For a convex polygon with n boundary vertices and interior angles $\alpha_1, \dots, \alpha_n$,*

$$\text{Area}(P) = R^2 \left(\sum_{j=1}^n \alpha_j - (n-2)\pi \right). \quad (11.7)$$

It is positive for a nonempty two-dimensional polygon, invariant under orthogonal transformations, and additive under finite geodesic polygonal dissections with disjoint interiors, with edges and vertices assigned area zero. It extends by such dissections to finite unions of spherical geodesic polygons.

Proof. Use a gnomonic chart to make the polygon an ordinary convex polygon. In a triangulation with boundary vertices only, summing all the triangle angles gives precisely the sum of boundary angles, and there are $n - 2$ triangles. This proves [equation \(11.7\)](#) and independence of the triangulation.

A refinement can introduce interior vertices or subdivide boundary edges. At an interior vertex the angles sum to 2π ; at an inserted straight boundary vertex the angle is π . If a planar triangulation has I interior vertices, B boundary vertices and T_Δ triangles, the finite Euler count is $T_\Delta = 2I + B - 2$. Thus the extra angles cancel exactly the extra triangle subtractions. A common finite refinement of two dissections proves additivity and consistency. Positivity follows by adding the positive triangle excesses. Orthogonal invariance follows from invariance of dot products and interior angles. For regions not contained in one open hemisphere, intersect with the eight closed coordinate octants, subdividing edges where necessary. Each octant lies in the open hemisphere whose normal has its three coordinate signs. Thus all the resulting pieces admit the preceding construction, and common refinement proves the same consistency. No topological compactness is needed for this explicit finite subdivision. $\square$

An octant triangle has three right angles, hence excess $\pi/2$. Eight octants therefore give

$$\text{Area}(S_R^2) = 4\pi R^2, \quad \text{Area}(\text{hemisphere}) = 2\pi R^2. \quad (11.8)$$

A lune of angle $\lambda \in (0, \pi)$ splits along the equator into two triangles with angles $\lambda, \pi/2, \pi/2$. Its area is $2R^2\lambda$. These values reproduce the usual geometric normalization of area.

For a circular cap of angular radius θ , the usual elementary chartwise primitive of the area density gives the canonical expression

$$A_{\text{cap}}(\theta) = 2\pi R^2(1 - \cos \theta), \quad 0 \leq \theta \leq \pi.$$

This is the extension of a specific classical analytic formula, obtained by evaluating the primitive $-\cos \theta$ of $\sin \theta$. It is not an assertion that arbitrary subsets have a surreal measure or that ordinary Darboux sums define this value. Our finite-additivity theorem concerns the geodesic polygonal class just constructed.

11.3 Solid angle of three arbitrary rays

Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in F^3$ be linearly independent vectors from an observation point, with lengths $A, B, C > 0$. Their spherical triangle has solid angle

$$\Omega = 2 \operatorname{atan2}(|\det[\mathbf{A} \ \mathbf{B} \ \mathbf{C}]|, ABC + C(\mathbf{A} \cdot \mathbf{B}) + A(\mathbf{B} \cdot \mathbf{C}) + B(\mathbf{C} \cdot \mathbf{A})). \quad (11.9)$$

Normalize each ray to a unit vector in [equation \(11.3\)](#); multiplying both arguments by the positive number ABC proves the formula. This is the branch-complete form of the classical plane-triangle solid-angle formula associated with van Oosterom and Strackee [7]. The formula is independent of positive rescaling of any individual ray. Replacing the absolute determinant by its signed value gives the oriented angle, with sign set by the ray order.

Solid angle is dimensionless and finite, whereas the corresponding area $R^2\Omega$ can have any surreal scale. A determinant that is infinitesimal relative to ray lengths does not by itself imply small solid angle: when the second argument is negative, the angle can instead be close to 2π .

12 Differential geometry and parallel transport

All derivatives in this section are with respect to explicit chart or curve parameters. Scalar parameters such as R are held fixed. These derivatives are not an intrinsic derivation of the scalar field $\mathbf{No}$. The calculations are identities between restricted-analytic germs and rational operations; no existence theorem for arbitrary differential equations is being invoked.

12.1 Connection, geodesics, and curvature

For a sphere centered at zero, its tangent plane at P is $T_P S_R^2 = \{X : P \cdot X = 0\}$. Orthogonally project ambient derivatives to obtain the connection

$$\nabla_X Y = D_X Y + \frac{X \cdot Y}{R^2} P. \quad (12.1)$$

Indeed differentiating $P \cdot Y = 0$ gives $P \cdot D_X Y = -X \cdot Y$, so the displayed correction makes the result tangent. The product rule shows metric compatibility; symmetry of the ambient derivative gives zero torsion.

For a unit tangent vector V at P , a unit-speed great-circle geodesic is

$$\gamma(s) = \cos(s/R)P + R \sin(s/R)V, \quad s/R \in \mathcal{O}. \quad (12.2)$$

Direct differentiation gives $\|\gamma'\| = 1$ and $\gamma'' = -\gamma/R^2$, hence $\nabla_{\gamma'}\gamma' = 0$. All minor geodesic segments have $0 \leq s/R \leq \pi$, so they are included. No oscillatory value at an infinitely large s/R is required.

With curvature convention $\mathcal{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z$, substitution of [equation \(12.1\)](#) and cancellation of commuting ambient derivatives gives

$$\mathcal{R}(X, Y)Z = \frac{1}{R^2}((Y \cdot Z)X - (X \cdot Z)Y). \quad (12.3)$$

Thus the Gaussian curvature is exactly $1/R^2$. In particular a sphere of infinite radius has positive infinitesimal curvature, not zero curvature. An infinitesimal sphere has infinite curvature. Both statements are literal field identities.

The exponential map at P , for $X \in T_P S_R^2$ with $r = \|X\|$ and r/R finite, is

$$\text{Exp}_P(X) = \cos(r/R)P + R \sin(r/R) \frac{X}{r},$$

with value P at $r = 0$. For $0 \leq r < \pi R$, it is bijective onto the sphere minus the antipode. Its inverse for $Q \neq -P$ is

$$\text{Log}_P(Q) = \frac{a}{\sin a}(Q - \cos a P), \quad a = \arccos(P \cdot Q/R^2), \quad (12.4)$$

with the removable value zero at $Q = P$. Taking norms and using the unique minor arc verifies both identities.

12.2 Explicit parallel transport

For unit directions $\mathbf{u} \neq -\mathbf{v}$ and $X \cdot \mathbf{u} = 0$, define

$$\mathcal{P}_{uv}(X) = X - \frac{X \cdot \mathbf{v}}{1 + \mathbf{u} \cdot \mathbf{v}}(\mathbf{u} + \mathbf{v}). \quad (12.5)$$

This is the parallel transport along the minor great-circle arc. To verify it, choose the tangent to the arc and its constant binormal as a basis of the initial tangent plane. The tangent rotates with the great circle and has zero covariant derivative; the binormal is ambient-constant and also has zero covariant derivative. The endpoint values of these two parallel fields are exactly [equation \(12.5\)](#). Alternatively, restricting [equation \(5.5\)](#) to $\mathbf{u}^\perp$ reduces algebraically to the same formula. It follows that transport is an isometry between tangent planes. This specifies the canonical transport by explicit analytic continuation along the parametrized arc. It does not assert uniqueness among arbitrary intrinsically differentiable fields: the locally constant counterexample in [section 15](#) explains that distinction. Within the definable restricted-analytic class, uniqueness follows because the two coefficients in the parallel frame have zero derivative and hence are constant. For a radius- R sphere the unit directions are used but tangent vectors may retain their physical lengths.

12.3 A quaternion proof of spherical holonomy

Use the positive orientation $\tau = d > 0$. The unit quaternion of the minimal rotation from $\mathbf{u}$ to $\mathbf{v}$ is

$$q_{uv} = \frac{(1 + z, \mathbf{u} \times \mathbf{v})}{\sqrt{2(1 + z)}}. \quad (12.6)$$

Define q_{vw} and q_{wu} cyclically. Quaternion multiplication gives the especially informative identity

$$q_{wu}q_{vw}q_{uv} = \frac{(M, d\mathbf{u})}{\sqrt{M^2 + D}}. \quad (12.7)$$

For a direct verification, multiply the three numerators using [equation \(5.6\)](#). The scalar component is $2M$ and the vector component is $2d\mathbf{u}$; the denominator is $\sqrt{8(1+x)(1+y)(1+z)} = 2\sqrt{M^2 + D}$. The simplifications use only unit norms and the vector triple-product identities.

Theorem 12.1 (Holonomy equals spherical excess). *Parallel transport around the positively oriented boundary $\mathbf{u} \rightarrow \mathbf{v} \rightarrow \mathbf{w} \rightarrow \mathbf{u}$ is rotation of the initial tangent plane through angle E modulo 2π . Equivalently,*

$$\text{holonomy phase} = \text{cis } E, \quad E = \frac{\text{Area}(\triangle)}{R^2}.$$

Proof. Each edge transport is the restriction of the corresponding quaternion rotation. By [equations \(11.4\)](#) and [\(12.7\)](#), their product is the quaternion $(\cos(E/2), \sin(E/2)\mathbf{u})$, which acts on $\mathbf{u}^\perp$ by rotation through E . The area equality is [definition 11.2](#). $\square$

This is a finite-triangle version of the curvature–area holonomy relation, proved without an unconstructed global integral. The observed tangent-plane rotation records E modulo 2π , while the area branch has already been fixed by $0 < E < 2\pi$. Reversing the boundary orientation changes the holonomy sign but not the unoriented area.

12.4 Curvature correction in normal coordinates

Let p, q be vectors in the tangent plane at P , with $r = \|p\|$, $s = \|q\|$, and $(r + s)/R$ infinitesimal. The squared intrinsic distance between their exponential images satisfies

$$\begin{aligned} d_R(\text{Exp}_P p, \text{Exp}_P q)^2 &= \|p - q\|^2 - \frac{r^2 s^2 - (p \cdot q)^2}{3R^2} \\ &\quad + O\left(\frac{(r + s)^6}{R^4}\right). \end{aligned} \quad (12.8)$$

The error estimate is absolute and uniform over the relative directions; it does not claim a small *relative* error when $p - q$ is exceptionally small.

To derive it, for $r, s \neq 0$ use the spherical cosine law with $\cos \alpha = (p \cdot q)/(rs)$. Expanding to degree four in $r/R, s/R$ gives

$$\cos(d_R/R) = 1 - \frac{\|p - q\|^2}{2R^2} + \frac{r^4 + s^4 + 6r^2 s^2 - 4(p \cdot q)(r^2 + s^2)}{24R^4} + O((r + s)^6/R^6).$$

The function $\arccos(1 - h)^2 = 2h + h^2/3 + O(h^3)$ has a real analytic extension through $h = 0$. Substitution yields [equation \(12.8\)](#). Expressions with $r = 0$ or $s = 0$ are obtained by their removable analytic values. Uniform real-germ remainder estimates transfer after rescaling by R . The coefficient $-1/(3R^2)$ explicitly records positive curvature.

13 Infinitesimal scales and geometric conditioning

The extra information carried by surreal coordinates is not a new algebraic identity for an ordinary triangle. It is the ability to retain exact rank, orientation, angle, and size when several different infinitesimal scales occur in the same configuration.

13.1 A valuation dictionary

Write v for the Hahn valuation, with $v(0) = +\infty$. For a nonzero vector $X \in F^3$,

$$v(\|X\|) = \min_j v(X_j). \quad (13.1)$$

Indeed, in $\sum_j X_j^2$ the summands of smallest valuation have positive leading coefficients, so their leading terms cannot cancel. Taking a positive square root halves the resulting valuation.

For unit vectors u, v , put $a = d_S(u, v)$. The chord bounds and their antipodal versions give

$$v(\|u - v\|) = v(a) \quad (a > 0), \quad (13.2)$$

$$v(\|u + v\|) = v(\pi - a) \quad (a < \pi). \quad (13.3)$$

For $0 < a < \pi$ one also has

$$v(\sin a) = v(a) + v(\pi - a). \quad (13.4)$$

To check the last assertion, when both endpoints are separated from a by positive real amounts all three valuations are zero. When a is infinitesimal, $\sin a/a = 1 + O(a^2)$ and $\pi - a$ is a unit; the case $\pi - a$ infinitesimal is identical after reflection. These exhaust the possibilities by standard parts.

Consequently a proper spherical triangle satisfies the exact scale laws

$$\begin{aligned} v(d) &= v(\sin b) + v(\sin c) + v(\sin \alpha) \\ &= v(\sin c) + v(\sin a) + v(\sin \beta) \\ &= v(\sin a) + v(\sin b) + v(\sin \gamma). \end{aligned} \quad (13.5)$$

A small determinant can therefore come from a small side, a nearly antipodal side, a small interior angle, or an angle close to π . Neither its standard part nor its valuation alone identifies which geometric mechanism is responsible.

13.2 A shape-independent local area estimate

The following estimate does not assume that a shrinking triangle has angles bounded away from zero. This matters over the surreals: the area of a triangle can be smaller than every ordinary power of its diameter.

Theorem 13.1 (Uniform gnomonic area comparison). *Let $p_0, p_1, p_2 \in F^2$ be noncollinear, let $\rho = \max_i \|p_i\|$, and suppose $0 < \rho \leq 1/4$. On the unit sphere take the vertices $u_i = (p_i, 1)/\sqrt{1 + \|p_i\|^2}$. Set*

$$H = \det[p_1 - p_0 \quad p_2 - p_0], \quad A_{\text{flat}} = \frac{|H|}{2}.$$

The spherical excess E of their minor-arc triangle satisfies

$$\left| \frac{E}{A_{\text{flat}}} - 1 \right| \leq 4\rho^2. \quad (13.6)$$

In particular, for infinitesimal ρ ,

$$E = \frac{|H|}{2}(1 + O(\rho^2)), \quad v(E) = v(H).$$

On the sphere of radius $R > 0$ the spherical area is $R^2 E$, so $v(\text{Area}) = 2v(R) + v(H)$, without any restriction on the size of R .

Proof. Put $A_i = \sqrt{1 + \|p_i\|^2}$ and

$$B = A_0 A_1 A_2 + (1 + p_0 \cdot p_1) A_2 + (1 + p_1 \cdot p_2) A_0 + (1 + p_2 \cdot p_0) A_1.$$

The determinant and denominator in the solid-angle formula are $d = |H|/(A_0 A_1 A_2)$ and $M = B/(A_0 A_1 A_2)$, respectively. Thus

$$E = 2 \operatorname{atan}(|H|/B). \quad (13.7)$$

Here $B > 0$; the triangle lies near the north pole, not near the hemispherical branch cut.

The inequalities $1 \leq A_i \leq 1 + \rho^2/2$ and $|p_i \cdot p_j| \leq \rho^2$ imply

$$|B - 4| \leq 8\rho^2, \quad B \geq 7/2, \quad |H| \leq 4\rho^2.$$

For example, the deviation of the product $A_0 A_1 A_2$ from 1 is at most $2\rho^2$, and each of the other three terms deviates from 1 by at most $2\rho^2$. For the lower bound one may use the sharper $B \geq 4 - 3\rho^2$.

Set $\zeta = |H|/B$, so $0 < \zeta \leq 2\rho^2$. The ordinary analytic inequality $1 - \zeta^2/3 \leq \operatorname{atan} \zeta / \zeta \leq 1$ holds here by restricted-analytic transfer. Therefore

$$\begin{aligned} \left| \frac{E}{A_{\text{flat}}} - 1 \right| &= \left| \frac{4}{B} \frac{\operatorname{atan} \zeta}{\zeta} - 1 \right| \\ &\leq \frac{16}{7} \rho^2 + \frac{32}{21} \rho^4 \leq 4\rho^2. \end{aligned}$$

Crucially, H was divided out only after the exact factorization of E ; no lower bound on $|H|/\rho^2$ was used. The valuation claims follow because the final multiplicative factor has standard part 1. $\square$

Remark 13.2. For $p_0 = 0$, $p_1 = (\varepsilon, 0)$, and $p_2 = (\varepsilon, \omega^{-\omega})$ with $\varepsilon = \omega^{-1}$, the ratio $|H|/\rho^2$ is smaller than every ordinary positive power of ε . The theorem still gives a relative, rather than merely absolute, estimate for the spherical area. A generic error bound $E = A_{\text{flat}} + O(\rho^4)$ would not give that information.

13.3 Stable vector data and unstable derived data

Proposition 13.3 (Three conditioning estimates). *For unit vectors u, v, u', v' ,*

$$|d_S(u, v) - d_S(u', v')| \leq \frac{\pi}{2} (\|u - u'\| + \|v - v'\|). \quad (13.8)$$

For nonzero X, Y ,

$$\left\| \frac{X}{\|X\|} - \frac{Y}{\|Y\|} \right\| \leq \frac{2\|X - Y\|}{\|X\|}. \quad (13.9)$$

Finally, let $a = d_S(u, v) \in (0, \pi)$ and $\eta = \|u - u'\| + \|v - v'\| < \sin a$. Then $u' \times v' \neq 0$ and

$$\left\| \frac{u \times v}{\|u \times v\|} - \frac{u' \times v'}{\|u' \times v'\|} \right\| \leq \frac{2\eta}{\sin a}. \quad (13.10)$$

Proof. The angular triangle inequality bounds the left side of equation (13.8) by $d_S(u, u') + d_S(v, v')$. Apply the lower chord bound to each summand. For equation (13.9), write the difference as

$$\frac{X - Y}{\|X\|} + Y \left(\frac{1}{\|X\|} - \frac{1}{\|Y\|} \right)$$

and use $|\|X\| - \|Y\|| \leq \|X - Y\|$. For the last estimate,

$$\|u \times v - u' \times v'\| \leq \|(u - u') \times v\| + \|u' \times (v - v')\| \leq \eta.$$

The strict inequality $\eta < \|u \times v\| = \sin a$ prevents the new cross product from vanishing; normalization gives the claimed bound. $\square$

Thus angular distance is uniformly stable under consistent perturbations of the unit vectors. The normal to their spanning plane is not uniformly stable near coincident or antipodal directions: the amplification factor $1/\sin a$ is then infinite. There is no contradiction between these statements; they concern different outputs.

If one perturbs only the scalar cosine as though it were independent of all other data, inverse cosine is also ill-conditioned near an endpoint. Away from the endpoints its derivative is

$$\frac{d}{dc} \arccos c = -\frac{1}{\sqrt{1-c^2}},$$

whereas for $h > 0$ infinitesimal,

$$\arccos(1-h) = \sqrt{2h} \left(1 + \frac{h}{12} + O(h^2)\right). \quad (13.11)$$

The square-root law follows by inverting $1 - \cos \theta = \theta^2/2 - \theta^4/24 + \dots$ after extracting $\sqrt{2h}$. It is a genuinely different scale from an ordinary linear perturbation estimate.

14 Worked configurations at several surreal scales

14.1 Infinite lengths with infinitesimal directions

The vectors $X = (\omega, 1, 0)$ and $Y = (\omega, 0, 0)$ have infinite lengths and the finite infinitesimal angle

$$\text{ang}(X, Y) = \text{atan}(\omega^{-1}) = \omega^{-1} - \frac{1}{3}\omega^{-3} + \frac{1}{5}\omega^{-5} - \dots.$$

This displayed series is strongly summable because its parameter is infinitesimal. No evaluation of $\sin \omega$ occurs.

For the flat triangle

$$P = 0, \quad Q = (\omega, 0, 0), \quad S = (\omega, \omega^{-\omega}, 0),$$

the area is exactly $\frac{1}{2}\omega^{1-\omega}$, while the angle at P is $\text{atan}(\omega^{-\omega-1})$. The base is infinite, the altitude is smaller than every ordinary negative power of ω , and the area is nevertheless a well-defined positive surreal number. A real standard-part picture alone does not encode this configuration.

14.2 Nearly parallel lines meet their closest points infinitely far away

Let $\varepsilon > 0$ be infinitesimal and let $h = \varepsilon^2$. Consider

$$L_1 = \{(t, 0, 0) : t \in F\}, \quad L_2 = \{(s, 1 + \varepsilon s, h) : s \in F\}.$$

Their squared distance between parameter points is

$$(t - s)^2 + (1 + \varepsilon s)^2 + h^2.$$

Its minimum is attained at $t = s = -1/\varepsilon$ and is h^2 . Hence the lines have distance h , although the standard parts of their *finite-coordinate* points lie on the real lines $y = 0, z = 0$ and $y = 1, z = 0$, whose distance is 1.

The discrepancy is not a failure of the distance formula. The minimizing points have infinite first coordinates and disappear from a finite-coordinate standard-part picture. Taking standard parts and minimizing over an entire affine line are different operations. With $h = 0$, the same lines intersect at a genuine affine point infinitely far from the origin.

14.3 Tangency splits at a square-root scale

Intersect the unit sphere with the line $(t, 1 - \varepsilon, 0)$, where $0 < \varepsilon \in \mathfrak{m}$. The exact solutions are

$$t = \pm \sqrt{2\varepsilon - \varepsilon^2} = \pm \sqrt{2\varepsilon} \left(1 - \frac{\varepsilon}{4} + O(\varepsilon^2)\right).$$

A displacement of valuation $v(\varepsilon)$ splits a double tangency into points separated at valuation $v(\varepsilon)/2$. Replacing $1 - \varepsilon$ by $1 + \varepsilon$ produces no intersection, while $\varepsilon = 0$ gives a single point. All three cases have the same ordinary standard-part tangency.

14.4 An almost planar tetrahedron with an infinite circumradius

Take vertices $0, e_1, e_2, (2, 0, \varepsilon)$ with $0 < \varepsilon \in \mathfrak{m}$. The tetrahedron has volume $\varepsilon/6$. Its circumcenter is

$$C = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\varepsilon} + \frac{\varepsilon}{2}\right),$$

since $2C \cdot e_i = 1$ for $i = 1, 2$ and $2C \cdot (2, 0, \varepsilon) = 4 + \varepsilon^2$. The circumradius is

$$\|C\| = \sqrt{\frac{1}{2} + \left(\frac{1}{\varepsilon} + \frac{\varepsilon}{2}\right)^2},$$

which is infinite. Nonzero volume, however small, is enough for uniqueness of the center; it is not enough for a finite conditioning constant.

14.5 A right spherical triangle and the flat limit of an infinite sphere

For positive infinitesimals ε, η , set

$$u = e_3, \quad v = \frac{(\varepsilon, 0, 1)}{A}, \quad w = \frac{(0, \eta, 1)}{B}, \quad A = \sqrt{1 + \varepsilon^2}, \quad B = \sqrt{1 + \eta^2}.$$

The angle at u is exactly $\pi/2$, the adjacent side angles are $c = \text{atan } \varepsilon$ and $b = \text{atan } \eta$, and

$$x = \frac{1}{AB}, \quad y = \frac{1}{B}, \quad z = \frac{1}{A}, \quad d = \frac{\varepsilon\eta}{AB}.$$

The exact excess is therefore

$$E = 2 \text{atan } \frac{\varepsilon\eta}{(A+1)(B+1)} = \frac{\varepsilon\eta}{2} - \frac{\varepsilon^3\eta + \varepsilon\eta^3}{8} + O(\varepsilon\eta(\varepsilon^2 + \eta^2)^2). \quad (14.1)$$

This is a relative expansion with an explicit common area factor. In the symmetric case $\varepsilon = \eta = h$ it becomes

$$E = \frac{h^2}{2} - \frac{h^4}{4} + \frac{7h^6}{48} + O(h^8). \quad (14.2)$$

Now choose $R = \omega$ and $h = \omega^{-1}$, and scale all vertices by R . The physical area is

$$R^2 E = \frac{1}{2} - \frac{1}{4\omega^2} + \frac{7}{48\omega^4} + O(\omega^{-6}),$$

and the physical adjacent side lengths are $\omega \operatorname{atan}(\omega^{-1}) = 1 - 1/(3\omega^2) + \dots$. The sphere has positive infinitesimal curvature ω^{-2} . A bounded patch is asymptotically planar, but its curvature and area corrections remain exact surreal information rather than being discarded.

14.6 A tiny determinant with almost hemispherical area

Let $0 < \varepsilon \in \mathfrak{m}$ and consider

$$u = (1, 0, 0), \quad v = (-1/2, \sqrt{3}/2, 0), \quad w = \frac{(-1/2, -\sqrt{3}/2, \varepsilon)}{\sqrt{1 + \varepsilon^2}}.$$

Writing $A = \sqrt{1 + \varepsilon^2}$, one obtains

$$d = \frac{\sqrt{3}\varepsilon}{2A}, \quad M = \frac{1}{2} - \frac{1}{A} < 0.$$

Thus $D = d^2$ is infinitesimal, but the correct branch gives

$$E = 2\pi - 2 \operatorname{atan} \frac{d}{-M} = 2\pi - 2\sqrt{3}\varepsilon + O(\varepsilon^3). \quad (14.3)$$

The triangle nearly fills a hemisphere rather than collapsing to zero area. Its polar triangle is small. Replacing $\operatorname{atan}2(d, M)$ by the principal $\operatorname{atan}(d/M)$ would give the wrong branch and even the wrong sign for the area. This example and [equation \(14.1\)](#) show why $d \rightarrow 0$ is not a complete description of spherical degeneration.

15 What the development does not silently assume

15.1 The intrinsic order topology is not ordinary real topology

Suppose F is non-Archimedean. Its infinitesimal ideal $\mathfrak{m}$ is both open and closed in the order topology: each infinitesimal has a sufficiently small infinitesimal neighborhood inside $\mathfrak{m}$, while for $x \notin \mathfrak{m}$ a neighborhood of radius $|x|/2$ stays outside $\mathfrak{m}$. Consequently the function

$$f(x) = \begin{cases} 0, & x \in \mathfrak{m}, \\ 1, & x \notin \mathfrak{m} \end{cases}$$

is locally constant and has derivative zero everywhere in the intrinsic ordered-field sense (derivatives with radii from F), but $f(0) \neq f(1)$. Local differentiability alone therefore does not give the ordinary global mean value theorem or uniqueness of an antiderivative.

There is no conflict with the use of restricted-analytic transfer. The externally specified set of all real infinitesimals is not a new definable predicate in the restricted-analytic language used here. Our formulas come from compatible real analytic germs or explicit algebraic definitions, not from arbitrary functions that happen to be locally constant on intrinsic neighborhoods. The same function is locally constant for the full fine topology of **No** as well, since **m** is also open there; [section 26](#) compares the two topologies.

15.2 Ordinary finite partitions do not automatically define an integral

Even the function $f(x) = x$ on $[0, 1] \subset F$ exposes a difficulty. For any partition with an ordinary finite number N of subintervals, put $\Delta_i = x_i - x_{i-1} > 0$. Its upper and lower Darboux sums satisfy

$$U - L = \sum_{i=1}^N \Delta_i^2 \geq \frac{1}{N},$$

by Cauchy–Schwarz and $\sum_i \Delta_i = 1$. Given any positive infinitesimal ε , this difference is always greater than ε , whatever ordinary finite N is chosen. Hence the usual Darboux criterion requiring gaps smaller than *every positive element of F* fails for these partitions. A mesh criterion without a requirement that the relevant partitions exist would instead become vacuous below infinitesimal mesh sizes.

Nevertheless the polynomial antiderivative $x^2/2$ is perfectly well-defined, as are our polygon-area formula and the selected analytic cap primitive. A theory of generalized integration must say which functional, which functions, and which summation or partition notion it uses. This report does not infer a countably additive area measure from finite-polygon additivity. It also does not claim completeness, ordinary topological compactness, arbitrary smooth-ODE existence, or a supremum-based length for every curve in surreal space.

15.3 Finite dimension does not remove size or support conditions

The ambient coordinate count is three, but the class **No**³ remains a proper class. The set-field construction in [section 2](#) allows every finite configuration in the report to be treated in a single honest field and every needed analytic evaluation to be interpreted there. Passing to a larger workspace preserves the answers. Neither this compatibility nor the finite dimension permits unrestricted summation of arbitrary families of surreals.

Likewise, ordinary-size angles do not imply ordinary-size arc lengths. For an infinite R and a positive real a , the minor arc has infinite length Ra . Conversely, a bounded physical arc on an infinite sphere has infinitesimal angular length and is governed by the local expansions above. Choosing a global oscillatory phase at infinite scalar arguments would add another analytic convention; it would not change these coordinate-defined directions or finite geometric angles.

Part II

The rotation group $\text{SO}(3, \mathbf{No})$

Part II is source 09, apart from the material printed once in [sections 2, 3 and 5](#). Its axis theorem, Rodrigues formula, quaternion action, spin map and Cayley chart are [theorem 5.3](#), [equation \(5.7\)](#),

[theorem 5.5](#) and [theorem 5.6](#). Vectors are written in plain type, and $G(F) = \mathrm{SO}(3, F)$ for a real closed field F .

Source 09 organizes its material specifically around the three-dimensional rotation group. Centralizers at half-turns, a globally valid matrix Cayley chart with its exceptional locus, infinitesimal conjugacy of finite subgroups, and the differential of the *rotation* exponential are developed together. The ordinary quaternion algebra and spin background is standard [8]. The surreal field and restricted-analytic background is due to Conway, Gonshor, van den Dries–Ehrlich and Ehrlich–Kaplan; the analytic constructions here rely specifically on [1, 3]. The perfectness of the infinitesimal $\mathrm{SO}(3)$ group has a published precedent: Bays–Peterzil explicitly record that every element is a commutator [9, §3.1]. [Theorem 19.3](#) gives an elementary, exact quaternion construction rather than presenting that fact as new.

16 Frames, Euler coordinates, and quaternion matrices

This section collects source 09’s further finite algebra of $G(F)$.

16.1 Products of reflections and transitivity

Reflection in a plane perpendicular to a unit vector u is $S_u = I - 2uu^T$, of determinant -1 . In an oriented perpendicular plane, two such reflections produce the usual doubled planar angle; the assertion is a finite matrix multiplication. Every rotation is a product of two plane reflections: take the two planes through its axis, choosing their normal lines by a half-angle square root on the algebraic circle. The half-turn and identity are included. Alternatively this follows from the explicit quaternion lifts of [section 5.2](#).

Given two unit vectors, complete each to an oriented orthonormal frame. The matrix sending the first frame to the second belongs to $G(F)$. Thus the orbits of $G(F)$ on F^3 are the origin and the spheres of each positive F -radius. The stabilizer of a unit vector n is

$$T_n = \{\mathrm{Rot}(n; c, s) : c^2 + s^2 = 1\} \cong \mathrm{SO}(2, F), \quad G(F)/T_n \cong S^2(F). \quad (16.1)$$

This homogeneous-space statement is algebraic and semialgebraic; it does not assert that these are ordinary real manifolds.

16.2 Euler coordinates and their degeneracy

Write $\mathrm{Rot}_z(c, s) = \mathrm{Rot}(e_3; c, s)$ and $\mathrm{Rot}_y(c, s) = \mathrm{Rot}(e_2; c, s)$. For every $A \in G(F)$ there is a decomposition

$$A = \mathrm{Rot}_z(c_\alpha, s_\alpha) \mathrm{Rot}_y(c_\beta, s_\beta) \mathrm{Rot}_z(c_\gamma, s_\gamma), \quad s_\beta \geq 0. \quad (16.2)$$

To construct it, put $n = Ae_3$, $c_\beta = n_3$, and $s_\beta = \sqrt{n_1^2 + n_2^2}$. When $s_\beta \neq 0$, choose $c_\alpha = n_1/s_\beta$, $s_\alpha = n_2/s_\beta$. Then the first two factors send e_3 to n , and their inverse times A fixes e_3 , giving the last factor. If $s_\beta = 0$, choose any suitable first factor and solve in the remaining stabilizer. The nonuniqueness there is the algebraic form of gimbal lock; infinitesimal s_β also makes these coordinates poorly conditioned. For actual surreals all three circle pairs admit finite angles, as shown in [theorem 5.4](#).

16.3 Quaternion coordinates

For $q = a + bi + cj + dk$ of norm one the rotation matrix is

$$\begin{pmatrix} 1 - 2(c^2 + d^2) & 2(bc - ad) & 2(bd + ac) \\ 2(bc + ad) & 1 - 2(b^2 + d^2) & 2(cd - ab) \\ 2(bd - ac) & 2(cd + ab) & 1 - 2(b^2 + c^2) \end{pmatrix}. \quad (16.3)$$

Composition of rotations is quaternion multiplication in the same order as matrix multiplication, with column-vector convention.

16.4 Complex matrices, projective coordinates, and a caution

The faithful algebra representation

$$a + bi + cj + dk \mapsto \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix} \quad \text{in } M_2(F(i)) \quad (16.4)$$

has determinant $N(q)$ and sends quaternion conjugation to conjugate transpose. Norm-one quaternions are exactly the corresponding $SU(2, F(i))$ matrices, where conjugation fixes F and sends i to $-i$.

Every nonzero quaternion can be normalized by its positive norm. Therefore

$$G(F) \cong \mathbb{H}_F^\times / F^\times \cong \mathbb{P}^3(F) \quad (16.5)$$

as a group with the quaternion-induced law, and as a semialgebraic space. The second statement concerns F -points: the norm has no zero on $F^4 \setminus \{0\}$. It is *not* an isomorphism of algebraic varieties $G \cong \mathbb{P}^3$. The algebraic projective model is the open subset $N \neq 0$; the norm quadric has points after algebraic closure. This distinction prevents an incorrect inference that the affine algebraic group is projective.

The quaternion quotient also explains why three dimensions are special. It is not necessary to choose an axis continuously to compose rotations. A unit quaternion carries a uniformly bounded four-coordinate representation, with only the unavoidable sign identification.

16.5 A quaternion extraction algorithm without a small divisor

Cayley coordinates become large near a half-turn. A uniformly bounded alternative works directly from A . Set

$$\begin{aligned} d_0 &= 1 + \operatorname{tr} A, & d_1 &= 1 + 2A_{11} - \operatorname{tr} A, \\ d_2 &= 1 + 2A_{22} - \operatorname{tr} A, & d_3 &= 1 + 2A_{33} - \operatorname{tr} A. \end{aligned}$$

By [equation \(16.3\)](#), these are $4a^2, 4b^2, 4c^2, 4d^2$, respectively, for either unit lift. They are nonnegative and sum to four. Choose a largest one, so $d_j \geq 1$, and choose the sign of the lift so that its j -th coordinate is $\sqrt{d_j}/2 \geq 1/2$. The other coordinates are recovered from

$$\begin{aligned} 4ab &= A_{32} - A_{23}, & 4ac &= A_{13} - A_{31}, & 4ad &= A_{21} - A_{12}, \\ 4bc &= A_{12} + A_{21}, & 4bd &= A_{13} + A_{31}, & 4cd &= A_{23} + A_{32}. \end{aligned}$$

All divisions are by a quantity of magnitude at least two. This is an exact ordered-field algorithm with four branches and positive square roots. Ties can be resolved by the first index. It does not depend on a floating-point tolerance, and it does not become singular when the rotation is infinitesimally close to a half-turn.

An axis itself cannot be extracted with comparable conditioning at the identity: there is no preferred axis there. The trace loses first-order information near identity, since $1 - c$ is quadratic in a small angle; the skew coordinates retain the linear part.

17 Conjugacy, centralizers, and maximal axis groups

Theorem 17.1 (Conjugacy and the half-turn exception). *For $A, B \in G(F)$:*

- (i) *They are conjugate in $G(F)$ if and only if $\text{tr } A = \text{tr } B$.*
- (ii) *If $A \neq I$ is not a half-turn and has axis Fn , then $C_{G(F)}(A) = T_n$.*
- (iii) *For a half-turn H_n ,*

$$C_{G(F)}(H_n) = \{B \in G(F) : Bn = \pm n\} \cong O(2, F) \cong T_n \rtimes C_2.$$

- (iv) *The normalizer of T_n is the same line stabilizer. Its quotient by T_n is C_2 , acting by inversion on T_n .*

Proof. Conjugation preserves trace and sends $\text{Rot}(n; c, s)$ to $\text{Rot}(Bn; c, s)$. Equal traces give equal c . The possible s 's differ only in sign; changing n to $-n$ absorbs that sign. Transitivity on unit vectors proves (i), including the half-turn case.

An element commuting with a nonidentity rotation preserves its one-dimensional fixed space, so sends n to n or $-n$. In the first case it is in T_n , which is abelian. In the second case it conjugates the planar rotation to its inverse. A nonidentity circle element equals its inverse exactly at the half-turn. This proves (ii) and (iii). In a frame ending in n , the line stabilizer has matrices $\text{diag}(B, \det B)$ with $B \in O(2, F)$. A half-turn about a perpendicular axis realizes its nontrivial component.

Finally, the common fixed space of all of T_n is Fn , so a normalizer must preserve that line. Every line-preserving rotation either fixes or reverses its orientation, and accordingly centralizes or inverts T_n . This proves (iv). $\square$

In particular, $G(F)$ has trivial center. The centralizers of non-half-turns also show that every nontrivial connected algebraic axis torus is conjugate to T_{e_3} . “Connected” in that sentence refers to the algebraic torus, not to its fine topological point set.

Corollary 17.2 (Commuting rotations). *Two nonidentity rotations commute only in the following cases: they have the same axis, or they are half-turns about perpendicular axes. Both possibilities do occur.*

Proof. If either is not a half-turn, use [theorem 17.1\(ii\)](#). For two half-turns, H_n preserves the line Fn precisely when m is parallel to n or perpendicular to it: its two eigenspaces are Fn and $n^\perp$. $\square$

Thus the three coordinate half-turns and the identity form a Klein four group. The slogan that commuting spatial rotations always share an axis is false, over both the reals and the surreals.

18 The standard-part extension

From now on suppose $F \supseteq \mathbb{R}$. Every matrix entry of a rotation belongs to $\mathcal{O}_F$, so standard part can be applied entrywise to every rotation, not only to a restricted subgroup.

Theorem 18.1 (Real rotation plus an infinitesimal rotation). *There is a split exact sequence*

$$1 \longrightarrow K_F \longrightarrow G(F) \xrightarrow{\text{st}} \text{SO}(3, \mathbb{R}) \longrightarrow 1, \quad K_F = G(F) \cap (I + M_3(\mathfrak{m}_F)). \quad (18.1)$$

Every $A \in G(F)$ has the unique decomposition

$$A = UA_0, \quad A_0 = \text{st}(A), \quad U = AA_0^{-1} \in K_F. \quad (18.2)$$

Consequently $G(F) \cong K_F \rtimes \text{SO}(3, \mathbb{R})$, with the usual conjugation action.

Proof. The equations $A^T A = I$, $\det A = 1$, and matrix multiplication use finitely many sums and products. Applying the ring homomorphism st preserves them. The inclusion of constant real matrices is a section. The kernel is exactly as displayed, and multiplying by $\text{st}(A)^{-1}$ gives the unique decomposition. The transported product on pairs, for $U, V \in K_F$ and $B, B' \in \text{SO}(3, \mathbb{R})$, is

$$(U, B)(V, B') = (U(BVB^{-1}), BB').$$

□

Proposition 18.2 (Exact coordinates on the kernel). *The Cayley map restricts to a bijection $\mathfrak{m}_F^3 \rightarrow K_F$, with multiplication $t \star u = (t + u + t \times u)/(1 - t \cdot u)$. The unit quaternions infinitesimally close to 1 map isomorphically onto K_F .*

Proof. For $t \in \mathfrak{m}_F^3$, equation (5.10) gives $\text{Cay}(t) - I \in M_3(\mathfrak{m}_F)$. Conversely, for $A \in K_F$, $1 + \text{tr } A$ has standard part four, so equation (5.12) is defined and infinitesimal. The denominator in the product law has standard part one. The distinguished spin lift is $(1, t)/\sqrt{1 + \|t\|^2}$, which has standard part 1. Among the two lifts only this one has that standard part, proving the isomorphism. □

The quaternion side of theorem 18.1 and proposition 18.2 is the split sequence $1 \rightarrow U_1 \rightarrow S^3(\mathbf{No}) \rightarrow S^3(\mathbb{R}) \rightarrow 1$ of the surquaternion report (`squat:sub:unitreduction`), whose kernel U_1 is the group of unit quaternions infinitesimally close to 1 (`squat:thm:rotation-log`). The spin map carries it isomorphically onto K .

When $F \neq \mathbb{R}$, the kernel is nontrivial: any nonzero $\varepsilon \in \mathfrak{m}_F$ gives $\text{Cay}(\varepsilon e_1) \neq I$. It is proper because a nonidentity constant real rotation has nontrivial standard part. Thus $\text{SO}(3, \mathbf{No})$ is *not abstractly simple*. Its simple algebraic group and simple Lie algebra are compatible with this fact.

The split extension is not a direct product. For a real rotation B ,

$$B \text{Cay}(t) B^{-1} = \text{Cay}(Bt), \quad (18.3)$$

and this action is visibly nontrivial. Two rotations with the same standard part can therefore interact at finer scales rather than differing by a central error.

The quotient has a useful metric interpretation. For $x \in \mathcal{O}_F^3$, $\text{st}(Ax) = \text{st}(A) \text{st}(x)$. Infinitesimal rotations are invisible on the real shadow of every finite vector. This conclusion need not hold for infinite vectors; section 27 quantifies the failure.

19 A perfect, uniquely divisible infinitesimal group

The word infinitesimal does not mean abelian. Nor does a Baker–Campbell–Hausdorff formula imply that the entire kernel is nilpotent. The following results make both points precise. Throughout this section, $F \supsetneq \mathbb{R}$ is real closed; the same finite proofs apply to the full surreal class.

Lemma 19.1 (No infinitesimal torsion). *If $U \in K_F$ and $U^n = I$ for an ordinary positive integer n , then $U = I$.*

Proof. Write $U = \text{Rot}(n_0; c, s)$ and regard $z = c + is$ as an element of $F(i)$. The trace gives $c \in 1 + \mathfrak{m}_F$. Since $s^2 = 1 - c^2$, also $s \in \mathfrak{m}_F$. The equation $U^n = I$ gives $z^n = 1$. The polynomial $X^n - 1$ already splits into distinct linear factors over the ordinary complex subfield. Hence z is an ordinary complex root of unity. Its standard part is one, so $z = 1$, and $U = I$. $\square$

Theorem 19.2 (Unique divisibility). *For every ordinary $n \geq 1$, the power map $U \mapsto U^n$ is a bijection of K_F . No nontrivial finite group is a homomorphic image of K_F .*

Proof. For $U = \text{Rot}(n_0; c, s)$, choose an n -th root w of $z = c + is$ in the algebraically closed field $F(i)$. Its squared norm satisfies $N(w)^n = N(z) = 1$, and $N(w) > 0$, so $N(w) = 1$. Its coordinates are bounded and $\text{st}(w)$ is an ordinary complex n -th root of unity. Dividing w by this root gives a circle element w_0 infinitesimally close to one with $w_0^n = z$. Rotation around n_0 by this algebraic angle gives a root in K_F .

For uniqueness, the identity case follows from lemma 19.1. Otherwise, if $V^n = U$, then V commutes with U . A nonidentity element of K_F is not a half-turn, so $V \in T_{n_0}$ by theorem 17.1. Two roots in that abelian group have quotient of finite order in K_F , and are equal. Finally, an image of a divisible group is divisible. In a finite image of order m , every m -th power is the identity, so surjectivity of that power map forces the image to be trivial. $\square$

Unique divisibility is a statement about solving one power equation at a time. It does not say that K_F is a vector space over $\mathbb{Q}$, because its group operation is not commutative.

Theorem 19.3 (Every infinitesimal rotation is one commutator). *For every $U \in K_F$ there are $V, W \in K_F$ with*

$$U = (V, W) := VWV^{-1}W^{-1}.$$

Thus $K_F = [K_F, K_F]$. In particular, it is nonabelian and neither solvable nor nilpotent, and it has no nontrivial solvable quotient.

Proof. The identity is immediate. Take the unique lift of $U \neq I$ near one, $q = (c, w) \in \text{Sp}(1, F)$. Then $0 < c < 1$, $c - 1 \in \mathfrak{m}_F$, and $w \neq 0$. Put

$$s = \left(\frac{1 - c}{2} \right)^{1/4}, \quad a = \sqrt{1 - s^2}, \quad p = (a, se_1), \quad r = (a, se_2).$$

These are unit quaternions near one. Direct quaternion multiplication gives

$$\text{Re}(p, r) = 1 - 2s^4 = c. \tag{19.1}$$

For completeness, a version before the equal-parameter specialization is

$$\text{Re}((a, se_1)(b, te_2)(a, -se_1)(b, -te_2)) = (a^2 + s^2)(b^2 + t^2) - 2s^2t^2.$$

Since (p, r) and q have norm one and the same scalar part, their imaginary vectors have the same nonzero norm. A unit quaternion h conjugates one vector to the other, by the transitivity and spin surjectivity already proved. Therefore

$$q = (hph^{-1}, hrh^{-1}).$$

Conjugation by a unit quaternion preserves infinitesimal imaginary coordinates: its matrix entries are bounded. Thus both factors are still near one. Applying p proves the claim in K_F . A nontrivial perfect group is neither solvable nor nilpotent; every homomorphic image is perfect, and a solvable perfect group is trivial. $\square$

The conclusion has a literature precedent [9, proof of Claim 3.3]; the proof above exposes the exact fourth-root scale used in a commutator construction. The two factors generally live at a *larger infinitesimal magnitude* than their commutator. There is no contradiction with the abelian leading layers established in section 23.

The whole group $G(F)$ likewise has commutator width one. Choose an algebraic square root B of a rotation A around the same axis, and let H be a half-turn about a perpendicular axis. Then $HBH^{-1} = B^{-1}$, so $(B, H) = B^2 = A$. This latter proof alone does not prove the infinitesimal claim, since the perpendicular half-turn is not infinitesimal.

20 Finite subgroups have no new surreal forms

Lemma 20.1 (Positive matrix square roots over a real closed field). *Every symmetric positive definite matrix over F has a unique symmetric positive definite square root. If $P - I \in M_d(\mathbf{m}_F)$, its positive square root and inverse square root are also infinitesimally close to I .*

Proof. The symmetric spectral theorem holds over any real closed field, by the argument of section 4.3: an eigenvector $z \neq 0$ over $F(i)$ has eigenvalue $(z^*Pz)/(z^*z) \in F$, and induction on perpendicular complements gives an orthonormal eigenbasis. (Source 09 gave the same argument in full.) For positive definite P , its eigenvalues λ_j are positive, so take $\sqrt{\lambda_j}$ in that basis. Any symmetric positive square root commutes with P , preserves its eigenspaces, and has only the eigenvalue $\sqrt{\lambda_j}$ on each such space; this proves uniqueness.

If every entry of $Y = P - I$ is infinitesimal, a unit eigenvector gives $|\lambda_j - 1| = |x^TYx| \leq d \max |Y_{ij}|$, by $\sum_i |x_i| \leq \sqrt{d}$. Thus $\lambda_j - 1$, $\sqrt{\lambda_j} - 1$, and $\lambda_j^{-1/2} - 1$ are infinitesimal. Conjugation by the bounded orthonormal eigenbasis preserves this property. $\square$

Theorem 20.2 (Infinitesimal conjugacy of finite subgroups). *Let $H \leq G(F)$ be finite. Standard part is injective on H , and there is $C \in K_F$ such that*

$$C^{-1}hC = \text{st}(h) \quad \text{for every } h \in H. \quad (20.1)$$

In particular, every finite surreal rotation group is conjugate to an ordinary real rotation group by a rotation with standard part I .

Proof. The kernel of $\text{st}|_H$ is finite and lies in the torsion-free group K_F , so it is trivial. Write $\rho(h) = \text{st}(h)$ and average the two representations:

$$B = \frac{1}{|H|} \sum_{h \in H} h\rho(h)^{-1}. \quad (20.2)$$

Each summand is infinitesimally close to I ; hence so is B , and its determinant has standard part one. In particular it is invertible and has positive determinant. Reindexing the finite sum shows $gB = B\rho(g)$ for every $g \in H$.

Set $P = B^T B$. This is symmetric positive definite and infinitesimally close to I . Orthogonality of both representations gives $\rho(g)^T P \rho(g) = P$, so P commutes with $\rho(g)$. By uniqueness of the positive square root, $P^{-1/2}$ commutes with all those matrices as well. Put

$$C = BP^{-1/2}.$$

Then $C^T C = I$, and both determinants in the product are positive, so $\det C = 1$. The preceding lemma gives $C \in K_F$. Finally $gC = C\rho(g)$, which is exactly [equation \(20.1\)](#). $\square$

This proof is finite averaging followed by algebraic polar decomposition. It uses neither an integral over the full group nor a completeness theorem. The same argument works in $O(d, F)$ for any fixed finite dimension, with the appropriate determinant statement.

20.1 Classification and what it does not claim

The familiar finite-subgroup classification of $SO(3, \mathbb{R})$ therefore applies: cyclic groups C_n , rotational dihedral groups D_n of order $2n$, and the tetrahedral, octahedral, and icosahedral groups A_4, S_4, A_5 , together with the trivial group. The word dihedral here denotes a group of *three-dimensional rotations*: the elements outside its principal cyclic subgroup are perpendicular half-turns, not spatial reflections.

One can see the numerical restriction by counting poles on the ordinary unit sphere. If $H \neq 1$, each nonidentity element fixes exactly two points. At a pole the stabilizer is cyclic. If the pole orbits have stabilizer orders $m_1, \dots, m_k \geq 2$, counting incident pairs gives

$$\sum_{j=1}^k (1 - 1/m_j) = 2 - 2/|H|. \quad (20.3)$$

It follows that $k = 2$ or 3 . For $k = 2$, both stabilizers have order $|H|$, giving a cyclic group. For $k = 3$, order the m_j 's; the only possibilities are

$$(2, 2, n), \quad (2, 3, 3), \quad (2, 3, 4), \quad (2, 3, 5),$$

with group orders $2n, 12, 24, 60$. The usual spherical realizations identify them with the dihedral and three polyhedral groups. This last identification is the classical finite rotation-group classification; the new field introduces no additional possibility because of [theorem 20.2](#).

Individual finite-order surreal rotations need not have real axes. A conjugate CHC^{-1} of a real finite group by a nonreal $C \in K_F$ generally has nonreal entries. The theorem says that these are conjugate realizations, not that every finite subgroup is literally contained in the constant matrices. It also says nothing about arbitrary infinite or set-generated subgroups.

21 The Lie algebra and angular velocity

The algebraic tangent space at the identity is computed with the formal dual number κ , satisfying $\kappa^2 = 0$. It is not a nonzero surreal infinitesimal: a field has no such nilpotents. Substituting $I + \kappa X$ in the orthogonality equation gives

$$(I + \kappa X)^T (I + \kappa X) = I + \kappa(X^T + X).$$

Thus

$$\mathfrak{so}(3, F) = \{X \in M_3(F) : X^T = -X\} = \{\widehat{u} : u \in F^3\}. \quad (21.1)$$

Its bracket and adjoint action are exactly

$$[\widehat{u}, \widehat{v}] = \widehat{u \times v}, \quad \text{Ad}_A(\widehat{u}) = \widehat{Au}. \quad (21.2)$$

This identifies the Lie algebra with the cross-product algebra of three-space.

Proposition 21.1 (Simplicity and the spin factor). *The Lie algebra $\mathfrak{so}(3, F)$ is simple. The differential of the spin map is $u \mapsto 2\widehat{u}$ on imaginary quaternions.*

Proof. Let a nonzero ideal contain $u \neq 0$ in cross-product coordinates. Then it contains $u \times v$ for every v . The image of this linear map is exactly $u^\perp$, by [equation \(3.5\)](#). Together with u , these vectors span F^3 , so the ideal is the entire algebra. The algebra is nonabelian. For the spin differential, conjugate x by $1 + \kappa u$: the linear term is $ux - xu = 2u \times x$. The imaginary-quaternion Lie bracket is itself $2u \times v$, consistent with the factor two. $\square$

For a differentiable F -parameter family of rotations $A(t)$, whenever the chosen derivative has the product rule, differentiation of $A^T A = I$ gives

$$\Omega_{\text{space}} = A' A^{-1} \in \mathfrak{so}(3, F), \quad \Omega_{\text{body}} = A^{-1} A' \in \mathfrak{so}(3, F), \quad \Omega_{\text{space}} = A \Omega_{\text{body}} A^{-1}.$$

These are the two angular-velocity conventions. Their vector representatives are related by $\omega_{\text{space}} = A \omega_{\text{body}}$. The algebraic identity remains true for an entrywise field derivation, but the existence or uniqueness of solutions of the resulting differential equation is an additional question. In particular, the Berarducci–Mantova derivation of surreal coefficients is not the variable derivative used in the fine differential of [section 25](#).

A further distinction is essential: $\mathfrak{so}(3, \mathbf{No})$ is a three-dimensional $\mathbf{No}$ -vector space, whereas $\mathfrak{so}(3, \mathfrak{m})$ is not closed under multiplication by arbitrary infinite scalars. The latter is the domain of the infinitesimal logarithm, not a replacement for the full algebraic tangent space.

22 Strong exponential, logarithm, and BCH

22.1 A support theorem, not a convergence assumption

Let Γ be a set-sized divisible ordered abelian group, and work in $F_\Gamma = \mathbb{R}((t^\Gamma))$, the Hahn field with well-ordered supports. It is real closed [\[1, 13\]](#). A common surreal workspace uses $t^\gamma = \omega^{-\gamma}$ for an ordered subgroup of surreal exponents. The rational span of the supports of finitely or set-many given normal forms supplies a set-sized divisible exponent group. This localization is larger than the finite-algebra subfield of [proposition 2.1](#); its purpose is to contain the strong sums.

Strong summability is the notion of [section 2](#): a well-ordered union of supports, each exponent in only finitely many members, and coefficientwise sums. It is independent of ordinary sequence convergence. The following all-lengths form of the positive-support lemma is `squat:lem:neumann` (Lemma 7.1 of the surquaternion report), with the same Higman argument; the repository’s `Surreal/HahnSeries/Neumann.lean` proves the support statements for other reports’ labels. Source 09’s statement and proof are kept because they add fixed-size matrix arguments.

Lemma 22.1 (All-length support control). *Let $S \subseteq \Gamma_{>0}$ be well ordered. The additive monoid S^* generated by S is well ordered, and every $\gamma \in \Gamma$ is the sum of only finitely many finite ordered words in S . Consequently every formal power series in finitely many commuting or noncommuting variables can be evaluated strongly at Hahn elements or fixed-size matrices whose entries have positive valuation. The nonconstant support lies in $S^* \setminus \{0\}$.*

Proof. Use Higman's finite-word lemma for the well-quasi-ordered alphabet S [5]. In any infinite sequence of words, an earlier word embeds into a later word with each letter no larger than the corresponding letter. Such an embedding makes the total sum no larger; unless the words are identical, either an extra positive letter or an increased letter makes it strictly larger. Therefore a strictly decreasing sequence of word sums is impossible, and infinitely many distinct words cannot have one fixed sum. This proves the two support assertions.

For the evaluation assertion take the finite union of the positive supports of all input entries. Each coefficient of a fixed total exponent involves only finitely many such words. The finite choices of variables and intermediate matrix indices add only finitely many contributions per word. Thus the formal evaluation is strongly summable, including all word lengths at once. $\square$

The same argument applies to a set-supported surreal calculation after localizing the supports. It justifies rearrangements in what follows; it does not assert that $n\gamma$ is cofinal in an arbitrary ordered value group.

Theorem 22.2 (The infinitesimal exponential and logarithm). *On a divisible Hahn workspace, and on the full surreal class by strong normal-form evaluation, the maps*

$$\text{Exp}_{\mathfrak{m}}(X) = \sum_{n \geq 0} \frac{X^n}{n!}, \quad X \in \mathfrak{so}(3, \mathfrak{m}), \quad (22.1)$$

$$\text{Log}_{\mathfrak{m}}(U) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (U - I)^n, \quad U \in K, \quad (22.2)$$

are inverse bijections $\mathfrak{so}(3, \mathfrak{m}) \leftrightarrow K$. They commute with conjugation by any rotation. Their transported multiplication is the strongly evaluated Baker–Campbell–Hausdorff law, not vector addition.

Proof. The support lemma makes both expressions meaningful. Formal one-variable identities show that exponential and logarithm are inverse on the ambient matrix ideals and groups $M_3(\mathfrak{m})$ and $I + M_3(\mathfrak{m})$. Substitution is legitimate because the nonconstant arguments have positive support.

For skew X , transposition gives $\text{Exp}_{\mathfrak{m}}(X)^T = \text{Exp}_{\mathfrak{m}}(-X)$. The product is I , and the result has standard part I , so its determinant is $+1$, not -1 . Conversely, if $U \in K$, then

$$\text{Log}_{\mathfrak{m}}(U)^T = \text{Log}_{\mathfrak{m}}(U^T) = \text{Log}_{\mathfrak{m}}(U^{-1}) = -\text{Log}_{\mathfrak{m}}(U).$$

The last identity is a formal identity in the one matrix $U - I$. Conjugation commutes with all finite products and with the supported sums. Finally, substituting X, Y in the formal noncommutative identity $\log(e^X e^Y) = \text{BCH}(X, Y)$ is justified by the same word-support argument. $\square$

In vector coordinates, if $\text{Exp}_{\mathfrak{m}}(\hat{u}) \text{Exp}_{\mathfrak{m}}(\hat{v}) = \text{Exp}_{\mathfrak{m}}(\hat{w})$, then

$$\begin{aligned} w &= u + v + \frac{1}{2}u \times v \\ &\quad + \frac{1}{12}(u \times (u \times v) + v \times (v \times u)) + \text{terms of total degree at least four.} \end{aligned} \quad (22.3)$$

The omitted terms are the uniquely specified formal BCH series. If every coordinate of u, v has valuation at least $\gamma > 0$, their sum has valuation at least 4γ . This is a finite-order error certificate, not an assertion of convergence through all positive surreal tolerances.

Proposition 22.3 (Exact bridge to Cayley coordinates). *For $t \in \mathfrak{m}^3$,*

$$\text{Log}_{\mathfrak{m}}(\text{Cay}(t)) = 2 \sum_{j \geq 0} \frac{(-1)^j}{2j+1} (t \cdot t)^j \hat{t}. \quad (22.4)$$

Its leading term is $2\hat{t}$. In particular, exponential coordinates and Cayley coordinates have the same valuation.

Proof. The commuting factors $I + \hat{t}$ and $I - \hat{t}$ give

$$\log((I + \hat{t})(I - \hat{t})^{-1}) = \log(I + \hat{t}) - \log(I - \hat{t}) = 2 \sum_{j \geq 0} \frac{\hat{t}^{2j+1}}{2j+1}.$$

Now use $\hat{t}^{2j+1} = (-t \cdot t)^j \hat{t}$. Every later term has strictly higher valuation than the first. $\square$

23 Valuation layers and noncommutativity at every scale

Use the valuation of [section 2](#): $v(x) = -\max \text{supp}_{\omega}(x)$, $v(0) = +\infty$, so $v(\omega^{-\gamma}) = \gamma$ and infinitesimals have positive valuation. For a vector or matrix, v is the minimum of the valuations of its entries. On a Hahn workspace this is the ordinary Hahn valuation with value group Γ . For all surreals the value group is a class; statements involving one layer still have finite formulas, and direct-sum notation is localized to set-sized workspaces.

Because all rotation entries have nonnegative valuation, and so do all entries of the inverse, $v(Ax) = v(x)$. The equality follows by applying the two inequalities from A and A^{-1} . Thus a rotation preserves not only the norm, but also the leading scale of every vector.

For $\gamma > 0$, define

$$K_{\geq \gamma} = \{U \in K : v(U - I) \geq \gamma\}, \quad K_{> \gamma} = \{U \in K : v(U - I) > \gamma\}. \quad (23.1)$$

These are normal subgroups of G . If t is the Cayley coordinate of $U \in K$, then

$$v(U - I) = v(t) = v(\text{Log}_{\mathfrak{m}} U). \quad (23.2)$$

For the first equality use the rational chart and its inverse, whose denominators have nonzero real standard parts. The second follows from [equation \(22.4\)](#).

Theorem 23.1 (Graded rotation algebra). *For $\alpha, \beta, \gamma > 0$,*

$$(K_{\geq \alpha}, K_{\geq \beta}) \subseteq K_{\geq \alpha + \beta}. \quad (23.3)$$

For surreals, the map

$$\ell_{\gamma}(U) = 2 \text{st}(\omega^{\gamma} t(U)) \quad (23.4)$$

induces an isomorphism $K_{\geq \gamma}/K_{> \gamma} \cong (\mathbb{R}^3, +)$. On a Hahn workspace replace $\omega^{-\gamma}$ by t^{γ} . The commutator between layers is the ordinary cross product:

$$\ell_{\alpha + \beta}((U, V)) = \ell_{\alpha}(U) \times \ell_{\beta}(V). \quad (23.5)$$

Conjugation on every layer factors through standard part and is the standard three-dimensional action of $\text{SO}(3, \mathbb{R})$.

Proof. The Cayley law is addition modulo terms of degree at least two. For inputs of valuation at least γ , scaling by ω^γ kills those terms under standard part. Thus ℓ_γ is a homomorphism with kernel $K_{>\gamma}$. It is onto, since $U = \text{Cay}(\frac{1}{2}\omega^{-\gamma}a)$ maps to $a \in \mathbb{R}^3$.

For the commutator it is convenient to use $U = e^X, V = e^Y$ with positive valuation. Formal multiplication gives

$$\text{Log}_m((e^X, e^Y)) = [X, Y] + \text{Lie terms containing at least three letters.}$$

Every nonzero term contains both X and Y . Terms after the bracket have valuation strictly larger than $\alpha + \beta$, whereas the bracket has at least that valuation. Since X has leading vector twice the Cayley vector, [equation \(3.6\)](#) proves [equations \(23.3\)](#) and [\(23.5\)](#) with the stated factor two. Lastly [equation \(18.3\)](#) gives $\ell_\gamma(AUA^{-1}) = \text{st}(A)\ell_\gamma(U)$. $\square$

On a set-sized workspace the associated graded Lie algebra is therefore

$$\bigoplus_{\gamma \in \Gamma_{>0}} \mathfrak{so}(3, \mathbb{R}) T^\gamma, \quad [aT^\alpha, bT^\beta] = (a \times b)T^{\alpha+\beta},$$

with vector notation for the coefficients. A leading bracket can vanish when the leading vectors are parallel; the theorem deliberately gives an inclusion of filtration levels, not equality of valuations in every case.

The quaternion form of this grading is `squat:thm:graded` (Theorem 8.6 of the surquaternion report), where the graded bracket of leading imaginary coefficients is $2X \times Y$. The two statements agree: the leading imaginary coefficient X of the unit-quaternion lift $(1, t)/\sqrt{1 + \|t\|^2}$ of U is the leading coefficient of its Cayley vector t , so $\ell_\gamma(U) = 2X$. Hence $\ell_{\alpha+\beta}((U, V)) = 2(2X \times Y) = \ell_\alpha(U) \times \ell_\beta(V)$; the factor two is that of the spin differential ([proposition 21.1](#)).

23.1 Why finite jets are nilpotent but the kernel is perfect

For any ordinary integer $N \geq 1$,

$$K_{\geq \gamma} / K_{> N\gamma}$$

is nilpotent of class at most N : an $(N + 1)$ -fold iterated commutator has valuation at least $(N + 1)\gamma > N\gamma$. This is a fixed-scale, finite-depth statement. The full group is the union over *all* positive scales. Its commutator construction can use parameters at scales such as $\gamma/2$, outside a chosen $K_{\geq \gamma}$. Theorem [19.3](#) therefore remains fully compatible with the nilpotence of these quotients.

It would be incorrect to conclude merely from the BCH description that the whole K is a pronilpotent group in an unspecified topology. In particular, one fixed sequence of multiples $n\gamma$ need not exhaust all valuation scales, and the whole group is perfect rather than residually nilpotent in the sense of its ordinary lower central series.

23.2 An imported structural perspective

For a set-sized proper real closed extension $F \supsetneq \mathbb{R}$, a theorem of Bays–Peterzil says that the pure group K_F , with parameters allowed, is bi-interpretable with the valued field $(F; +, \cdot, \mathcal{O}_F)$ [[9](#), Theorem 1.1 and Remark 1.2]. Thus the exact infinitesimal group carries much more information than an abelian tangent space. This is an imported model-theoretic theorem, not proved here, and its statement is about set-sized structures. We do not automatically replace its universe by the full proper class of surreals.

24 Finite angles and canonical infinite phases

24.1 Finite-angle trigonometry from strong sums

For $\theta = r + \varepsilon \in \mathcal{O}$, with $r \in \mathbb{R}$, $\varepsilon \in \mathfrak{m}$, define

$$\text{cis}(\theta) = e^{ir} \sum_{k \geq 0} \frac{(i\varepsilon)^k}{k!} = \cos \theta + i \sin \theta. \quad (24.1)$$

Only the infinitesimal part is strongly expanded; e^{ir} is an ordinary complex constant. This is the finite-angle construction used in the repository [11] and in the restricted-analytic framework [3]. The formal exponential identities prove addition, parity, Pythagoras, and the usual finite trigonometric identities. In particular cis is a homomorphism from $(\mathcal{O}, +)$ to the algebraic surcomplex unit circle. It agrees with the real-centered evaluation of equation (2.3).

Source 09's theorem on finite angles has two halves, both printed in Part I. The map cis is onto the unit circle with kernel exactly $2\pi\mathbb{Z}$: this is theorem 2.2, whose second proof is source 09's. Every $A \in G(\mathbf{No})$ has a representation $A = \text{Rot}(n; \cos \theta, \sin \theta)$ with $0 \leq \theta \leq \pi$, unique for $A \neq I$, with the axis oriented uniquely for $0 < \theta < \pi$: this is theorem 5.4, whose proof is source 09's.

Consequently the stabilizer of any chosen oriented unit axis has the group presentation

$$T_n \cong \mathcal{O}/(2\pi\mathbb{Z}). \quad (24.2)$$

Its finite angles are surreal, not necessarily ordinary real. A rotation can have an infinitesimal angle, or differ infinitesimally from an ordinary angle. There is no need to attach an infinite angle to represent any rotation.

24.2 The Ehrlich–Kaplan global extension does exist

It would nevertheless be wrong to say that no canonical global surreal trigonometry is available. The initial-integer-part construction of Ehrlich–Kaplan [3, §11] supplies one. In terms of normal forms, write

$$\mathbf{No}^\uparrow = \{x : x \text{ has only positive normal-form exponents}\}, \quad \mathbf{Oz} = \mathbf{No}^\uparrow + \mathbb{Z}.$$

The class $\mathbf{Oz}$ is the initial integer part, often called the omnific integers. It is not the class of ordinary ordinal numbers. The zero element is included in $\mathbf{No}^\uparrow$.

The corresponding phase has a particularly transparent expression. For the three-part decomposition $\theta = \theta^\uparrow + r + \varepsilon$, set

$$\text{cis}_{\text{EK}}(\theta) = \text{cis}(r + \varepsilon) = \text{Cos } \theta + i \text{Sin } \theta. \quad (24.3)$$

This is exactly reduction by $2\pi\mathbf{Oz}$: a real multiple of a purely infinite normal form is still purely infinite, so

$$2\pi\mathbf{Oz} = \mathbf{No}^\uparrow + 2\pi\mathbb{Z}, \quad \ker \text{cis}_{\text{EK}} = 2\pi\mathbf{Oz}. \quad (24.4)$$

The decomposition into purely infinite and finite parts is additive. Thus the global phase is a surjective additive-to-multiplicative homomorphism. This construction is canonical for the specified initial-integer-part normalization; it is not an assertion that every conceivable global phase convention agrees with it.

For example,

$$\text{Cos } \omega = 1, \quad \text{Sin } \omega = 0, \quad \text{cis}_{\text{EK}}(\omega + \pi/2) = i.$$

The infinite part is removed before evaluating the finite phase. No divergent Taylor series at ω is being summed. On a fixed axis one therefore has

$$(\mathbf{No}, +)/(2\pi\mathbf{Oz}) \cong T_n, \quad \theta \mapsto \text{Rot}(n; \cos \theta, \sin \theta). \quad (24.5)$$

The finite and global presentations describe the *same* axis subgroup, with different parameter domains and period classes.

The associated spin parameter is

$$q_n(\theta) = \cos(\theta/2) + n \sin(\theta/2), \quad \ker q_n = 4\pi\mathbf{Oz}.$$

An ordinary 2π rotation has spin value -1 , while a purely infinite parameter such as ω has spin value $+1$. Indeed, multiplication by any nonzero real preserves $\mathbf{No}^\uparrow$, so the purely infinite parts of $2\pi\mathbf{Oz}$ and $4\pi\mathbf{Oz}$ coincide; their ordinary integer parts differ. This prevents misleading interpretations of an infinite parameter as an ordinary number of physical revolutions.

25 The global rotation exponential and its differential

25.1 A radial map on the whole Lie algebra

For $w \in \mathbf{No}^3$, put $r = \|w\|$. Define

$$\mathcal{E}(\hat{w}) = \begin{cases} I + \frac{\sin r}{r} \hat{w} + \frac{1 - \cos r}{r^2} \hat{w}^2, & r \neq 0, \\ I, & r = 0. \end{cases} \quad (25.1)$$

Equivalently, it is the spin image of $\cos(r/2) + (w/r) \sin(r/2)$. This is a well-defined global map to $G(\mathbf{No})$, and is equivariant under conjugation. It agrees with the strong matrix exponential for infinitesimal w ; finite w are evaluated using finite trigonometry, not by assuming strong summability of the uncentered matrix power series.

The map is already onto on the finite ball $\|w\| \leq \pi$, by [theorem 5.4](#). On any fixed line of generators it is a homomorphism:

$$\mathcal{E}(\alpha \hat{n}) \mathcal{E}(\beta \hat{n}) = \mathcal{E}((\alpha + \beta) \hat{n}).$$

It is not a homomorphism from the whole additive Lie algebra. The additive domain is abelian while its image is not; the BCH correction is nonzero even for small independent generators.

Theorem 25.1 (Complete fibers of the rotation exponential). *Let $A \neq I$ have the principal representation $A = \text{Rot}(n; \cos \theta, \sin \theta)$, $0 < \theta \leq \pi$. Then*

$$\mathcal{E}^{-1}(A) = \{(\widehat{(\theta + 2\pi z)})n : z \in \mathbf{Oz}\}. \quad (25.2)$$

At $\theta = \pi$ either choice of axis orientation gives the same set. The identity fiber is

$$\mathcal{E}^{-1}(I) = \{0\} \cup \{\hat{w} : \|w\| \in (2\pi\mathbf{Oz})_{>0}\}. \quad (25.3)$$

Restricting to finite generators replaces $\mathbf{Oz}$ by $\mathbb{Z}$.

Proof. If $\mathcal{E}(\hat{w}) \neq I$, its unique fixed line is Fw , so $w = \lambda n$. Oddness of sine and evenness of cosine show that $\mathcal{E}(\lambda \hat{n}) = \text{Rot}(n; \text{Cos } \lambda, \text{Sin } \lambda)$ for signed λ . Equality of circle values is exactly congruence modulo $2\pi \mathbf{Oz}$, proving [equation \(25.2\)](#). When $\theta = \pi$, negating λ does not change that congruence class. For the identity, the radial circle value must be one, which is precisely the condition in [equation \(25.3\)](#). The finite part of $\mathbf{Oz}$ is $\mathbb{Z}$, giving the last assertion. $\square$

The identity fiber is a union of spheres, not a subgroup of the additive three-dimensional Lie algebra. It must not be called the kernel of a global group homomorphism. Only a fixed-axis restriction has a kernel.

25.2 Fine differentiability

For a function on a surreal finite-dimensional coordinate space, the fine Fréchet derivative at x is a **No**-linear map L such that for every positive surreal η , some positive surreal δ satisfies

$$0 < \|h\| < \delta \implies \|f(x+h) - f(x) - Lh\| < \eta \|h\|.$$

Equivalent finite-dimensional coordinate norms give the same definition. The quantifiers range over all positive surreal scales. This derivative is not defined by ordinary real sequences.

The global phase satisfies the usual local derivative identities $D \text{Sin} = \text{Cos}$, $D \text{Cos} = -\text{Sin}$. Indeed, for infinitesimal h the global addition formulas reduce the increment to $\text{Sin } h$ and $\text{Cos } h$, whose strong Taylor expansions have bounded quadratic remainders. For any fixed positive surreal tolerance, one can choose a still smaller radius. The purely infinite part of the base point causes no obstruction to this local argument. For $w \neq 0$, differentiating the positive square root gives $Dr_w(h) = w \cdot h/r$; the binomial expansion at one justifies its local quadratic remainder even when r is infinite or infinitesimal. Here a quadratic remainder means a bound $C \|h\|^2$ on some neighborhood, with a fixed nonnegative surreal C depending on the base point; C need not be an ordinary real. Choosing a radius smaller than $\eta/(C+1)$ then gives the required fine derivative estimate.

Theorem 25.2 (Differential and exact critical radii). *Right-translate the differential of $\mathcal{E}$ back to the Lie algebra. For $w \neq 0$, $r = \|w\|$, one obtains*

$$(\mathcal{E}_w(h))\mathcal{E}(\hat{w})^{-1} = \widehat{\mathcal{J}(w)h}, \quad \mathcal{J}(w) = I + \frac{1 - \text{Cos } r}{r^2} \hat{w} + \frac{r - \text{Sin } r}{r^3} \hat{w}^2. \quad (25.4)$$

Here $\mathcal{E}$ is viewed as a function of the vector w , and $\mathcal{J}(0) = I$. For $r \neq 0$,

$$\det \mathcal{J}(w) = \frac{4 \text{Sin}^2(r/2)}{r^2}. \quad (25.5)$$

The nonzero critical radii are exactly $r \in (2\pi \mathbf{Oz})_{>0}$, and the fine differential has rank one there. Otherwise it has rank three.

Proof. Write $w = rn$ and split $h = h_{\parallel} + h_{\perp}$ along n . A radial increment changes only the angle and gives $\mathcal{J}(w)h_{\parallel} = h_{\parallel}$. For a perpendicular increment, $Dr(h_{\perp}) = 0$ and $Dn(h_{\perp}) = h_{\perp}/r$. Differentiate $A = cI + (1-c)nn^T + s\hat{n}$, with $c = \text{Cos } r$, $s = \text{Sin } r$, and multiply by A^T . The skew matrix that results has vector

$$\mathcal{J}(w)h_{\perp} = \frac{\text{Sin } r}{r} h_{\perp} + \frac{1 - \text{Cos } r}{r} n \times h_{\perp}. \quad (25.6)$$

This calculation can also be made with $n = e_3$, then transported by [equation \(21.2\)](#). Since $\hat{w}^2$ is zero on the axis and $-r^2 I$ on the plane, [equation \(25.6\)](#) is precisely [equation \(25.4\)](#). The derivative

identities preceding the theorem provide quadratic remainders; finite sums, products, and inverses with nonzero denominators preserve that property locally. At zero the small-generator expansion is $\mathcal{E}(\hat{h}) = I + \hat{h} + O(\|\hat{h}\|^2)$, giving $\mathcal{J}(0) = I$.

On the perpendicular plane, the determinant of [equation \(25.6\)](#) is

$$\frac{\sin^2 r + (1 - \cos r)^2}{r^2} = \frac{2(1 - \cos r)}{r^2} = \frac{4 \sin^2(r/2)}{r^2}.$$

The axis contributes one. Over an ordered field the planar numerator vanishes exactly when $\sin r = 0$ and $\cos r = 1$, equivalently $r \in 2\pi\mathbf{Oz}$. At such a point the whole perpendicular plane is killed and the axis survives, proving the rank statement. $\square$

The singular values of the right-trivialized derivative are

$$1, \quad \frac{2 |\sin(r/2)|}{r}, \quad \frac{2 |\sin(r/2)|}{r}. \quad (25.7)$$

This follows directly from [equation \(25.6\)](#), whose symmetric square is the scalar in [equation \(25.5\)](#) on the perpendicular plane. Thus a noncritical infinite generator can still be very poorly conditioned transversely. For example, at $r = \omega + \pi$ the transverse singular values are $2/(\omega + \pi)$, positive but infinitesimal.

At $w = \omega e_1$, the image is I and the differential has rank one. At the ordinary half-turn radius $r = \pi$, by contrast, the differential is invertible. The ambiguity of a principal logarithm at half-turns is a branch issue, not a critical point of the exponential there. Also note the factor of two relative to the radial unit-quaternion exponential: its imaginary parameter has half the spatial rotation angle. The four-dimensional quaternion radial map in [13] has a different domain and different rank counts.

No inverse function theorem in an unspecified surreal analytic category is inferred solely from nonsingularity of [equation \(25.4\)](#). The formulas and ranks are direct differential statements. Nor are they chain rules for a field derivation acting on the coefficients.

26 Three topologies, and what the spin cover means

This section separates the most important analytical ambiguity. Algebraic identities do not depend on a topology, but connectedness, convergence, compactness, and fundamental groups do.

26.1 The full fine topology

On $\mathbf{No}^d$ use neighborhoods $\|x - a\|_\infty < \rho$, with every positive surreal ρ allowed. On matrices and $G(\mathbf{No})$ use the corresponding finite-coordinate topology. For the full class these are neighborhood schemes, or ordinary topologies in a suitably larger-universe formulation with the size restrictions retained.

Theorem 26.1 (Set-sized discreteness in the full surreal topology). *Every set-sized subset of $\mathbf{No}^d$, and hence of $G(\mathbf{No})$, is closed and discrete. Every convergent set-indexed net is eventually equal to its limit. Every set-indexed Cauchy net is eventually constant. Nevertheless $G(\mathbf{No})$ has no isolated points.*

Proof. For a point a and a set-sized subset S , form the set of positive distances $\|s - a\|_\infty$ for $s \in S \setminus \{a\}$. The separated surreal cut with left set $\{0\}$ and that right set has a positive separator ρ

smaller than every listed distance. If the list is empty, any positive ρ works. Its ball meets S in at most $\{a\}$. This proves both closedness and discreteness.

A set-indexed net has a set-sized range. Applying the preceding argument at its limit proves eventual equality. For a Cauchy net, apply the cut argument to all positive pairwise distances in its range; these form a set. A Cauchy tail of diameter smaller than the resulting positive separator has only one value. Finally, $\text{Cay}(te_1) \rightarrow I$ as a neighborhood assertion with t arbitrarily small in $\mathbf{No}$, and $\text{Cay}(te_1) \neq I$ for $t \neq 0$. Translations give no isolated points anywhere in the group. $\square$

The general size phenomenon is already emphasized in the repository's foundations report [14]; the proof here specializes it to rotation coordinates. In a universe-relative version, "set-sized" means small relative to the universe in which the whole surreal carrier is large. It cannot be dropped from the statement.

In particular, an ordinary real interval admits no nonconstant fine-continuous map into $G(\mathbf{No})$: its image is a set, is discrete, and must be connected. An ordinary irrational-angle cyclic subgroup is a countable closed discrete subset of $G(\mathbf{No})$, even though its standard part can be dense in an ordinary real circle. Strongly summed exponential partial sums cannot converge in this fine topology unless they are eventually equal to their limit. This is why the support theorem, rather than a real convergence argument, was required.

26.2 The intrinsic topology of a set-sized workspace

Let $F \supsetneq \mathbb{R}$ be a set-sized real closed field. Its *intrinsic* order topology permits only radii from F . If $F \subseteq \mathbf{No}$, this is not its topology induced from the full surreal class: the latter is discrete by [theorem 26.1](#), while the intrinsic order topology has no isolated points. The distinction applies to $G(F)$ as well.

Even intrinsically, F is totally separated. Its infinitesimal ideal $\mathfrak{m}_F$ is clopen: a positive infinitesimal radius proves openness at zero, and a point with noninfinitesimal magnitude has a ball disjoint from $\mathfrak{m}_F$. Translates and nonzero scalar multiples of $\mathfrak{m}_F$ are clopen. For $a \neq b$, the clopen set $a + |b - a|\mathfrak{m}_F$ contains a but not b . Coordinate projections therefore separate any two points of $G(F)$ by a clopen set. Ordinary real-parameter paths into $G(F)$ are constant, and its ordinary fundamental group at every point is trivial.

This topology is not locally compact. To verify it rather than merely infer it from non-Archimedeaness, note that $\mathbb{R} \subset F$ is closed and discrete: near a real point choose an infinitesimal radius, near a finite nonreal point choose less than its distance to its standard part, and near an infinite point choose a ball avoiding all finite elements. Hence every subset of $\mathbb{R}$ is closed in F . Every neighborhood of the identity in $G(F)$ contains $\{\text{Cay}(ae_1/n) : n \geq 1\}$ for a sufficiently small $a > 0$ in F . This is an infinite closed discrete subset. It is closed even at the boundary of the Cayley chart, since its trace stays a positive F -distance from -1 , by [equation \(5.14\)](#). A compact neighborhood would contain it as a closed subset, which is impossible.

On a Hahn workspace, intrinsic valuation convergence is still meaningful and can be nontrivial. In $\mathbb{R}((t^{\mathbb{Q}}))$, for instance, $t^n \rightarrow 0$, since the ordinary integers are cofinal in $\mathbb{Q}$. In a higher-rank group that conclusion can fail for a given positive exponent. These are statements about the workspace, not convergence in the full surreal fine topology.

26.3 The standard-part quotient topology is discrete

The subgroup K_F is open and closed in the intrinsic topology, since it is the intersection of finitely many conditions $A_{ij} - I_{ij} \in \mathfrak{m}_F$. Thus all fibers of standard part are open. It follows that the quotient topology on

$$G(F)/K_F \cong \mathrm{SO}(3, \mathbb{R}) \quad (26.1)$$

is *discrete*, not the usual real Lie-group topology. Standard part into the usual $\mathrm{SO}(3, \mathbb{R})$ is continuous, but it is not a quotient map onto that usual topology. The constant section from the usual real Lie group is not continuous. The same neighborhood conclusions hold for the full surreal class.

Accordingly, a topological reading of the split extension must use the discrete topology on its real factor. Alternatively, one can deliberately choose the coarser topology pulled back from the usual real group by standard part. That topology has the usual real group as its quotient, but does not separate distinct members of one infinitesimal fiber. These are different choices, not competing proofs about one topology.

26.4 Semialgebraic paths, compactness, and the spin fundamental group

Over a real closed field F , semialgebraic topology restricts maps and homotopies to semialgebraic ones with parameter intervals $[0, 1]_F$. Closed and bounded semialgebraic sets are definably compact, and the matrix equations show that $G(F)$ and $\mathrm{Sp}(1, F)$ have that property. This is not ordinary compactness of all open covers.

There is an explicit semialgebraic path from the identity to every rotation. Choose a unit quaternion lift q with nonnegative scalar part and put

$$q(u) = \frac{(1 - u) + uq}{\sqrt{N((1 - u) + uq)}}, \quad 0 \leq u \leq 1 \text{ in } F. \quad (26.2)$$

The numerator never vanishes: for $0 \leq u < 1$ its scalar part is strictly positive, and at $u = 1$ it is q . Its spin image is the required path. Thus $G(F)$ is semialgebraically path connected even when its ordinary topological connected components are singletons. The parameter interval and the allowed notion of homotopy have changed.

The spin map is a semialgebraic two-sheeted covering. Around the identity use the non-half-turn open set and choose the lift with positive scalar part; the other lift has negative scalar part. Both branches are continuous and semialgebraic by the explicit square-root formulas, and translation supplies such neighborhoods at every point. This is also a local two-sheeted covering for the intrinsic topology of F -points.

Theorem 26.2 (The category in which the familiar fundamental group survives). *For every set-sized real closed field F ,*

$$\pi_1^{\mathrm{sa}}(G(F), I) \cong \mathbb{Z}/2\mathbb{Z}, \quad \pi_1^{\mathrm{sa}}(\mathrm{Sp}(1, F), 1) = 0.$$

In the semialgebraic category the spin map is the simply connected double cover. For a proper real closed extension of $\mathbb{R}$, this is not the ordinary fundamental group of the intrinsic topological space.

Proof. The equations of G , the unit three-sphere, and the spin map are defined over the rationals. The semialgebraic homotopy comparison and real-closed-field extension theorem of Baro–Otero [10, Corollaries 4.3–4.4] identifies their semialgebraic homotopy groups with the ordinary homotopy groups of their real realizations. Over $\mathbb{R}$ the map is the antipodal double cover $S^3 \rightarrow \mathbb{RP}^3$, giving the displayed groups. The local branches just constructed identify the transferred covering. For a proper extension, total separation has already shown that all ordinary loops are constant. $\square$

Likewise, $\pi_2^{\text{sa}}G(F) = 0$ and $\pi_3^{\text{sa}}G(F) \cong \mathbb{Z}$, by the same comparison. A claim about a “surreal universal cover” is meaningful only after specifying this category and, for the full class, the compatible universe or workspace interpretation. The fine topology does not make $G(\mathbf{No})$ into an ordinary real Lie group.

27 Worked examples at surreal scales

27.1 A tiny angle with a macroscopic displacement

Take $\varepsilon = \omega^{-1}$ and

$$A = \text{Cay}(\varepsilon e_3) = \begin{pmatrix} \frac{1-\varepsilon^2}{1+\varepsilon^2} & -\frac{2\varepsilon}{1+\varepsilon^2} & 0 \\ \frac{2\varepsilon}{1+\varepsilon^2} & \frac{1-\varepsilon^2}{1+\varepsilon^2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (27.1)$$

It belongs to K , is not the identity, and has angle $2 \operatorname{atan} \varepsilon = 2\varepsilon - \frac{2}{3}\varepsilon^3 + \dots$, where the arctangent is the strong local series. Acting on Le_1 gives

$$A(Le_1) - Le_1 = \left(-\frac{2L\varepsilon^2}{1+\varepsilon^2}, \frac{2L\varepsilon}{1+\varepsilon^2}, 0 \right)^T.$$

For $L = \omega$, the transverse displacement has standard part two. For $L = \omega^2$, it is of order ω , although the matrix still has standard part I . An infinitesimal relative error need not be a negligible absolute error at an infinite spatial scale.

Proposition 27.1 (Exact displacement formula). *If $A = \text{Rot}(n; c, s)$ and $x_\perp = x - (n \cdot x)n$, then*

$$\|Ax - x\|^2 = 2(1 - c) \|x_\perp\|^2. \quad (27.2)$$

For a finite angle θ , this is $\|Ax - x\| = 2 |\sin(\theta/2)| \|x_\perp\|$.

Proof. The axis component cancels. On the plane, $(A - I)x_\perp = (c - 1)x_\perp + s(n \times x_\perp)$. The two vectors are perpendicular with equal norms, so the squared norm is $((c - 1)^2 + s^2) \|x_\perp\|^2 = 2(1 - c) \|x_\perp\|^2$. $\square$

27.2 A near-half-turn with an infinite Cayley coordinate

Let $B = \text{Cay}(\omega e_3)$. Then

$$c = -1 + \frac{2}{1 + \omega^2}, \quad s = \frac{2\omega}{1 + \omega^2}, \quad \theta = \pi - 2 \operatorname{atan}(\omega^{-1}).$$

This is not a half-turn, although its standard part is one. The matrix entries are bounded; its Cayley coordinate alone is infinite. Its quaternion lift has scalar part $1/\sqrt{1 + \omega^2}$ and third imaginary part $\omega/\sqrt{1 + \omega^2}$. The largest-coordinate extraction algorithm selects the latter rather than dividing by a tiny scalar coordinate.

27.3 Infinitesimally different axes

The unit vector

$$n = \frac{(1, \varepsilon, \varepsilon^2)^T}{\sqrt{1 + \varepsilon^2 + \varepsilon^4}}$$

defines a half-turn $H_n = 2nn^T - I$ with the same standard part as H_{e_1} . The half-turns are distinct because their fixed lines differ. Their difference is visible already in the off-diagonal entry $(H_n)_{12} = 2\varepsilon/(1 + \varepsilon^2 + \varepsilon^4)$. This is axis information that a real standard-part calculation discards.

27.4 Noncommutativity first appears at a finer scale

For positive infinitesimals ε, δ , let $U = \text{Cay}(\varepsilon e_1)$, $V = \text{Cay}(\delta e_2)$. Then

$$(U, V) = I + 4\varepsilon\delta\widehat{e}_3 + \text{terms divisible by } \varepsilon\delta \text{ of total degree at least three.} \quad (27.3)$$

In valuation language, the remainder has valuation at least $\min(2v(\varepsilon) + v(\delta), v(\varepsilon) + 2v(\delta))$, strictly above the displayed term. This follows either by rational multiplication or by the BCH bracket. The two rotations commute to first order but not exactly.

27.5 Infinite generators and local sensitivity

Put $w = \omega e_1$ and $z = \pi e_2$. Equation [equation \(25.1\)](#) gives $\mathcal{E}(\widehat{w}) = I$ and $\mathcal{E}(\widehat{z}) = H_{e_2}$. Yet

$$\|w + z\| = \sqrt{\omega^2 + \pi^2} = \omega + \frac{\pi^2}{2\omega} + O(\omega^{-3}),$$

so the global phase removes ω and leaves an infinitesimal angle. Consequently

$$\mathcal{E}(\widehat{w + z}) = I + \frac{\pi^2}{2\omega}\widehat{e}_1 + O(\omega^{-2}),$$

not H_{e_2} . Here $O(\omega^{-k})$ means an entrywise bound by a finite real constant times that scale; it is not an unspecified real limit. This illustrates why global radial exponentiation is not additive across noncommuting directions, even when one generator has trivial image.

27.6 A nonreal copy of the octahedral group

Let H_0 be the 24 signed permutation matrices of determinant one, and set $C = \text{Cay}(\varepsilon e_1 + \varepsilon^2 e_2)$. Then

$$H = CH_0C^{-1}$$

is an exact 24-element subgroup with generally nonreal entries. Its standard part is H_0 , and conjugation by $C^{-1} \in K$ restores the ordinary real group. The multiplication table is unchanged; the coordinate realization has been deformed at several infinitesimal scales.

Part III

Computation, limits, and provenance

28 Exact computation and formalization routes

The two sources give complementary implementation plans, one for the geometry and one for the rotation group. Both are printed; neither describes existing software.

28.1 A coordinate-to-invariants algorithm (source 04)

For three proposed vertices P_1, P_2, P_3 on a sphere of center O and radius $R > 0$, an exact workflow is as follows.

1. Verify $\|P_i - O\|^2 = R^2$ and set $u_i = (P_i - O)/R$. Compute $x = u_2 \cdot u_3$, $y = u_3 \cdot u_1$, $z = u_1 \cdot u_2$, and $\tau = u_1 \cdot (u_2 \times u_3)$.
2. Check $\tau \neq 0$ for a proper triangle. Retain its sign for orientation, and put $d = |\tau|$. Verify the redundant identity $d^2 = 1 + 2xyz - x^2 - y^2 - z^2$ as a useful certificate.
3. Compute the side angles with $a = \text{atan2}(\|u_2 \times u_3\|, x)$ and cyclic variants; compute $\alpha = \text{atan2}(d, x - yz)$ and cyclic variants. All these calls have strictly positive first argument.
4. Compute $M = 1 + x + y + z$ and $E = 2 \text{atan2}(d, M)$. Return physical side lengths Ra, Rb, Rc and area $R^2 E$. Centers, polar vertices, and transport matrices are obtained from the explicit formulas above, with each denominator checked before division.

This is a mathematical algorithm over a field with exact operations, order decisions, positive square roots, and finite-angle evaluation. It is not a claim that every surreal input has a finite machine representation. A general Conway normal form has set-sized support, and a generic element need not come with effective coefficient or support oracles. An actual computer algebra implementation must name its representable subclass and its supported comparison procedure.

For exact finite descriptions, it is useful to separate an algebraic certificate from a displayed numerical approximation. For example, $\tau \neq 0$ can be certified by a leading coefficient even when $\text{st}(\tau) = 0$. Near tangency, an exact discriminant sign should not be replaced by an ordinary floating-point tolerance. Truncated Hahn expansions must retain enough support to decide the leading nonzero term of the predicate being tested; an unresolved cancellation is an unknown sign, not a zero.

28.2 An implementation hierarchy for rotations (source 09)

There is no finite data structure representing every surreal number with a decidable exact equality test. A practical rotation package must name its scalar backend. The finite algebra in this report works over any exact ordered field with the required square roots; analytic operations require more.

For a backend of rational functions in one formal positive infinitesimal, Cayley coordinates are especially convenient: composition, inverse, and the action on vectors are rational formulas. Unit quaternions can be kept projectively, avoiding square roots until normalization is needed. A bounded Puiseux or Hahn-series backend can additionally track leading exponents, standard parts, and finite-order BCH truncations. General supports need an explicit effective representation; a

mathematical support theorem is not an algorithm for deciding equality of arbitrary input names.

A robust division of responsibilities is the following.

1. The scalar layer supplies exact arithmetic, order, positive square roots, and, when supported, valuation and standard part.
2. The finite rotation layer supplies matrix, quaternion, algebraic axis-angle, and Cayley representations, with all exceptional branches.
3. The supported-series layer supplies infinitesimal exp/log and BCH with declared error valuations and support certificates.
4. The phase layer distinguishes finite angles from the chosen global Ehrlich–Kaplan phase; it never treats an infinite argument as an uncentered convergent Taylor input.

The exact rational law on $\mathbf{m}^3$ can often replace BCH entirely. BCH is useful for logarithmic generators and graded errors, not necessary merely to multiply two rotations.

28.3 A layered Lean development for the geometry (source 04)

At source 04’s pin, the repository’s `Surreal/Algebra/Geometry.lean` works with two-coordinate complexifications and field-valued moduli [12]. Its approach suggests a clean separation for the three-dimensional extension. The following are proposed layers, not assertions that these modules already exist or have been compiled. At the merge pin 50cb709 no such module exists.

Layer	Assumptions	Principal obligations
Vector algebra	Commutative ring	Dot/cross identities; determinants; Gram polynomials
Ordered geometry	Ordered field; positive square roots	Norm inequalities; projections; affine incidence; intersections
Orthogonal algebra	Real closed field when needed	Principal axes; axis–angle cosine/sine data; quaternion action
Spherical algebra	Unit vectors; nonzero determinant	Tangent and normal formulas; cosine/sine laws; reconstruction
Finite angles	Compatible restricted-analytic structure	Phase bijection; inverse branches; angle inequalities
Area and transport	Proper minor-arc polygons	Excess branches; finite additivity; edge transport; holonomy
Scale analysis	Valuation and standard part	Noncancellation; asymptotic bounds; degeneration certificates

An appropriate vector type is $\text{Fin}(3) \rightarrow F$ or a three-coordinate structure. Length should initially be an F -valued function, not a real-valued `NormedSpace` norm. Theorems about matrices and finite sums can be proved before introducing any surreal construction. Angles should not enter the algebraic spherical-law layer until the circle-phase bridge is available. Proper-class statements are then obtained externally from compatible set-sized workspaces, rather than by pretending that all of `No` is a small type in one universe.

The exceptional SSA family in section 9.2 is a useful specification test: a solver returning only a list of at most two triangles is mathematically incomplete unless it explicitly excludes the case $\cos b = \sin b \cos \alpha = 0$. Its output type should distinguish no solution, finitely many solutions, and a parametrized family.

28.4 A Lean development plan for the rotation group (source 09)

The repository’s formalization ledger is the authority for which existing statements have actually been matched to compiled Lean declarations [15]. This report does not add a compiled Lean development, and its proofs must not be marked proved in that ledger solely because they appear here or pass symbolic tests.

The natural finite-algebra modules would begin with F^3 , the hat map, and equations (3.5) and (3.6), then prove the distance-isometry theorem and axis theorem. Next come the Hamilton action, spin kernel and explicit surjectivity, the matrix Cayley equivalence, and the centralizer cases. A standard-part interface then gives the split extension and the kernel chart. Finite subgroup rigidity can be formalized using a finite sum and the positive matrix-root interface, without invoking Hahn series.

The torsion-free and unique-divisibility proofs are also finite algebra over a real closed extension of $\mathbb{R}$. The single-commutator construction reduces to one checked quaternion identity, positive fourth roots, and the already proved transitivity theorem. These are particularly attractive next targets: they deliver significant group structure without depending on a general surreal analytic inverse function theorem.

For the analytic modules, formal noncommutative series must be evaluated only after establishing joint strong summability. The derivative theorem requires a fine Fréchet interface in three coordinates and local quadratic remainders; it cannot be obtained merely by substituting surreal symbols into a theorem whose hypotheses require a real Banach space. Semialgebraic fundamental groups are a further, independent formalization layer.

28.5 The companion checks

Each source came with an exact SymPy script. Both are kept byte-identical, under the prefixed names given here.

`code/04-three-space-verify_identities.py` (source 04) checks selected identities and expansions: the vector and Gram identities, the spherical-excess polynomial, the quaternion holonomy numerator, the chart metrics, the local curvature coefficient, and the worked examples. Its recorded run, `data/04-three-space-verification.txt`, passed all 19 test groups. These checks can detect a sign, coefficient, or transcription error in the displayed formulas. They do not prove the analytic transfer theorem, establish Hahn summability, discharge every branch condition, or formally verify the geometry. Those distinctions are part of the mathematical specification, not merely software caveats.

`code/09-so3-verify.py` (source 09) checks the hat identities, quaternion multiplication and rotation formulas, orthogonality and determinant of the Cayley chart, its composition law, the scalar commutator identity used in theorem 19.3, selected leading commutators, and the Jacobian calculation in an axis-adapted frame. It also checks all 576 products of the 24-element octahedral group and an exact rational averaging intertwiner. Its recorded run, `data/09-so3-verification.txt`, passed all 24 checks. Rational specialization tests supplement, but do not replace, symbolic identities. These checks can catch sign errors, factor-of-two errors, and incorrect matrix branches. They do not prove real closedness of $\mathbf{No}$, the Neumann support lemma, a global topology theorem, or a Lean kernel judgment. Those claims rest on the mathematical proofs and explicitly imported results stated in the text.

Both recorded runs used Python 3.13.5 and SymPy 1.14.0. For this merge both scripts were rerun with Python 3.14.4 and SymPy 1.14.0. The rerun reproduced 19/19 groups and 24/24 checks;

only the version line and the timing line differ. Neither script writes a file.

29 Non-claims and limits, by source

A merged report must not silently upgrade either source. This section records every limitation, disclaimer and priority caveat stated by either manuscript. Most also appear at the point of use.

29.1 Source 04 (Part I)

- G1. No priority claim is made, and no named open problem is claimed to be resolved. This covers in particular the uniform gnomonic estimate ([theorem 13.1](#)), the exceptional-family classification ([proposition 9.2](#)), and their combination into a surreal development. Repeated classical identities are not presented as new results merely because their arguments are surreal.
- G2. Established surreal analytic machinery and classical identities are cited: [1, 2, 3, 4, 6, 7]. The proofs supply the precise extensions used.
- G3. The symbolic checks test selected polynomial identities and formal coefficients only. They do not verify the whole development or its foundations ([section 28.5](#)).
- G4. No Lean formalization is claimed. The layers of [section 28.3](#) are proposals, not assertions that the modules exist or have been compiled.
- G5. Real-valued metric completeness, ordinary Riemann integration, and a global oscillatory differential equation are not imported without additional hypotheses.
- G6. The finite theory does not assert that no distinguished global sine or cosine exists ([remark 2.3](#)). It needs no phase for infinite inputs. The zero-centred sine series at ω is not strongly summable, whatever global definition is chosen.
- G7. The distance is field-valued, not an ordinary real-valued metric, and F^3 is not made a Banach space over $\mathbb{R}$.
- G8. The “double cover” of [section 5](#) is an algebraic two-to-one homomorphism, not a topological covering assertion in the fine topology.
- G9. The spherical-coordinate metric and Jacobian are finite identities in restricted-analytic coordinate derivatives. They are not an integral transformation theorem for every function on F^3 .
- G10. The minimizing property of minor arcs concerns finite chains of minor great-circle arcs. No supremum over arbitrary partitions, no real-valued length metric, and no unrestricted class of nondefinable curves is introduced.
- G11. Area is a finitely additive functional on geodesic polygons. The cap formula extends one specific analytic formula. It asserts neither that arbitrary subsets have a surreal measure nor that Darboux sums define it, and no countably additive area measure is inferred.
- G12. Derivatives in [section 12](#) are taken with respect to external chart or curve parameters. They are not an intrinsic derivation of **No**, and no existence theorem for arbitrary differential equations is invoked.
- G13. Parallel transport is canonical by explicit analytic continuation. It is unique within the definable restricted-analytic class, not among arbitrary intrinsically differentiable fields.

- G14. The normal-coordinate estimate [equation \(12.8\)](#) is absolute. It claims no small relative error when $p - q$ is exceptionally small.
- G15. The coordinate-to-invariants algorithm does not claim that every surreal input has a finite machine representation. An implementation must name its representable subclass and comparison procedure.
- G16. Completeness, ordinary topological compactness, existence for arbitrary smooth ODEs, and a supremum-based length for every curve are not claimed. Finite dimension does not permit unrestricted summation of arbitrary families of surreals. Ordinary-size angles do not imply ordinary-size arc lengths, and a global phase at infinite arguments would be an additional convention.
- G17. The repository comparison is not an exhaustive audit of every file, or of the published work on non-Archimedean geometry. The repository is a research source, not a substitute for published foundations.
- G18. No completed repository integration, general-purpose surreal computer algebra, or formal verification is asserted.

29.2 Source 09 (Part II)

- S1. The manuscript is an AI-assisted research exposition, subject to mathematical review. It makes no claim of new priority and no claim of complete machine-checked formalization.
- S2. No new priority is claimed for classical quaternion theory or for the perfectness of the infinitesimal rotation group. Bays–Peterzil record that every element is a commutator [9, §3.1], and [theorem 19.3](#) gives an elementary exact construction of that fact.
- S3. The quaternion spin description, standard-part reduction of unit quaternions, infinitesimal exponential and logarithm, and valuation filtration in the surquaternion report, finite angles in the trigonometry report, and the foundations report’s distinction of strong summation from fine convergence are starting points, not discoveries of the source.
- S4. The localization result, [proposition 2.1](#), does not assert that the subfield is closed under strong sums, or that its intrinsic topology is the one induced from **No**.
- S5. The homogeneous-space statement [equation \(16.1\)](#) is algebraic and semialgebraic, not a statement about ordinary real manifolds.
- S6. $G(F) \cong \mathbb{P}^3(F)$ is a statement about F -points, not an isomorphism of algebraic varieties.
- S7. Connectedness and absolute simplicity of the algebraic group imply neither abstract simplicity of $G(F)$ nor connectedness of its fine topology.
- S8. Unique divisibility of K_F does not make it a $\mathbb{Q}$ -vector space.
- S9. [Theorem 20.2](#) gives conjugate realizations. It does not say that finite subgroups are literally constant matrices, and it says nothing about infinite or set-generated subgroups.
- S10. The dual number κ is formal, not a surreal infinitesimal. For an angular-velocity equation, existence and uniqueness of solutions is a separate question. The Berarducci–Mantova derivation is not the variable derivative of [section 25](#).
- S11. The BCH truncation valuations are finite-order error certificates, not convergence through all positive surreal tolerances. The support lemma does not assert that $n\gamma$ is cofinal.

- S12. [Theorem 23.1](#) gives an inclusion of filtration levels, not equality of valuations in every case. The BCH description does not make K pronilpotent in an unspecified topology.
- S13. The Bays–Peterzil bi-interpretability theorem is imported, not proved, and it concerns set-sized structures. It is not automatically transferred to the full class.
- S14. The Ehrlich–Kaplan phase is canonical for its specified normalization. This is not an assertion that every conceivable global phase convention agrees with it.
- S15. The identity fiber of the global rotation exponential is not a kernel, and that exponential is not a homomorphism on the additive Lie algebra.
- S16. No inverse function theorem is inferred from nonsingularity of [equation \(25.4\)](#), and the derivative formulas are not chain rules for a field derivation.
- S17. The size restriction in [theorem 26.1](#) cannot be dropped. The spin fundamental group $\mathbb{Z}/2\mathbb{Z}$ is semialgebraic, not the ordinary fundamental group of the intrinsic or fine space. Definable compactness is not ordinary compactness. The fine topology does not make $G(\mathbf{No})$ an ordinary real Lie group, and a “surreal universal cover” is meaningful only after a category is specified.
- S18. No finite data structure represents every surreal number with decidable equality, and a support theorem is not an equality algorithm.
- S19. No compiled Lean development is added. The proofs must not be marked proved in the formalization ledger solely because they appear here or pass symbolic tests.
- S20. The checks do not prove real closedness of $\mathbf{No}$, the Neumann support lemma, a topology theorem, or a Lean kernel judgment ([section 28.5](#)).
- S21. There is no claim to classify all abstract subgroups or automorphisms, to construct a Haar measure on the full class, or to obtain a physical model merely from a change of scalars.
- S22. The repository was not rebuilt for the source, and the source changed no repository file.

29.3 The merge

- M1. The merge adds no mathematical result. Each statement comes from one of the sources, and each statement proved by both is marked in [section 1.1](#). Merge-written remarks only relate statements to each other or to other reports (for example the factor-two comparison after [theorem 23.1](#)). [Theorem 5.4](#) and [theorem 5.1](#) combine clauses that the two sources stated separately; the proofs printed are the sources’ proofs.
- M2. The renamings of [section 1.4](#) change symbols, not statements.
- M3. [Theorem 5.5](#) is cited from the surquaternion report and not reproved. Its proof there is outside the scope of this report’s review.
- M4. The report has not been refereed. None of its labels is mapped to a Lean declaration.

30 Conclusion and directions

The first two paragraphs are source 04’s conclusion, and the rest is source 09’s.

Three-dimensional surreal geometry has a large, rigorous core that is both familiar and genuinely more informative than its real standard-part picture. Affine incidence, orthogonality, quadrics,

rotations, and finite volumes are ordered-field algebra. Canonical finite angles supply the usual trigonometric language for all directions, even when coordinates and physical lengths are infinite. Spherical geometry then admits exact sine and cosine laws, reconstruction criteria, centers, a finite-polygon area functional, and a curvature–holonomy identity.

The indispensable extra care concerns branch choices, degeneracy, and the meaning of analysis. A zero standard part is not a zero determinant; a small determinant need not mean small spherical area; a locally constant, intrinsically differentiable function need not be globally constant; and a formal integral symbol is not a measure. With those distinctions made explicit, infinitesimal geometry becomes a calculus of exact scales rather than a collection of heuristic limiting pictures.

The correct analogue of the ordinary spatial rotation group is not a new multiplication invented for surreal coordinates: it is the positive-definite special orthogonal class group $\mathrm{SO}(3, \mathbf{No})$. Its axes, algebraic angles, quaternion double cover, and rational charts have exact finite formulas. Every matrix entry is bounded by one, but the entries can retain arbitrarily fine surreal information.

The strongest structural contrast with the ordinary group is the extension $K \rtimes \mathrm{SO}(3, \mathbb{R})$. The real quotient does not exhaust the geometry. The kernel has a full hierarchy of cross-product layers, yet the entire kernel is perfect and uniquely divisible. It contains no finite torsion and no nontrivial finite quotient. Finite subgroups have no genuinely new forms: finite averaging and polar normalization conjugate them to real subgroups.

Finite angles suffice to represent every rotation. The canonical global phase also exists, but its period class is $2\pi \mathbf{Oz}$, not just ordinary multiples of 2π . The resulting radial rotation exponential has explicitly described fibers and an exact Jacobian. Infinite generators can be critical or severely ill-conditioned without producing unbounded rotation matrices.

Finally, the phrase “like $\mathrm{SO}(3)$ ” has a category-dependent answer. The algebraic group is three-dimensional and connected. Over a set-sized real closed field it is semialgebraically connected and definably compact, with spin fundamental group $\mathbb{Z}/2\mathbb{Z}$. In the non-Archimedean intrinsic topology it is totally separated and not locally compact. In the full surreal fine topology, every set-sized subset is closed and discrete. None of these assertions contradicts the others.

Further work should therefore specify a category before posing a problem. Substantial directions include normal-subgroup classification compatible with convex valuations; effective multi-scale rotation algorithms with support and conditioning certificates; integration of the finite proofs into the repository’s Lean development; and comparison of semialgebraic, valued-field, and fine analytic representation theories. We do not claim to classify all abstract subgroups or automorphisms, construct a Haar measure on the full class, or obtain a physical model merely from a change of scalars.

A Formula and convention reference for Part II

Object	Convention or formula
Rotation group	$G(F) = \{A : A^T A = I, \det A = 1\}$, acting on column vectors.
Hat map	$\hat{u}v = u \times v$, with the standard right-handed cross product.
Rodrigues	$\mathrm{Rot}(n; c, s) = cI + (1 - c)nn^T + s\hat{n}$.
Half-turn	$H_n = 2nn^T - I$; its centralizer is a line stabilizer, not just an axis circle.

Object	Convention or formula
Spin map	$p(a, u) = (a^2 - \ u\ ^2)I + 2uu^T + 2a\hat{u}$, for $a^2 + \ u\ ^2 = 1$.
Cayley chart	$\text{Cay}(t) = ((1 - t \cdot t)I + 2tt^T + 2\hat{t})/(1 + t \cdot t)$.
Cayley product	$t \star u = (t + u + t \times u)/(1 - t \cdot u)$, where the denominator is nonzero.
Infinitesimal kernel	$K = \ker \text{st}$, identified with all $\mathfrak{m}^3$ by the Cayley chart.
Logarithmic leading term	$\text{Log}_{\mathfrak{m}} \text{Cay}(t) = 2\hat{t} + \text{higher terms}$.
Valuation	$v(\omega^{-\gamma}) = \gamma$; infinitesimals have $v > 0$.
Graded coordinate	$\ell_{\gamma}(U) = 2 \text{st}(\omega^{\gamma} t(U))$.
Group commutator	$(U, V) = UVU^{-1}V^{-1}$.
Lie bracket	$[\hat{u}, \hat{v}] = \widehat{u \times v}$.
Finite phase period	$\ker \text{cis} = 2\pi\mathbb{Z}$, on $\mathcal{O}$.
Global phase period	$\ker \text{cis}_{\text{EK}} = 2\pi\mathbf{Oz}$, on $\mathbf{No}$.
Global exponential	Radial map $\mathcal{E}$ of equation (25.1) ; its identity fiber is not an additive kernel.
Fine Jacobian	Right-trivialized as $(d\mathcal{E}_w h)\mathcal{E}_w^{-1} = \widehat{\mathcal{J}(w)h}$.
Critical radii	$r \in (2\pi\mathbf{Oz})_{>0}$, rank one; $r = 0$ is regular.

B Source, dependency, and verification ledger

B.1 Pins and files

Item	Value
Source 04 pin	d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0
Source 09 pin	dcf86662b574988e75d08d515d3a2a5ba7bbc0d7
Placement	5fe7f8d (both manuscripts placed, source 04's text as the starting file)
Merge written at	50cb709
Source 04 files	code/04-three-space-verify_identities.py, data/04-three-space-verification.txt, data/04-three-space-requirements.txt
Source 09 files	code/09-so3-verify.py, data/09-so3-verification.txt, data/09-so3-requirements.txt

The manuscripts themselves, their PDFs, READMEs and source 09's checksum list are not shipped. The code and data files are byte-identical to the delivered ones.

B.2 Source 04: inspected sources and dependencies

At its pin, source 04 specifically inspected the trigonometry report and the Lean module `Surreal/Algebra/Geometry.lean`. The repository is a research source, not a substitute for published foundations. The source comparison is not an exhaustive audit of every file or of all published work on non-Archimedean geometry.

The main dependencies of Part I are these. Real closedness and the restricted-analytic Hahn-field model theorem are imported from [1], with support-combinatorial cautions informed by its erratum

[2]. The canonical global extension available from an initial integer part is attributed to [3]; Part I needs only its finite restriction. The classical spherical identities are recorded in [6]; their algebraic proofs and sign conditions are provided here. The solid-angle expression is classical and attributed to [7]. No priority claim is made for the remaining formulations, including the uniform gnomonic estimate, the exceptional-family classification, or their combination into a surreal development. The proofs are mathematical proofs, not Lean proof terms.

B.3 Source 09: inspected sources and dependency ledger

At its pin, source 09 reviewed the documentation index, the introductory status of the formalization ledger, the surquaternion README and the relevant surquaternion source sections. Its bibliography separates the report directories identified through that index from the source sections actually inspected. It did not rebuild the repository.

Result family	Essential assumptions	Evidence in this report
Isometries, axes, spin, Cayley, centralizers	Real closed ordered field; finite matrices	Direct finite proofs and selected symbolic identities; spin cited from <code>squat:thm:rotations</code> .
Standard-part splitting	Real closed $F \supseteq \mathbb{R}$	Direct ring-homomorphism proof.
Torsion-free and unique divisibility	Real closed $F \supseteq \mathbb{R}$	Algebraic circle and root proofs.
Perfect infinitesimal group	Proper real closed extension; fourth roots	Explicit commutator construction; known precedent [9].
Finite-subgroup rigidity	Finite group, real closed $F \supseteq \mathbb{R}$	Averaging and positive polar factor, fully proved.
Strong exp/log and BCH	Set-supported normal forms or a divisible Hahn workspace	Support proof using Higman's lemma; formal identities.
Global phase	Ehrlich–Kaplan initial-integer-part normalization	Imported normalization, explicitly expressed and analyzed.
Rotation exponential differential	Fine variable derivatives and local Taylor remainder bounds	Direct radial/perpendicular calculation.
Full fine discreteness	Full surreal cut property; small index sets	Direct separating-radius proof.
Semialgebraic homotopy	Set-sized real closed field	Imported comparison theorem [10] and explicit spin branches.
Machine formalization	A compiled proof assistant development	Not supplied; no such verification is claimed.

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